R and clinical/lab data

2026 edition

Matteo Miotto

2026-05-01

Contents

Welcome **5**

1 Dataframes, vectors and numbers **7**

1.1 Dataframes 7

1.2 Import file 7

1.3 Vectors 10

1.4 Save a dataframe 21

1.5 Home exercise 21

2 Strings, Factors and Dataframe Handling **25**

2.1 What we will see today 25

2.2 Strings and Factors 25

2.3 Dataframe Handling 41

2.4 Save your results 60

2.5 Home exercise 61

2.6 Bonus: how the clinical data were created 62

3 Data visualization **63**

3.1 Data visualization in R with ggplot2 63

3.2 Geom 64

3.3 Labs 79

3.4 Scales 80

3.5 Facet 86

3.6 Theme 88

3.7 Save the plots 90
3.8 Advanced examples 91

4 Statistics **95**
4.1 Categorical data 96
4.2 Contingency table 96
4.3 Chi-squared test 99
4.4 Comparing continuous variables 102
4.5 PCA 112
4.6 Linear models 121

5 For loops and If/Else statements **127**
5.1 For loops 127
5.2 If/Else statements 131
5.3 Putting it all together: reading files in a loop 132

6 Extra: Table 1 **135**
6.1 Load necessary packages 135
6.2 Load table1_df 136
6.3 Select column of interest 137
6.4 Define group levels 138
6.5 Detect variable types 139
6.6 Continuous variables 140
6.7 Categorical variables 143
6.8 Combine everything 144
6.9 Save it 146

Welcome

Welcome to this course, here you can find the notebooks used in the PhD course of 2026.

Subscribe to receive notification when a new content is uploaded! NO SPAM, just updates.

Stay updated

Get notified when new material is published.

Powered by Buttondown

Chapter 1

Dataframes, vectors and numbers

1.1 Dataframes

The usual data you will deal with are tabular (rows x columns) data, usually improperly called "Excel files".

The standard formats for raw data files in data analyses are *.csv*, *.tsv* and *.txt* files; these should be our starting point.

1.2 Import file

The first thing to do is to import the file into R session. We can do it with several functions, but the standard one for *.csv* files is `read.csv`.

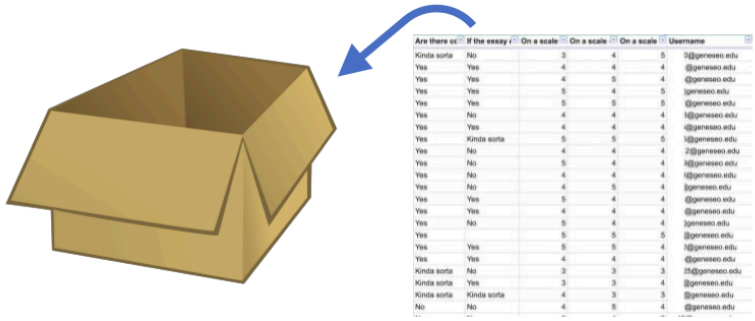
Let's see how to do so:

```
my_data <- read.csv("data/input_df.csv")
```

1.2.0.1 Notes about variables

Variables are the "memory" of an R session, they represent box where R stores data/values you want.

You can think them as boxes that have a name on them:



Are there co	If the essay i	On a scale	On a scale	On a scale	Username
Kinda sorta	No	3	4	5	3@geneseo.edu
Yes	Yes	4	4	4	@geneseo.edu
Yes	Yes	4	5	4	6@geneseo.edu
Yes	Yes	5	4	5	)geneseo.edu
Yes	Yes	5	5	5	6@geneseo.edu
Yes	No	4	4	4	1@geneseo.edu
Yes	Yes	4	4	4	4@geneseo.edu
Yes	Kinda sorta	5	5	5	1@geneseo.edu
Yes	No	4	4	4	2@geneseo.edu
Yes	No	5	4	4	8@geneseo.edu
Yes	No	4	4	4	1@geneseo.edu
Yes	No	4	5	4	)geneseo.edu
Yes	Yes	5	4	4	@geneseo.edu
Yes	Yes	4	4	4	@geneseo.edu
Yes	No	5	4	4	)geneseo.edu
Yes		5	5	5	)geneseo.edu
Yes	Yes	5	5	4	1@geneseo.edu
Yes	Yes	4	4	4	@geneseo.edu
Kinda sorta	No	3	3	3	75@geneseo.edu
Kinda sorta	Yes	3	3	4	)geneseo.edu
Kinda sorta	Kinda sorta	4	3	3	)geneseo.edu
No	No	4	5	4	@geneseo.edu

There are important **rules** about variables:

- In each variable you can store one single item (a number, a string, a dataframe, a list, ...)
- variable names can start only with a letter (uppercase or lowercase)
- variable names can contain **only** letters, numbers, underscores (_) and dots (.). **No spaces, no dash (-) or any other special characters**
- there could be only ONE variable with a specific name, saving an object with the same of another results in the substitution of the first one
- suggestion: variable names should be meaningful to you (and to the ones you share the code with): e.g. `x` is not a good name, while `patients_glucose_data` is preferred

1.2.1 First steps on a dataframe

When importing a dataframe, there is a fundamental step to do: *inspection*. Inspection helps us to understand how the dataframe is structured, which type

of data we have, which columns etc.

To do it, we use 3 functions that you need to remember (tatoo yourself): `summary`, `str` and `head`. Let's see them:

```
summary(my_data)
```

```
     sample        timepoint           genotype
 ↪  treatment
 Min.   :  1.00   Length:192         Length:192
 ↪  Length:192
 1st Qu.: 48.75   Class :character   Class :character   Class
 ↪  :character
 Median : 96.50   Mode  :character   Mode  :character   Mode
 ↪  :character
 Mean   : 96.50
 ↪
 3rd Qu.:144.25
 ↪
 Max.   :192.00
 ↪
      area        intensity
 Min.   :3000   Min.   : 9130286
 1st Qu.:3259   1st Qu.:16283904
 Median :3517   Median :18894103
 Mean   :3508   Mean   :19419175
 3rd Qu.:3769   3rd Qu.:22090201
 Max.   :3988   Max.   :29812114
```

```
str(my_data)
```

```
'data.frame':   192 obs. of  6 variables:
 $ sample   : int  1 2 3 4 5 6 7 8 9 10 ...
 $ timepoint: chr  "P1" "P1" "P1" "P1" ...
 $ genotype : chr  "wt" "wt" "wt" "wt" ...
 $ treatment: chr  "n" "n" "n" "n" ...
 $ area     : num  3125 3500 3773 3538 3936 ...
 $ intensity: num  16015465 19449815 21118210 18100075 20718554
 ↪  ...
```

We can see how many rows and columns we have, as well as which columns, the data type of each and the first entries.

Lastly, `head` is the function that shows us the first 6 entries of the dataframe, as a table:

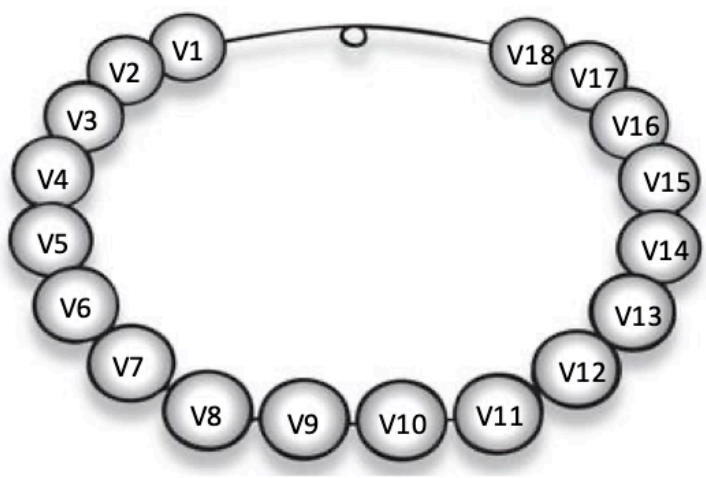
```r
head(my_data)
```

```
  sample timepoint genotype treatment     area intensity
1      1        P1       wt         n 3124.709  16015465
2      2        P1       wt         n 3499.559  19449815
3      3        P1       wt         n 3773.000  21118210
4      4        P1       wt         n 3538.320  18100075
5      5        P1       wt         n 3936.014  20718554
6      6        P1       wt         n 3646.389  19507110
```

1.3 Vectors

Dataframes are organized and read by columns. Each column is a **vector** in R.

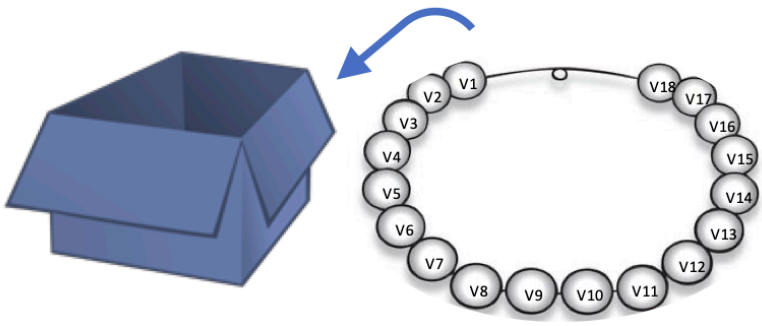
A vector is a collection of data of the **same type**; you can think of a vector as a pearl necklace, with each pearl being a single value:

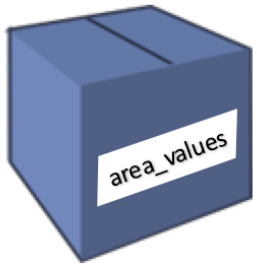


1.3.1 Accessing dataframe columns

For exercise purposes, we can create a new variable storing the values of the area column of the dataframe. To extract a single column from a dataframe, we use the structure `dataframe_name$column_name`.

```
area_values <- my_data$area
```





1.3.2 Accessing values of a vector: slicing

Slicing is the operation of extract values out of a variable (in this case a vector). It is done to answer these questions:

- *Which is the first value of the vector?*
- *Which is the value at position n of a vector?*
- *Which are the values from position n to m?*

There are different ways in which we can slice a vector, and today we will see indexing (through positions).

The syntax is:

- `variable_name[position]` to get one single value

- `variable_name[start:end]` to get elements from `start` to `end`

REMEMBER: In R, the first element is at index 1.

```
# extract one single value

first_area_value <- area_values[1]
print(first_area_value)
```

```
[1] 3124.709
```

```
# extract range of values
from_fifth_to_ninth_areas <- area_values[5:9]
print(from_fifth_to_ninth_areas)
```

```
[1] 3936.014 3646.389 3073.109 3717.063 3887.006
```

Q: *What 5:9 does in R?* A: It creates a vector containing numbers from 5 to 9, that's why it works:

```
print(5:9)
```

```
[1] 5 6 7 8 9
```

Q: *And to get last element?*

A: We need to know the lenght of the vector and use it as index. To get the length of a vector we use the function `length`:

```
length_area <- length(area_values)
print(length_area)
```

```
[1] 192
```

```
last_area <- area_values[length_area]
print(last_area)
```

```
[1] 3808.121
```

Exercise 1.1. Which is the 5th value of the intensity column of the dataframe?

Solution

We can do it in many steps:

```
intensity_values <- my_data$intensity

fifth_intensity <- intensity_values[5]

print(fifth_intensity)
```

```
[1] 20718554
```

Or in just one step:

```
fifth_intensity <- my_data$intensity[5]

print(fifth_intensity)
```

```
[1] 20718554
```

1.3.2.1 Replace a value by index

We can replace of a vector using indexes.

Note: new values should be of the same type (numbers for numbers, strings for strings ecc), otherwise everything will be changed to strings (most of the time).

To change a single value, we can type `variable_name[position] <- new_value`; to change multiple values `variable_name[start:end] <- new_values`.

1.3.3 Numbers in R

Now that we know how to get and extract numbers from numeric vectors, let's see what we can do with numbers in R.

1.3.3.1 Declare a new numeric variable

To declare a new numeric variable, we use the statement `variable_name <- number`:

```
conversion_factor <- 1000

print(conversion_factor)
```

```
[1] 1000
```

```
print(typeof(conversion_factor))
```

```
[1] "double"
```

Decimal point numbers wants the dot `.` as decimal separator:

```
decimal_n <- 215.6
print(decimal_n)
```

```
[1] 215.6
```

```
print(typeof(decimal_n))
```

```
[1] "double"
```

1.3.3.2 Arithmetical operations

We can perform all sort of matematical operation:

```
# sum

print(2 + 5)
```

```
[1] 7
```

```
# subtraction

print(7 - 2.5)
```

```
[1] 4.5
```

```r
# multiplication
print(12 * 5)
```

```
[1] 60
```

```r
# division
print(105 / 43)
```

```
[1] 2.44186
```

```r
# exponential
print(5 ** 2)
```

```
[1] 25
```

```r
print(5 ^ 3)
```

```
[1] 125
```

```r
# square root
print(sqrt(12))
```

```
[1] 3.464102
```

We can combine them all to do more complex operations. For example, you can resolve this equation $x^2 - 7x + 12 = 0$. We know that the formula to resolve this equation is: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$, so we can reconstruct them in R:

```r
a <- 1
b <- -7
c <- 12

x1 <- (-b + sqrt(b**2 - 4*a*c)) / (2 * a)
x2 <- (-b - sqrt(b**2 - 4*a*c)) / (2 * a)

print(x1)
```

```
[1] 4
```

```r
print(x2)
```

```
[1] 3
```

Exercise 1.2. Which are the last 3 values of the intensity column of the dataframe?

Solution

We can do it in many steps:

```r
intensity_values <- my_data$intensity

length_intensity <- length(intensity_values)

third_to_last_index <- length_intensity - 2

last_3_intensities <-
 ↪  intensity_values[third_to_last_index:length_intensity]

print(last_3_intensities)
```

```
[1] 22965783 22027811 22338574
```

1.3.4 Vector-number operations

What we have seen so far on arithmetical operations between single numbers, can be applied also to numerical vector x single number operations.

In our data, area is expressed in µm^2 and we want to convert it into mm^2. To do so, we should divide each value of the area by a factor of 1,000,000; the cool thing about R is that this is done automatically when we use the statement `numeric_vector <operand> number`:

```r
division_factor = 1e6

areas_mm2 <- area_values / division_factor

print(head(areas_mm2))
```

```
[1] 0.003124709 0.003499559 0.003773000 0.003538320 0.003936014
↪  0.003646389
```

```
print(head(area_values))
```

```
[1] 3124.709 3499.559 3773.000 3538.320 3936.014 3646.389
```

This is true for ANY arithmetical operations:

```
print(head(intensity_values))
```

```
[1] 16015465 19449815 21118210 18100075 20718554 19507110
```

```
print(head(intensity_values + 10))
```

```
[1] 16015475 19449825 21118220 18100085 20718564 19507120
```

1.3.5 Operations between vectors

It is possible also to perform element-wise operations between vectors.

Let's load a new dataset with some patient data, and calculate the BMI of each patient.

```
patient_df <- read.csv("data/patient_data.csv")

str(patient_df)
```

```
'data.frame':   100 obs. of  3 variables:
 $ patient_id: int  1 2 3 4 5 6 7 8 9 10 ...
 $ weight    : num  75.8 61.8 91.7 66.4 80.4 ...
 $ height    : num  186 212 188 176 174 ...
```

```
summary(patient_df)
```

```
   patient_id         weight           height
 Min.   :  1.00   Min.   : 49.79   Min.   : 91.88
 1st Qu.: 25.75   1st Qu.: 68.25   1st Qu.:159.20
 Median : 50.50   Median : 73.91   Median :173.15
```

```
 Mean   : 50.50   Mean   : 75.18   Mean   :174.96
 3rd Qu.: 75.25   3rd Qu.: 84.61   3rd Qu.:185.90
 Max.   :100.00   Max.   :112.18   Max.   :253.13
```

```r
head(patient_df)
```

```
  patient_id   weight   height
1          1 75.82265 185.6147
2          2 61.77137 212.2587
3          3 91.65377 188.3604
4          4 66.38117 176.2986
5          5 80.42689 174.0509
6          6 79.32015 173.7906
```

BMI formula: $\frac{\text{weight in kg}}{(\text{height in m})^2}$

To do so, we should transform the height in m, and then perform the operation. We will save the results as a **new column** of the dataframe; this can be done with the syntax `dataframe_variable$new_column_name <- vector` or `dataframe_variable$["new_column_name"] <- vector` (we will see on day 2 the strings).

```r
# Get height in m
patient_df$height_in_m <- patient_df$height / 100

print(head(patient_df))
```

```
  patient_id   weight   height height_in_m
1          1 75.82265 185.6147    1.856147
2          2 61.77137 212.2587    2.122587
3          3 91.65377 188.3604    1.883604
4          4 66.38117 176.2986    1.762986
5          5 80.42689 174.0509    1.740509
6          6 79.32015 173.7906    1.737906
```

```r
patient_df["BMI"] <- patient_df$weight / (patient_df$height_in_m
 ↪  ** 2)

print(head(patient_df))
```

```
  patient_id   weight   height height_in_m      BMI
1          1 75.82265 185.6147    1.856147 22.00768
```

```
2          2 61.77137 212.2587    2.122587 13.71059
3          3 91.65377 188.3604    1.883604 25.83279
4          4 66.38117 176.2986    1.762986 21.35733
5          5 80.42689 174.0509    1.740509 26.54902
6          6 79.32015 173.7906    1.737906 26.26220
```

1.3.6 Sumamry statistics of numeric vectors

Usually, when dealing with numeric data we want to have some summary statistics on a specific data (e.g. mean values, median, quartile, standard deviation, …).

In R there are many built-in functions that can help us in doing so:

- `mean` to calculate the mean
- `sd` to calculate the standard deviation
- `variance` to calculate the variance
- `median` to calculate the median
- `quantile` to calculate the nth quantile of a distribution
- `round` to round decimal values to n decimal places
- `sum` to sum all the values of the vector
- many others

They all have in common the syntax: `name_of_the_function(vector)`.

```r
intensity_mean <- mean(intensity_values)
print(intensity_mean)
```

```
[1] 19419175
```

```r
height_sd <- sd(patient_df$height)
print(height_sd)
```

```
[1] 26.73183
```

```
BMI_median <- median(patient_df$BMI)
print(BMI_median)
```

```
[1] 25.23683
```

```
first_quartile_weight <- quantile(patient_df$weight, probs =
 ↪  0.25)
print(first_quartile_weight)
```

```
     25%
68.25118
```

```
rounded_BMI <- round(patient_df$BMI, 1)
print(head(rounded_BMI))
```

```
[1] 22.0 13.7 25.8 21.4 26.5 26.3
```

Exercise 1.3. Scale BMI values (formula: $scaled_i = \frac{x_i - \overline{x}}{sd(x)}$)

Solution

Here is a step-by-step solution:

```
# calculate BMI mean
mean_BMI <- mean(patient_df$BMI)

# calculate BMI sd
sd_BMI <- sd(patient_df$BMI)

# Calculate scaled BMIs
patient_df$scaled_BMI <- (patient_df$BMI - mean_BMI) / sd_BMI

print(head(patient_df))
```

```
  patient_id   weight   height height_in_m      BMI  scaled_BMI
1          1 75.82265 185.6147    1.856147 22.00768 -0.44542706
2          2 61.77137 212.2587    2.122587 13.71059 -1.29224320
3          3 91.65377 188.3604    1.883604 25.83279 -0.05502959
4          4 66.38117 176.2986    1.762986 21.35733 -0.51180319
5          5 80.42689 174.0509    1.740509 26.54902  0.01807031
6          6 79.32015 173.7906    1.737906 26.26220 -0.01120341
```

That is exactly what the formula `scale` do underneath:

```
# Calculate scaled BMIs
patient_df$scaled_BMI_function <- scale(patient_df$BMI)

print(head(patient_df))
```

```
  patient_id   weight   height height_in_m      BMI  scaled_BMI
1          1 75.82265 185.6147    1.856147 22.00768 -0.44542706
2          2 61.77137 212.2587    2.122587 13.71059 -1.29224320
3          3 91.65377 188.3604    1.883604 25.83279 -0.05502959
4          4 66.38117 176.2986    1.762986 21.35733 -0.51180319
5          5 80.42689 174.0509    1.740509 26.54902  0.01807031
6          6 79.32015 173.7906    1.737906 26.26220 -0.01120341
  scaled_BMI_function
1         -0.44542706
2         -1.29224320
3         -0.05502959
4         -0.51180319
5          0.01807031
6         -0.01120341
```

1.4 Save a dataframe

We are satisfied with this preliminary edit of the patients' data, so we can save the data to a dedicated file.

We use `write.csv` function:

```
write.csv(patient_df, # what we want to save
          file = "output/patients_data_BMI.csv", # name of the
           ↪  file
          row.names = FALSE) # we'll see row names in day 2
```

1.5 Home exercise

For next time, if you want, you can try to do this exercise:

1. Starting from patient data (raw), load the file
2. Inspect it

3. You know that you have to give drug A to each patient so that the final concentration is 10 mg/kg (mg of drug every kg of patient). Calculate how much drug you should give to each patient.
4. Given that a single stock of drug A is 5 g, how many stocks you have to order?

Solution

```r
# 1. Load the file
patient_df <- read.csv("data/patient_data.csv")
```

```r
# 2. Inspect it
head(patient_df)
```

```
  patient_id   weight   height
1          1 75.82265 185.6147
2          2 61.77137 212.2587
3          3 91.65377 188.3604
4          4 66.38117 176.2986
5          5 80.42689 174.0509
6          6 79.32015 173.7906
```

```r
str(patient_df)
```

```
'data.frame':   100 obs. of  3 variables:
 $ patient_id: int  1 2 3 4 5 6 7 8 9 10 ...
 $ weight    : num  75.8 61.8 91.7 66.4 80.4 ...
 $ height    : num  186 212 188 176 174 ...
```

```r
summary(patient_df)
```

```
   patient_id         weight           height      
 Min.   :  1.00   Min.   : 49.79   Min.   : 91.88  
 1st Qu.: 25.75   1st Qu.: 68.25   1st Qu.:159.20  
 Median : 50.50   Median : 73.91   Median :173.15  
 Mean   : 50.50   Mean   : 75.18   Mean   :174.96  
 3rd Qu.: 75.25   3rd Qu.: 84.61   3rd Qu.:185.90  
 Max.   :100.00   Max.   :112.18   Max.   :253.13  
```

```r
# 3. Calculate drug for each patient

patient_df$drugA_quantity_mg <- patient_df$weight * 10
head(patient_df)
```

```
  patient_id   weight   height drugA_quantity_mg
1          1 75.82265 185.6147          758.2265
2          2 61.77137 212.2587          617.7137
3          3 91.65377 188.3604          916.5377
4          4 66.38117 176.2986          663.8117
5          5 80.42689 174.0509          804.2689
6          6 79.32015 173.7906          793.2015
```

```r
# 4. How many stocks you have to order?

total_drug_A_mg <- sum(patient_df$drugA_quantity_mg)

total_drug_A_g <- total_drug_A_mg / 1000

stock_weight <- 5

number_of_stocks <- total_drug_A_g/stock_weight

print(number_of_stocks)
```

```
[1] 15.03539
```

To be precise, we can use `ceiling` function that round to the upper integer:

```r
number_of_stocks <- ceiling(number_of_stocks)

print(number_of_stocks)
```

```
[1] 16
```

Chapter 2

Strings, Factors and Dataframe Handling

2.1 What we will see today

Today we will look at some patient data, and we will answer these questions:

- Are data consistent?
- Which is the mean and sd values of glucose in each treatment group? How many patients we have per treatment group?
- Which group has the highest mean creatinine? (2 different datasets)

2.2 Strings and Factors

Today we will work with a clinical dataset containing information about patients enrolled in a multi-site trial. Let's import it and, as always, **inspect it first**:

```r
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.3     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.2
```

```
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to
```

```r
clinical_data <- read.csv("data/clinical_data.csv")
```

```r
str(clinical_data)
```

```
'data.frame':   240 obs. of  15 variables:
 $ patient_id      : chr  "PT-001" "PT-002" "PT-003" "PT-004" ...
 $ site            : chr  "Milan" "Rome" "Rome" "Turin " ...
 $ visit_date      : chr  "2023-07-20" "2020-08-15" "2023-10-15"
 ↪  "2020-03-06" ...
 $ age             : int  40 40 59 46 58 53 51 34 43 60 ...
 $ sex             : chr  "Male" "Male" "Female" "Male" ...
 $ smoking         : chr  "never" "Curent" "former" "Curent" ...
 $ treatment       : chr  "Drug_A" "Drug_B" "Drug_A" "Drug_A" ...
 $ stage           : chr  "II" "I" "I" "I" ...
 $ weight_kg       : num  85.9 100.9 65 95.9 75.7 ...
 $ height_cm       : num  173 185 166 182 168 ...
 $ sbp_mmhg        : int  164 143 152 142 160 150 137 146 151 159
 ↪  ...
 $ dbp_mmhg        : int  101 102 98 88 96 107 96 106 94 104 ...
 $ glucose_mgdl    : int  113 105 110 120 131 78 108 93 106 110
 ↪  ...
 $ cholesterol_mgdl: int  172 239 146 232 192 232 163 199 200 185
 ↪  ...
 $ crp_ngml        : num  5.89 6.13 7.81 3.35 21.72 ...
```

```r
summary(clinical_data)
```

```
  patient_id            site            visit_date
 ↪  age
 Length:240         Length:240         Length:240         Min.
 ↪  :26.0
 Class :character   Class :character   Class :character   1st
 ↪  Qu.:49.0
 Mode  :character   Mode  :character   Mode  :character   Median
 ↪  :57.5
                                                          Mean
 ↪  :57.3
                                                          3rd
 ↪  Qu.:65.0
```

```
                                                                  Max.
↪  :85.0
    sex              smoking           treatment
↪  stage
Length:240         Length:240         Length:240
↪  Length:240
Class :character   Class :character   Class :character   Class
↪  :character
Mode  :character   Mode  :character   Mode  :character   Mode
↪  :character



  weight_kg        height_cm        sbp_mmhg        dbp_mmhg
↪
Min.   : 39.40   Min.   :148.9   Min.   :115.0   Min.   : 73.00
↪
1st Qu.: 65.35   1st Qu.:162.2   1st Qu.:139.0   1st Qu.: 90.75
↪
Median : 75.65   Median :168.5   Median :146.5   Median : 97.00
↪
Mean   : 76.87   Mean   :169.2   Mean   :146.7   Mean   : 96.11
↪
3rd Qu.: 87.83   3rd Qu.:174.8   3rd Qu.:153.2   3rd Qu.:102.00
↪
Max.   :132.90   Max.   :200.1   Max.   :192.0   Max.   :120.00
↪
 glucose_mgdl   cholesterol_mgdl    crp_ngml
Min.   : 60.0   Min.   :113.0    Min.   : 1.92
1st Qu.: 98.0   1st Qu.:189.8    1st Qu.: 5.30
Median :109.0   Median :207.0    Median : 8.37
Mean   :108.8   Mean   :207.9    Mean   :10.35
3rd Qu.:118.0   3rd Qu.:226.0    3rd Qu.:13.02
Max.   :149.0   Max.   :307.0    Max.   :42.09
```

```r
head(clinical_data)
```

```
  patient_id   site visit_date age    sex smoking treatment stage
↪  weight_kg
1     PT-001  Milan 2023-07-20  40   Male   never    Drug_A    II
↪  85.9
2     PT-002   Rome 2020-08-15  40   Male  Curent    Drug_B     I
↪  100.9
3     PT-003   Rome 2023-10-15  59 Female  former    Drug_A     I
↪  65.0
```

```
4     PT-004 Turin  2020-03-06  46   Male  Curent    Drug_A     I
 ↪  95.9
5     PT-005  Turin 2023-03-30  58   Male  FORMER    Drug_B   III
 ↪  75.7
6     PT-006  Milan 2023-04-15  53 Female  Curent   Control    IV
 ↪  69.6
  height_cm sbp_mmhg dbp_mmhg glucose_mgdl cholesterol_mgdl
 ↪  crp_ngml
1     172.7      164      101          113              172
 ↪  5.89
2     185.3      143      102          105              239
 ↪  6.13
3     165.6      152       98          110              146
 ↪  7.81
4     181.5      142       88          120              232
 ↪  3.35
5     168.1      160       96          131              192
 ↪  21.72
6     174.3      150      107           78              232
 ↪  32.42
```

You will notice that several columns contain text: `patient_id`, `sex`, `treatment`, `stage`, `smoking`, `site`, `diagnosis_code`. In R, text data can be represented in two fundamentally different ways: as **strings** (also called *characters*) or as **factors**. Let's understand the difference.

A **string** (or character) is simply a sequence of text. R treats it as a raw piece of text and does not impose any structure on it.

A **factor** is a *categorical variable*: it has a fixed, predefined set of allowed values called **levels**. Under the hood, R stores factors as integers (one integer per level) and keeps a table that maps integers to labels. This makes factors memory-efficient and very useful for statistical modelling (e.g., in regression, factors are automatically handled as categorical predictors) and for plotting purposes.

When to use strings and when factors?

Use **string** when…	Use **factor** when…
Text is free-form or unpredictable	The variable takes a limited set of known values

Use **string** when…	Use **factor** when…
You need to manipulate the text	Order matters (e.g., stage I < II < III < IV)
Unique identifiers (patient IDs)	Input to statistical models

As some of our columns fall in the conditions for factors, let's convert them with the function `factor`:

```
clinical_data$sex       <- factor(clinical_data$sex)
clinical_data$treatment <- factor(clinical_data$treatment)
clinical_data$site      <- factor(clinical_data$site)
clinical_data$smoking   <- factor(clinical_data$smoking)

# For stage, the categories have a natural ORDER: we declare an
 ↪  ordered factor
clinical_data$stage <- factor(clinical_data$stage,
                              levels = c("I", "II", "III",
                               ↪  "IV"),
                              ordered = TRUE)

str(clinical_data)
```

```
'data.frame':   240 obs. of  15 variables:
 $ patient_id      : chr  "PT-001" "PT-002" "PT-003" "PT-004" ...
 $ site            : Factor w/ 4 levels "Milan","Rome",..: 1 2 2
 ↪  4 3 1 2 2 1 2 ...
 $ visit_date      : chr  "2023-07-20" "2020-08-15" "2023-10-15"
 ↪  "2020-03-06" ...
 $ age             : int  40 40 59 46 58 53 51 34 43 60 ...
 $ sex             : Factor w/ 2 levels "Female","Male": 2 2 1 2
 ↪  2 1 2 1 2 2 ...
 $ smoking         : Factor w/ 7 levels "Curent","current",..: 5
 ↪  1 3 1 4 1 3 6 5 4 ...
 $ treatment       : Factor w/ 3 levels "Control","Drug_A",..: 2
 ↪  3 2 2 3 1 2 1 3 2 ...
 $ stage           : Ord.factor w/ 4 levels "I"<"II"<"III"<..: 2
 ↪  1 1 1 3 4 1 1 3 2 ...
 $ weight_kg       : num  85.9 100.9 65 95.9 75.7 ...
 $ height_cm       : num  173 185 166 182 168 ...
 $ sbp_mmhg        : int  164 143 152 142 160 150 137 146 151 159
 ↪  ...
 $ dbp_mmhg        : int  101 102 98 88 96 107 96 106 94 104 ...
 $ glucose_mgdl    : int  113 105 110 120 131 78 108 93 106 110
 ↪  ...
```

```
$ cholesterol_mgdl: int  172 239 146 232 192 232 163 199 200 185
↪  ...
$ crp_ngml        : num  5.89 6.13 7.81 3.35 21.72 ...
```

Now `summary` gives us much more useful information on those columns:

```
summary(clinical_data)
```

```
 patient_id            site     visit_date             age
↪  sex
Length:240         Milan :96   Length:240         Min.   :26.0
↪  Female:118
Class :character   Rome  :77   Class :character   1st Qu.:49.0
↪  Male  :122
Mode  :character   Turin :31   Mode  :character   Median :57.5
↪
                   Turin :36                      Mean   :57.3
↪
                                                  3rd Qu.:65.0
↪
                                                  Max.   :85.0
↪

        smoking     treatment  stage      weight_kg
↪  height_cm
Curent      :35   Control:74   I  :76   Min.   : 39.40   Min.
↪  :148.9
current     :28   Drug_A :88   II :73   1st Qu.: 65.35   1st
↪  Qu.:162.2
former      :44   Drug_B :78   III:53   Median : 75.65   Median
↪  :168.5
FORMER      :46                IV :38   Mean   : 76.87   Mean
↪  :169.2
never       :26                         3rd Qu.: 87.83   3rd
↪  Qu.:174.8
Never       :28                         Max.   :132.90   Max.
↪  :200.1
never smoker:33
↪
   sbp_mmhg        dbp_mmhg       glucose_mgdl
↪  cholesterol_mgdl
Min.   :115.0   Min.   : 73.00   Min.   : 60.0   Min.   :113.0
↪
1st Qu.:139.0   1st Qu.: 90.75   1st Qu.: 98.0   1st Qu.:189.8
↪
```

```
 Median :146.5   Median : 97.00   Median :109.0   Median :207.0
↪
 Mean   :146.7   Mean   : 96.11   Mean   :108.8   Mean   :207.9
↪
 3rd Qu.:153.2   3rd Qu.:102.00   3rd Qu.:118.0   3rd Qu.:226.0
↪
 Max.   :192.0   Max.   :120.00   Max.   :149.0   Max.   :307.0
↪

    crp_ngml
 Min.   : 1.92
 1st Qu.: 5.30
 Median : 8.37
 Mean   :10.35
 3rd Qu.:13.02
 Max.   :42.09
```

Notice how `summary` now shows the count per level for factors (e.g. how many Male vs Female), while before it just said `character`.

Moreover, we can see that in smoking column, we have different labels with the same meaning (never, Never, never smoker for never). R (as well as any programming language) is case sensitive, so it treats those 3 as different labels, even if for us they mean the same thing.

IMPORTANT: labels should be consistent across datasets when you fill them.

As we want to edit those to be consistent, we need to change the column back to be a character, with `as.character` function:

```
clinical_data$smoking   <- as.character(clinical_data$smoking)
str(clinical_data)
```

```
'data.frame':   240 obs. of  15 variables:
 $ patient_id      : chr  "PT-001" "PT-002" "PT-003" "PT-004" ...
 $ site            : Factor w/ 4 levels "Milan","Rome",..: 1 2 2
↪  4 3 1 2 2 1 2 ...
 $ visit_date      : chr  "2023-07-20" "2020-08-15" "2023-10-15"
↪  "2020-03-06" ...
 $ age             : int  40 40 59 46 58 53 51 34 43 60 ...
 $ sex             : Factor w/ 2 levels "Female","Male": 2 2 1 2
↪  2 1 2 1 2 2 ...
 $ smoking         : chr  "never" "Curent" "former" "Curent" ...
 $ treatment       : Factor w/ 3 levels "Control","Drug_A",..: 2
↪  3 2 2 3 1 2 1 3 2 ...
```

```
 $ stage           : Ord.factor w/ 4 levels "I"<"II"<"III"<..: 2
 ↪  1 1 1 3 4 1 1 3 2 ...
 $ weight_kg       : num  85.9 100.9 65 95.9 75.7 ...
 $ height_cm       : num  173 185 166 182 168 ...
 $ sbp_mmhg        : int  164 143 152 142 160 150 137 146 151 159
 ↪  ...
 $ dbp_mmhg        : int  101 102 98 88 96 107 96 106 94 104 ...
 $ glucose_mgdl    : int  113 105 110 120 131 78 108 93 106 110
 ↪  ...
 $ cholesterol_mgdl: int  172 239 146 232 192 232 163 199 200 185
 ↪  ...
 $ crp_ngml        : num  5.89 6.13 7.81 3.35 21.72 ...
```

We can now proceed.

2.2.1 Operations on strings

The main package for string operations in R (and the one we will use throughout the course) is `stringr`, which is part of the `tidyverse`. All its functions start with `str_` to make them easy to remember.

2.2.1.1 Character substitution

Let's start by looking at how to change characters in a string: from "Curent" to "current" using `str_replace(vector, old_pattern, new_pattern)`.

```r
print(unique(clinical_data$smoking))
```

```
[1] "never"        "Curent"       "former"       "FORMER"
 ↪  "Never"
[6] "never smoker" "current"
```

```r
clinical_data$smoking <- str_replace(clinical_data$smoking,
 ↪  "Curent", "current")
print(unique(clinical_data$smoking))
```

```
[1] "never"        "current"      "former"       "FORMER"
 ↪  "Never"
[6] "never smoker"
```

Then we can unify "FORMER" and "former" by transforming the former to lowercase with `str_to_lower`:

```
print(unique(clinical_data$smoking))
```

```
[1] "never"        "current"      "former"       "FORMER"
 ↪  "Never"
[6] "never smoker"
```

```
clinical_data$smoking <- str_to_lower(clinical_data$smoking)
print(unique(clinical_data$smoking))
```

```
[1] "never"        "current"      "former"       "never smoker"
```

We have converted both FORMER to former and Never to never. We lastly have to change "never smoker" to "never".

Exercise 2.1. Change "never smoker" to "smoker" and make smoker as factor.

Solution

```
clinical_data$smoking <- str_replace(clinical_data$smoking,
 ↪  "never smoker", "never")
clinical_data$smoking <- factor(clinical_data$smoking, levels =
 ↪  c("never", "former", "current"))

summary(clinical_data)
```

```
  patient_id            site     visit_date             age
 ↪  sex
 Length:240         Milan :96   Length:240         Min.   :26.0
 ↪  Female:118
 Class :character   Rome  :77   Class :character   1st Qu.:49.0
 ↪  Male  :122
 Mode  :character   Turin :31   Mode  :character   Median :57.5
 ↪
                    Turin :36                      Mean   :57.3
 ↪
                                                   3rd Qu.:65.0
 ↪
                                                   Max.   :85.0
 ↪
    smoking     treatment  stage      weight_kg        height_cm
 ↪
 never  :87   Control:74   I  :76   Min.   : 39.40   Min.
 ↪  :148.9
```

```
 former :90   Drug_A :88   II :73   1st Qu.: 65.35   1st
 ↪  Qu.:162.2
 current:63   Drug_B :78   III:53   Median : 75.65   Median
 ↪  :168.5
                           IV :38   Mean   : 76.87   Mean
 ↪  :169.2
                                    3rd Qu.: 87.83   3rd
 ↪  Qu.:174.8
                                    Max.   :132.90   Max.
 ↪  :200.1
    sbp_mmhg        dbp_mmhg       glucose_mgdl
 ↪  cholesterol_mgdl
 Min.   :115.0   Min.   : 73.00   Min.   : 60.0   Min.   :113.0
 ↪
 1st Qu.:139.0   1st Qu.: 90.75   1st Qu.: 98.0   1st Qu.:189.8
 ↪
 Median :146.5   Median : 97.00   Median :109.0   Median :207.0
 ↪
 Mean   :146.7   Mean   : 96.11   Mean   :108.8   Mean   :207.9
 ↪
 3rd Qu.:153.2   3rd Qu.:102.00   3rd Qu.:118.0   3rd Qu.:226.0
 ↪
 Max.   :192.0   Max.   :120.00   Max.   :149.0   Max.   :307.0
 ↪
    crp_ngml
 Min.   : 1.92
 1st Qu.: 5.30
 Median : 8.37
 Mean   :10.35
 3rd Qu.:13.02
 Max.   :42.09
```

For sake of completeness, we could have done all of this with a single function `replace_values` (for new versions of R):

```
clinical_data$smoking <-
 ↪  replace_values(as.character(clinical_data$smoking), c("never
 ↪  smoker", "Never") ~ "never",
                                          c("Curent") ~ "current",
                                          c("FORMER") ~ "former")

clinical_data$smoking = factor(clinical_data$smoking)
```

If it doesn't work, you could use:

```
clinical_data = mutate(smoking = case_when(
    smoking %in% c("never smoker", "Never") ~ "never",
    smoking %in% c("FORMER", "Former", "Fomer") ~ "former",
    smoking == "Curent" ~ "current",
    .default = smoking),
    .data = clinical_data
    )

clinical_data$smoking = factor(clinical_data$smoking)
```

Q: *Why there are 2 Turin?* A: Let's check in details the possible labels of that column with function `levels`:

```
print(levels(clinical_data$site))
```

```
[1] "Milan"  "Rome"   "Turin"  "Turin "
```

Here's the hitch, some have a trailing space after Turin. We can get rid of them using `str_trim` function:

```
print(levels(clinical_data$site))
```

```
[1] "Milan"  "Rome"   "Turin"  "Turin "
```

```
clinical_data$site <- factor(str_trim(clinical_data$site))

print(levels(clinical_data$site))
```

```
[1] "Milan" "Rome"  "Turin"
```

```
print(summary(clinical_data))
```

```
  patient_id           site     visit_date             age
 ↪  sex
 Length:240         Milan:96   Length:240         Min.   :26.0
 ↪  Female:118
 Class :character   Rome :77   Class :character   1st Qu.:49.0
 ↪  Male  :122
 Mode  :character   Turin:67   Mode  :character   Median :57.5
 ↪
```

```
                                                        Mean   :57.3
↪
                                                        3rd Qu.:65.0
↪
                                                        Max.   :85.0
↪
   smoking     treatment  stage      weight_kg        height_cm
↪
current:63   Control:74   I  :76   Min.   : 39.40   Min.
↪  :148.9
former :90   Drug_A :88   II :73   1st Qu.: 65.35   1st
↪  Qu.:162.2
never  :87   Drug_B :78   III:53   Median : 75.65   Median
↪  :168.5
                          IV :38   Mean   : 76.87   Mean
↪  :169.2
                                   3rd Qu.: 87.83   3rd
↪  Qu.:174.8
                                   Max.   :132.90   Max.
↪  :200.1
   sbp_mmhg        dbp_mmhg       glucose_mgdl
↪  cholesterol_mgdl
Min.   :115.0   Min.   : 73.00   Min.   : 60.0   Min.   :113.0
↪
1st Qu.:139.0   1st Qu.: 90.75   1st Qu.: 98.0   1st Qu.:189.8
↪
Median :146.5   Median : 97.00   Median :109.0   Median :207.0
↪
Mean   :146.7   Mean   : 96.11   Mean   :108.8   Mean   :207.9
↪
3rd Qu.:153.2   3rd Qu.:102.00   3rd Qu.:118.0   3rd Qu.:226.0
↪
Max.   :192.0   Max.   :120.00   Max.   :149.0   Max.   :307.0
↪
   crp_ngml
Min.   : 1.92
1st Qu.: 5.30
Median : 8.37
Mean   :10.35
3rd Qu.:13.02
Max.   :42.09
```

2.2.1.2 Other useful functions to handle strings

A few more `stringr` functions you will find very useful:

Function	What it does
`str_detect(x, pattern)`	Returns TRUE/FALSE: does the string contain the pattern?
`str_starts(x, pattern)`	Does the string *start* with the pattern?
`str_ends(x, pattern)`	Does the string *end* with the pattern?
`str_pad(x, width, side, pad)`	Pads a string to a given width
`str_sub(x, start, end)`	Extracts a substring by position

2.2.2 Operations on multiple columns

2.2.2.1 Split a column and merge multiple columns

Sometimes a single column contains multiple pieces of information that could be split into separate columns. Look at `visit_date`:

```
head(clinical_data$visit_date)
```

```
[1] "2023-07-20" "2020-08-15" "2023-10-15" "2020-03-06"
 ↪  "2023-03-30"
[6] "2023-04-15"
```

We can see that the year of visit varies a lot, maybe we should take it into account. We can split this into 3 separate columns using `separate` function:

```
clinical_data <- separate(clinical_data,
                          col = visit_date,         # column to
                           ↪  split
                          into = c("visit_year", "visit_month",
                           ↪  "visit_day"), # new column names
                          sep = "-",                # separator
                          remove = FALSE)           # keep the
                           ↪  original column

head(clinical_data)
```

```
  patient_id  site visit_date visit_year visit_month visit_day
 ↪  age    sex
1     PT-001 Milan 2023-07-20       2023          07        20
 ↪  40   Male
```

```
2     PT-002  Rome 2020-08-15       2020          08        15
 ↪  40   Male
3     PT-003  Rome 2023-10-15       2023          10        15
 ↪  59 Female
4     PT-004 Turin 2020-03-06       2020          03        06
 ↪  46   Male
5     PT-005 Turin 2023-03-30       2023          03        30
 ↪  58   Male
6     PT-006 Milan 2023-04-15       2023          04        15
 ↪  53 Female
  smoking treatment stage weight_kg height_cm sbp_mmhg dbp_mmhg
 ↪  glucose_mgdl
1   never    Drug_A    II      85.9     172.7      164      101
 ↪  113
2 current    Drug_B     I     100.9     185.3      143      102
 ↪  105
3  former    Drug_A     I      65.0     165.6      152       98
 ↪  110
4 current    Drug_A     I      95.9     181.5      142       88
 ↪  120
5  former    Drug_B   III      75.7     168.1      160       96
 ↪  131
6 current   Control    IV      69.6     174.3      150      107
 ↪  78
  cholesterol_mgdl crp_ngml
1              172     5.89
2              239     6.13
3              146     7.81
4              232     3.35
5              192    21.72
6              232    32.42
```

The opposite operation — merging multiple columns into one — is done with `unite`:

```
# example: unite back together
clinical_data <- unite(clinical_data,
                       col = "visit_year_month",         # name
                        ↪  of the new column
                       visit_year, visit_month,           #
 ↪  columns to merge
                       sep = "-",                       #
                        ↪  separator
                       remove = FALSE)                  # keep
                        ↪  original columns
```

```r
head(clinical_data)
```

```
  patient_id  site visit_date visit_year_month visit_year
 ↪  visit_month visit_day
1     PT-001 Milan 2023-07-20          2023-07       2023
 ↪  07        20
2     PT-002  Rome 2020-08-15          2020-08       2020
 ↪  08        15
3     PT-003  Rome 2023-10-15          2023-10       2023
 ↪  10        15
4     PT-004 Turin 2020-03-06          2020-03       2020
 ↪  03        06
5     PT-005 Turin 2023-03-30          2023-03       2023
 ↪  03        30
6     PT-006 Milan 2023-04-15          2023-04       2023
 ↪  04        15
  age    sex smoking treatment stage weight_kg height_cm sbp_mmhg
 ↪  dbp_mmhg
1  40   Male   never    Drug_A    II      85.9     172.7      164
 ↪  101
2  40   Male current    Drug_B     I     100.9     185.3      143
 ↪  102
3  59 Female  former    Drug_A     I      65.0     165.6      152
 ↪  98
4  46   Male current    Drug_A     I      95.9     181.5      142
 ↪  88
5  58   Male  former    Drug_B   III      75.7     168.1      160
 ↪  96
6  53 Female current   Control    IV      69.6     174.3      150
 ↪  107
  glucose_mgdl cholesterol_mgdl crp_ngml
1          113              172     5.89
2          105              239     6.13
3          110              146     7.81
4          120              232     3.35
5          131              192    21.72
6           78              232    32.42
```

2.2.3 Column and row names

Every dataframe has **column names** (the variable names) and **row names** (by default a number, but we can change it).

To retrieve column names:

```
colnames(clinical_data)
```

```
 [1] "patient_id"       "site"             "visit_date"
 ↪  "visit_year_month"
 [5] "visit_year"       "visit_month"      "visit_day"
 ↪  "age"
 [9] "sex"              "smoking"          "treatment"
 ↪  "stage"
[13] "weight_kg"        "height_cm"        "sbp_mmhg"
 ↪  "dbp_mmhg"
[17] "glucose_mgdl"     "cholesterol_mgdl" "crp_ngml"
```

To rename specific columns, we can assign a new vector of names to `colnames`, or use the tidyverse function `rename` (we will see this with `dplyr` below):

```
clinical_data = rename(clinical_data, "diastolic_bp_mmhg" =
 ↪  "dbp_mmhg", "systolic_bp_mmhg" = "sbp_mmhg")

colnames(clinical_data)
```

```
 [1] "patient_id"        "site"              "visit_date"
 [4] "visit_year_month"  "visit_year"        "visit_month"
 [7] "visit_day"         "age"               "sex"
[10] "smoking"           "treatment"         "stage"
[13] "weight_kg"         "height_cm"         "systolic_bp_mmhg"
[16] "diastolic_bp_mmhg" "glucose_mgdl"      "cholesterol_mgdl"
[19] "crp_ngml"
```

For row names:

```
# by default, row names are row numbers
head(rownames(clinical_data))
```

```
[1] "1" "2" "3" "4" "5" "6"
```

```r
# we could set patient_id as row names
rownames(clinical_data) <- clinical_data$patient_id

head(rownames(clinical_data))
```

```
[1] "PT-001" "PT-002" "PT-003" "PT-004" "PT-005" "PT-006"
```

However, **be careful** with row names: they are not a proper column of the dataframe, so they don't follow along easily in many tidyverse operations. It is generally better to keep your identifiers as a regular column.

2.3 Dataframe Handling

The main verbs we will learn today are:

- `mutate` — add or modify columns
- `select` — choose which columns to keep
- `filter`/`filter_out` — keep/remove rows that satisfy a condition
- `group_by` + `summarize` / `reframe` — aggregate data by groups
- `left_join` / `inner_join` — merge two dataframes

2.3.1 Create new columns with mutate

`mutate` creates new columns or modifies existing ones.

A key feature of `dplyr` is the **pipe operator** `|>` (or `%>%` in older tidyverse style). It passes the result of one operation directly to the next, making code much more readable:

You can read `|>` as *"and then"*.

```r
# Add a BMI column (we already know this formula!)
clinical_data <- clinical_data |> # take clinical_data and then
  mutate(height_m = height_cm / 100,
         BMI = weight_kg / height_m^2,
         BMI = round(BMI, 1))

head(clinical_data)
```

```
       patient_id  site visit_date visit_year_month visit_year
 ↪  visit_month
PT-001     PT-001 Milan 2023-07-20          2023-07       2023
 ↪  07
PT-002     PT-002  Rome 2020-08-15          2020-08       2020
 ↪  08
PT-003     PT-003  Rome 2023-10-15          2023-10       2023
 ↪  10
PT-004     PT-004 Turin 2020-03-06          2020-03       2020
 ↪  03
PT-005     PT-005 Turin 2023-03-30          2023-03       2023
 ↪  03
PT-006     PT-006 Milan 2023-04-15          2023-04       2023
 ↪  04
       visit_day age    sex smoking treatment stage weight_kg
 ↪  height_cm
PT-001        20  40   Male   never    Drug_A    II      85.9
 ↪  172.7
PT-002        15  40   Male current    Drug_B     I     100.9
 ↪  185.3
PT-003        15  59 Female  former    Drug_A     I      65.0
 ↪  165.6
PT-004        06  46   Male current    Drug_A     I      95.9
 ↪  181.5
PT-005        30  58   Male  former    Drug_B   III      75.7
 ↪  168.1
PT-006        15  53 Female current   Control    IV      69.6
 ↪  174.3
       systolic_bp_mmhg diastolic_bp_mmhg glucose_mgdl
 ↪  cholesterol_mgdl
PT-001              164               101          113
 ↪  172
PT-002              143               102          105
 ↪  239
PT-003              152                98          110
 ↪  146
PT-004              142                88          120
 ↪  232
PT-005              160                96          131
 ↪  192
PT-006              150               107           78
 ↪  232
       crp_ngml height_m  BMI
PT-001     5.89    1.727 28.8
PT-002     6.13    1.853 29.4
```

```
PT-003     7.81    1.656 23.7
PT-004     3.35    1.815 29.1
PT-005    21.72    1.681 26.8
PT-006    32.42    1.743 22.9
```

`mutate` can also create columns based on conditions, using `if_else` or `case_when`:

```
# Classify BMI into categories
clinical_data <- clinical_data |>
  mutate(BMI_class = case_when(
    BMI < 18.5 ~ "Underweight",
    BMI < 25   ~ "Normal",
    BMI < 30   ~ "Overweight",
    BMI >= 30  ~ "Obese"
  ),
  BMI_class = factor(BMI_class, labels = c("Underweight",
   ↪  "Normal", "Overweight", "Obese"), ordered = T)
  )

summary(clinical_data$BMI_class)
```

```
Underweight      Normal  Overweight       Obese
         67          58         102          13
```

2.3.2 select

`select` chooses which **columns** to keep (or drop). This is useful when working with wide dataframes where you only need a subset of variables.

```
# keep only demographic and clinical variables
demo_cols <- clinical_data |>
  select(patient_id, site, age, sex, treatment, stage, BMI,
   ↪  BMI_class)

head(demo_cols)
```

```
       patient_id  site age    sex treatment stage  BMI
 ↪  BMI_class
PT-001     PT-001 Milan  40   Male    Drug_A    II 28.8
 ↪  Overweight
```

```
PT-002     PT-002  Rome  40   Male    Drug_B     I 29.4
 ↪  Overweight
PT-003     PT-003  Rome  59 Female    Drug_A     I 23.7
 ↪  Underweight
PT-004     PT-004 Turin  46   Male    Drug_A     I 29.1
 ↪  Overweight
PT-005     PT-005 Turin  58   Male    Drug_B   III 26.8
 ↪  Overweight
PT-006     PT-006 Milan  53 Female   Control    IV 22.9
 ↪  Underweight
```

To **drop** specific columns, use `-` before the column name:

```
colnames(clinical_data)
```

```
 [1] "patient_id"        "site"              "visit_date"
 [4] "visit_year_month"  "visit_year"        "visit_month"
 [7] "visit_day"         "age"               "sex"
[10] "smoking"           "treatment"         "stage"
[13] "weight_kg"         "height_cm"         "systolic_bp_mmhg"
[16] "diastolic_bp_mmhg" "glucose_mgdl"      "cholesterol_mgdl"
[19] "crp_ngml"          "height_m"          "BMI"
[22] "BMI_class"
```

```
# drop the helper/intermediate columns we created
clinical_data <- clinical_data |>
  select(-visit_year_month)

colnames(clinical_data)
```

```
 [1] "patient_id"        "site"              "visit_date"
 [4] "visit_year"        "visit_month"       "visit_day"
 [7] "age"               "sex"               "smoking"
[10] "treatment"         "stage"             "weight_kg"
[13] "height_cm"         "systolic_bp_mmhg"  "diastolic_bp_mmhg"
[16] "glucose_mgdl"      "cholesterol_mgdl"  "crp_ngml"
[19] "height_m"          "BMI"               "BMI_class"
```

`select` also supports handy helper functions:

```
# select all columns that start with "diag"
clinical_data |> select(starts_with("visit")) |> head()
```

```
       visit_date visit_year visit_month visit_day
PT-001 2023-07-20       2023          07        20
PT-002 2020-08-15       2020          08        15
PT-003 2023-10-15       2023          10        15
PT-004 2020-03-06       2020          03        06
PT-005 2023-03-30       2023          03        30
PT-006 2023-04-15       2023          04        15
```

```r
# select all columns that contain "bp"
clinical_data |> select(contains("bp")) |> head()
```

```
       systolic_bp_mmhg diastolic_bp_mmhg
PT-001              164               101
PT-002              143               102
PT-003              152                98
PT-004              142                88
PT-005              160                96
PT-006              150               107
```

2.3.3 Boolean Operators

A **boolean** (or logical) is a data type that can only take two values: `TRUE` or `FALSE`. It is the backbone of filtering and conditional logic in R.

A boolean is the answer to a yes/no question.

For example: *"Is this patient in stage III or IV?"* → `TRUE` or `FALSE`.

2.3.3.1 Comparison operators

These operators compare two values and return a boolean:

Operator	Meaning	Example	Result
`==`	equal to	`3 == 3`	`TRUE`
`!=`	not equal to	`3 != 4`	`TRUE`
`>`	greater than	`5 > 3`	`TRUE`
`<`	less than	`2 < 1`	`FALSE`
`>=`	greater or equal	`3 >= 3`	`TRUE`
`<=`	less or equal	`2 <= 1`	`FALSE`

```r
# simple examples
print(10 > 5)
```

```
[1] TRUE
```

```r
print(10 == 10)
```

```
[1] TRUE
```

```r
print(10 != 10)
```

```
[1] FALSE
```

```r
print(3.14 >= 3.14)
```

```
[1] TRUE
```

When applied to a **vector**, the comparison is done **element-wise** and returns a boolean vector:

```r
ages <- clinical_data$age
head(ages)
```

```
[1] 40 40 59 46 58 53
```

```r
is_over_60 <- ages > 60
head(is_over_60)
```

```
[1] FALSE FALSE FALSE FALSE FALSE FALSE
```

Under the hood, TRUE is stored as a 1, while FALSE as 0. This is useful when we want to get the number of TRUE values of a vector. How? By summing up all the values:

```r
sum(is_over_60)
```

```
[1] 84
```

2.3.3.2 Logical operators

We often need to combine multiple conditions. This is done with **logical operators**:

Operator	Meaning	Example
`&`	AND — both conditions must be true	`age > 50 & sex == "Female"`
`\\|`	OR — at least one condition must be true	`stage == "III" \\| stage == "IV"`
`!`	NOT — negates a condition	`!is.na(crp_ngml)`

```r
# patients who are Female AND over 60
female_over60 <- (clinical_data$sex == "Female") &
 ↪  (clinical_data$age > 60)
sum(female_over60)
```

```
[1] 49
```

```r
# patients in advanced stage (III or IV)
advanced_stage <- (clinical_data$stage == "III") |
 ↪  (clinical_data$stage == "IV")
sum(advanced_stage)
```

```
[1] 91
```

```r
# NOT: patients who are NOT in the Control group
not_control <- clinical_data$treatment != "Control"
sum(not_control)
```

```
[1] 166
```

```r
# equivalent using !
not_control_2 <- !(clinical_data$treatment == "Control")
sum(not_control_2)
```

```
[1] 166
```

A very useful operator when checking membership in a list of values is `%in%`:

```
# patients from Milan or Rome (equivalent to == "Milan" | ==
 ↪  "Rome")
big_sites <- clinical_data$site %in% c("Milan", "Rome")
sum(big_sites)
```

```
[1] 173
```

::: .{exercise #bool-sum}

How many patients have been diagnosed as stage I, in Turin in 2018?

:::

Solution

```
stageI_turin_2018 <- (clinical_data$site == "Turin") &
 ↪  (clinical_data$stage == "I") & (clinical_data$visit_year ==
 ↪  2018)
sum(stageI_turin_2018)
```

```
[1] 4
```

2.3.4 filter and filter out

`filter` keeps only the **rows** that satisfy one or more conditions:

```
# Keep only patients treated with Drug_A
drug_a_patients <- clinical_data |>
  filter(treatment == "Drug_A")

nrow(drug_a_patients)
```

```
[1] 88
```

```
head(drug_a_patients)
```

```
       patient_id  site visit_date visit_year visit_month
 ↪  visit_day age    sex
PT-001     PT-001 Milan 2023-07-20       2023          07
 ↪  20  40   Male
PT-003     PT-003  Rome 2023-10-15       2023          10
 ↪  15  59 Female
PT-004     PT-004 Turin 2020-03-06       2020          03
 ↪  06  46   Male
PT-007     PT-007  Rome 2020-09-09       2020          09
 ↪  09  51   Male
PT-010     PT-010  Rome 2023-09-27       2023          09
 ↪  27  60   Male
PT-012     PT-012  Rome 2025-03-20       2025          03
 ↪  20  52   Male
       smoking treatment stage weight_kg height_cm
 ↪  systolic_bp_mmhg
PT-001   never    Drug_A    II      85.9     172.7
 ↪  164
PT-003  former    Drug_A     I      65.0     165.6
 ↪  152
PT-004 current    Drug_A     I      95.9     181.5
 ↪  142
PT-007  former    Drug_A     I      69.7     169.7
 ↪  137
PT-010  former    Drug_A    II      93.4     186.3
 ↪  159
PT-012   never    Drug_A   III      84.3     168.8
 ↪  147
       diastolic_bp_mmhg glucose_mgdl cholesterol_mgdl crp_ngml
 ↪  height_m  BMI
PT-001               101          113              172     5.89
 ↪  1.727 28.8
PT-003                98          110              146     7.81
 ↪  1.656 23.7
PT-004                88          120              232     3.35
 ↪  1.815 29.1
PT-007                96          108              163     5.41
 ↪  1.697 24.2
PT-010               104          110              185     5.30
 ↪  1.863 26.9
PT-012                96           86              222    10.45
 ↪  1.688 29.6
         BMI_class
PT-001  Overweight
PT-003 Underweight
```

```
PT-004  Overweight
PT-007 Underweight
PT-010  Overweight
PT-012  Overweight
```

```r
# Patients in advanced stage AND not in control group
advanced_treated <- clinical_data |>
  filter((stage == "III" | stage == "IV") & treatment !=
   ↪  "Control")

nrow(advanced_treated)
```

```
[1] 76
```

```r
# Using %in% for multiple values
milan_or_rome <- clinical_data |>
  filter(site %in% c("Milan", "Rome"))

nrow(milan_or_rome)
```

```
[1] 173
```

To **filter out** (i.e., *exclude* rows matching a condition), use `filter_out` (for new versions of R and tidyverse):

```r
# Exclude patients from Naples
no_milan <- clinical_data |>
  filter_out(site %in% c("Milan"))

nrow(no_milan)
```

```
[1] 144
```

If it doesn't work, you could use:

```r
no_milan = clinical_data %>%
    filter(!(site == "Milan"))

nrow(no_milan)
```

```
[1] 144
```

Pipes shine when you chain multiple operations:

```
# Female patients, over 55, from Milan, keep only key columns
subset_females <- clinical_data |>
  filter(sex == "Female", age > 55, site == "Milan") |>
  select(patient_id, age, stage, treatment, BMI, glucose_mgdl)

head(subset_females)
```

```
       patient_id age stage treatment  BMI glucose_mgdl
PT-011     PT-011  65    II   Control 30.5           94
PT-014     PT-014  71     I   Control 21.5           86
PT-018     PT-018  72   III    Drug_A 25.2           83
PT-037     PT-037  58    II    Drug_B 26.8          108
PT-043     PT-043  60   III    Drug_B 25.9          109
PT-045     PT-045  65    II   Control 27.8           97
```

```
nrow(subset_females)
```

```
[1] 27
```

Note: inside `filter`, multiple conditions separated by commas are treated as AND.

Exercise 2.2. From the full dataset, extract patients who:

- are in stage II or III,
- have a glucose level between 90 and 130 mg/dL (inclusive),
- are assigned to Drug_A or Drug_B.

Keep only `patient_id`, `site`, `stage`, `treatment`, and `glucose_mgdl`.

Solution

```
exercise_result <- clinical_data |>
  filter(stage %in% c("II", "III"),
         glucose_mgdl >= 90,
         glucose_mgdl <= 130,
         treatment %in% c("Drug_A", "Drug_B")) |>
  select(patient_id, site, stage, treatment, glucose_mgdl)

head(exercise_result)
```

```
       patient_id  site stage treatment glucose_mgdl
PT-001     PT-001 Milan    II    Drug_A          113
PT-009     PT-009 Milan   III    Drug_B          106
PT-010     PT-010  Rome    II    Drug_A          110
PT-016     PT-016 Turin    II    Drug_B           96
PT-020     PT-020 Turin   III    Drug_B           98
PT-026     PT-026 Turin   III    Drug_A          122
```

```r
nrow(exercise_result)
```

```
[1] 68
```

2.3.5 group_by, summarize and reframe

One of the most powerful operations in data analysis is **grouping** data and computing summaries per group. In R, this is done with `group_by` + `summarize`.

How it works:

1. `group_by(column)` tells R: *"split the dataframe into groups based on the values of this column"*
2. `summarize(...)` computes one summary value per group

```
Original data:                 After group_by(treatment):

patient  treatment  glucose       GROUP: Control
PT-001   Control    105           PT-001  Control  105
PT-002   Drug_A     118    --->   PT-007  Control  98
PT-003   Drug_B     97            ...
PT-007   Control    98     --->   GROUP: Drug_A
...                               PT-002  Drug_A  118
                                  ...
```

We can perform several operation with this combo, and answer many questions. For example "*Which is the mean and sd values of glucose in each treatment group? How many patients we have per treatment group?*".

```r
# Mean glucose per treatment group
glucose_by_treatment <- clinical_data |>
  group_by(treatment) |>
```

```
  summarize(mean_glucose = mean(glucose_mgdl),
            sd_glucose   = sd(glucose_mgdl),
            n_patients   = n())

glucose_by_treatment
```

```
# A tibble: 3 x 4
  treatment mean_glucose sd_glucose n_patients
  <fct>            <dbl>      <dbl>      <int>
1 Control           109.       15.9         74
2 Drug_A            110.       14.9         88
3 Drug_B            108.       16.0         78
```

You can also group by multiple columns at once.

"*For each treatment/sex combination, which is the mean BMI and the median of the age?*"

```
# Summary by treatment AND sex
summary_by_group <- clinical_data |>
  group_by(treatment, sex) |>
  summarize(mean_BMI    = round(mean(BMI), 2),
            median_age  = median(age),
            n           = n(),
            .groups = "drop")   # good practice: drop grouping
             ↪  after summarize

summary_by_group
```

```
# A tibble: 6 x 5
  treatment sex    mean_BMI median_age     n
  <fct>     <fct>     <dbl>      <dbl> <int>
1 Control   Female     27.1       59      39
2 Control   Male       28.5       56      35
3 Drug_A    Female     25.6       58      43
4 Drug_A    Male       27.2       55      45
5 Drug_B    Female     25.2       59.5    36
6 Drug_B    Male       26.8       56      42
```

`summarize` returns **one row per group**. If you need to return **multiple rows per group** (e.g., all quantiles), use `reframe` instead:

```r
# Get quartiles of CRP per treatment group
crp_quantiles <- clinical_data |>
  group_by(treatment) |>
  reframe(quantile_value = quantile(crp_ngml, probs = c(0.25,
   ↪  0.5, 0.75)),
          quantile_name  = c("Q1", "median", "Q3"))

crp_quantiles
```

```
# A tibble: 9 x 3
  treatment quantile_value quantile_name
  <fct>              <dbl> <chr>
1 Control             5.18 Q1
2 Control             8.39 median
3 Control            13.8  Q3
4 Drug_A              5.25 Q1
5 Drug_A              7.3  median
6 Drug_A             11.4  Q3
7 Drug_B              5.36 Q1
8 Drug_B              8.52 median
9 Drug_B             12.7  Q3
```

Another useful function: `count`, which is a shortcut for `group_by + summarize(n = n())`:

```r
# Count patients per site and stage
clinical_data |>
  count(site, stage, sort = TRUE)
```

```
    site stage  n
1  Milan    II 35
2   Rome     I 33
3  Milan     I 23
4  Turin    II 22
5   Rome   III 21
6  Turin     I 20
7  Milan   III 19
8  Milan    IV 19
9   Rome    II 16
10 Turin   III 13
11 Turin    IV 12
12  Rome    IV  7
```

Exercise 2.3. For each combination of `site` and `sex`, compute: - the number

of patients, - the mean systolic blood pressure, - the proportion of hypertensive patients (hint: hypertension is when systolic blood pression is > 140).

Solution

```
bp_summary <- clinical_data |>
    mutate(hypertension = systolic_bp_mmhg > 140) |>
    group_by(site, sex) |>
    summarize(
        n_patients          = n(),
        mean_systolic_bp_mmhg    = round(mean(systolic_bp_mmhg),
         ↪  1),
        prop_hypertensive   = round(mean(hypertension), 3),
        .groups = "drop"
    )

bp_summary
```

```
# A tibble: 6 x 5
  site  sex    n_patients mean_systolic_bp_mmhg prop_hypertensive
  <fct> <fct>       <int>                 <dbl>             <dbl>
1 Milan Female         48                  146.             0.708
2 Milan Male           48                  149.             0.75
3 Rome  Female         39                  144.             0.667
4 Rome  Male           38                  148.             0.763
5 Turin Female         31                  143.             0.581
6 Turin Male           36                  150.             0.889
```

2.3.6 Join / Merge

In real-world analyses, data about the same subjects often lives in **separate files**: demographic data here, lab results there, outcomes somewhere else. To combine them, we **join** them on a shared key column (usually a patient ID).

Let's load the separate lab results file:

```
lab_data <- read.csv("data/lab_results.csv")
str(lab_data)
```

```
'data.frame':   160 obs. of  6 variables:
 $ patient_id     : chr  "PT-002" "PT-003" "PT-004" "PT-005" ...
```

```
 $ hemoglobin_gdl : num  15 11.8 16 15.4 14 16.2 10.5 15.6 12.4
 ↪  14.5 ...
 $ wbc_10e9l      : num  7.49 7.46 7.09 6.63 6.99 9.26 4.79 5.65
 ↪  8.39 7.87 ...
 $ platelets_10e9l: int  270 252 233 192 195 188 163 269 165 263
 ↪  ...
 $ alt_ul         : int  56 27 20 16 61 51 50 18 47 51 ...
 $ creatinine_mgdl: num  1.58 1.09 1.17 1.06 1.15 1.06 1.21 1.08
 ↪  1.12 1.03 ...
```

```r
head(lab_data)
```

```
  patient_id hemoglobin_gdl wbc_10e9l platelets_10e9l alt_ul
 ↪  creatinine_mgdl
1     PT-002           15.0      7.49             270     56
 ↪  1.58
2     PT-003           11.8      7.46             252     27
 ↪  1.09
3     PT-004           16.0      7.09             233     20
 ↪  1.17
4     PT-005           15.4      6.63             192     16
 ↪  1.06
5     PT-007           14.0      6.99             195     61
 ↪  1.15
6     PT-008           16.2      9.26             188     51
 ↪  1.06
```

The key column is `patient_id`. Not all patients have lab results. Depending on how we want to handle this, we choose different types of join:

Join type	What it does
`left_join(x, y)`	Keep **all rows from x**, add y columns where available (NA if no match)
`right_join(x, y)`	Keep all rows from y
`inner_join(x, y)`	Keep only rows that exist in **both** x and y
`full_join(x, y)`	Keep all rows from both, NA where no match

Let's see all approaches:

```r
# left_join: keep all patients, add lab data where available
left_df <- clinical_data |>
  left_join(lab_data, by = "patient_id")

nrow(left_df)
```

```
[1] 240
```

```r
# patients without lab data will have NA in lab columns
sum(is.na(left_df$hemoglobin_gdl))
```

```
[1] 80
```

Function `is.na()` returns TRUE when a value is NA, and FALSE if a value is an acutal value. Here an example:

```r
v1 = c(1, NA, 4) # create a vector with one NA

is.na(v1) # returns a vector with TRUE in position 2, and FALSE
 ↪  elsewhere
```

```
[1] FALSE  TRUE FALSE
```

```r
summary(left_df)
```

```
  patient_id           site     visit_date         visit_year
 ↪
 Length:240         Milan:96   Length:240         Length:240
 ↪
 Class :character   Rome :77   Class :character   Class
 ↪  :character
 Mode  :character   Turin:67   Mode  :character   Mode
 ↪  :character




 visit_month         visit_day              age           sex
 ↪  smoking
 Length:240         Length:240         Min.   :26.0   Female:118
 ↪  current:63
```

```
Class :character   Class :character   1st Qu.:49.0   Male  :122
↪  former :90
Mode  :character   Mode  :character   Median :57.5
↪  never  :87
                                       Mean   :57.3
↪
                                       3rd Qu.:65.0
↪
                                       Max.   :85.0
↪

  treatment  stage      weight_kg        height_cm
↪  systolic_bp_mmhg
Control:74   I  :76   Min.   : 39.40   Min.   :148.9   Min.
↪  :115.0
Drug_A :88   II :73   1st Qu.: 65.35   1st Qu.:162.2   1st
↪  Qu.:139.0
Drug_B :78   III:53   Median : 75.65   Median :168.5   Median
↪  :146.5
             IV :38   Mean   : 76.87   Mean   :169.2   Mean
↪  :146.7
                      3rd Qu.: 87.83   3rd Qu.:174.8   3rd
↪  Qu.:153.2
                      Max.   :132.90   Max.   :200.1   Max.
↪  :192.0

diastolic_bp_mmhg  glucose_mgdl   cholesterol_mgdl    crp_ngml
↪
Min.   : 73.00    Min.   : 60.0   Min.   :113.0    Min.   : 1.92
↪
1st Qu.: 90.75    1st Qu.: 98.0   1st Qu.:189.8    1st Qu.: 5.30
↪
Median : 97.00    Median :109.0   Median :207.0    Median : 8.37
↪
Mean   : 96.11    Mean   :108.8   Mean   :207.9    Mean   :10.35
↪
3rd Qu.:102.00    3rd Qu.:118.0   3rd Qu.:226.0    3rd Qu.:13.02
↪
Max.   :120.00    Max.   :149.0   Max.   :307.0    Max.   :42.09
↪

   height_m          BMI              BMI_class   hemoglobin_gdl
↪
Min.   :1.489   Min.   :16.00   Underweight: 67   Min.   : 9.20
↪
```

```
 1st Qu.:1.623   1st Qu.:23.77   Normal     : 58   1st Qu.:12.40
↪
 Median :1.685   Median :26.85   Overweight :102   Median :13.25
↪
 Mean   :1.692   Mean   :26.71   Obese      : 13   Mean   :13.50
↪
 3rd Qu.:1.748   3rd Qu.:29.80                     3rd Qu.:14.70
↪
 Max.   :2.001   Max.   :40.70                     Max.   :17.00
↪
                                                   NA's   :80
↪
   wbc_10e9l      platelets_10e9l     alt_ul
↪  creatinine_mgdl
 Min.   : 2.310   Min.   : 60.0   Min.   : 10.00   Min.   :0.630
↪
 1st Qu.: 5.518   1st Qu.:191.8   1st Qu.: 32.75   1st Qu.:1.018
↪
 Median : 7.000   Median :232.0   Median : 46.50   Median :1.140
↪
 Mean   : 6.758   Mean   :231.7   Mean   : 52.20   Mean   :1.138
↪
 3rd Qu.: 7.902   3rd Qu.:271.0   3rd Qu.: 61.00   3rd Qu.:1.252
↪
 Max.   :11.380   Max.   :391.0   Max.   :161.00   Max.   :1.580
↪
 NA's   :80       NA's   :80      NA's   :80       NA's   :80
↪
```

```r
# inner_join: keep only the 90 patients who have both clinical
↪  and lab data
matched_only <- clinical_data |>
  inner_join(lab_data, by = "patient_id")

nrow(matched_only)
```

```
[1] 160
```

In most clinical research scenarios, `left_join` is the most common choice: you start with your full patient list and enrich it with additional data, accepting that some patients may lack certain measurements (NAs).

Exercise 2.4.

1. Perform a `left_join` of `clinical_data` with `lab_data`.

2. Among patients who have lab data available, compute the mean hemoglobin and mean creatinine per treatment group.
3. Which group has the highest mean creatinine?

Solution

```
# 1. Left join
joined_df <- clinical_data |>
  left_join(lab_data, by = "patient_id")

# 2. Summarize (filter out NA first)
lab_summary <- joined_df |>
  filter(!is.na(hemoglobin_gdl)) |> # filter for what is NOT NA
  ↪  in hemoglobin
  group_by(treatment) |>
  summarize(mean_hemoglobin   = round(mean(hemoglobin_gdl), 2),
            mean_creatinine   = round(mean(creatinine_mgdl), 3),
            n                 = n(),
            .groups = "drop")

lab_summary
```

```
# A tibble: 3 x 4
  treatment mean_hemoglobin mean_creatinine     n
  <fct>               <dbl>           <dbl> <int>
1 Control              13.7            1.16    51
2 Drug_A               13.5            1.13    62
3 Drug_B               13.2            1.13    47
```

```
# 3. Highest mean creatinine
lab_summary |> arrange(desc(mean_creatinine))
```

```
# A tibble: 3 x 4
  treatment mean_hemoglobin mean_creatinine     n
  <fct>               <dbl>           <dbl> <int>
1 Control              13.7            1.16    51
2 Drug_B               13.2            1.13    47
3 Drug_A               13.5            1.13    62
```

2.4 Save your results

After all this work, let's save the enriched and cleaned dataset:

```r
# Save the left-joined dataset with all derived variables
final_data <- clinical_data |>
  left_join(lab_data, by = "patient_id")

write.csv(final_data,
          file = "output/clinical_data_enriched.csv",
          row.names = FALSE)
```

2.5 Home exercise

Try to do the following on your own:

1. Load `clinical_data_enriched.csv` from the output folder.
2. Clean up: keep only the columns `patient_id`, `site`, `sex`, `age`, `treatment`, `stage`, `BMI`, `BMI_class`, `glucose_mgdl`, `cholesterol_mgdl`, `crp_ngml`, `hemoglobin_gdl`, `creatinine_mgdl`.
3. Restrict to patients who have complete data (no NAs) — hint: use function `drop_na()`, see this.
4. For each stage, compute: number of patients, mean age, mean BMI, and proportion of obese patients.
5. Save the summary table to `output/summary_by_stage.csv`.

```r
library(tidyverse)

# 1. Load the enriched dataset
clinical_data_enriched <-
 ↪  read.csv("output/clinical_data_enriched.csv")

# 2. Keep only the columns of interest
clean_data <- clinical_data_enriched |>
  select(patient_id, site, sex, age, treatment, stage,
         BMI, BMI_class, glucose_mgdl, cholesterol_mgdl,
         crp_ngml, hemoglobin_gdl, creatinine_mgdl)

# 3. Restrict to patients with complete data (no NAs)
clean_data <- clean_data |>
  drop_na()

cat("Patients with complete data:", nrow(clean_data), "\n")
```

```
Patients with complete data: 160
```

```r
# 4. Summary by stage
summary_by_stage <- clean_data |>
  group_by(stage) |>
  summarize(
    n_patients       = n(),
    mean_age         = round(mean(age), 1),
    mean_BMI         = round(mean(BMI), 2),
    prop_obese       = round(mean(BMI_class == "Obese"), 3),
    .groups = "drop"
  )

print(summary_by_stage)
```

```
# A tibble: 4 x 5
  stage n_patients mean_age mean_BMI prop_obese
  <chr>      <int>    <dbl>    <dbl>      <dbl>
1 I             47     55.2     27.2      0.043
2 II            53     57.1     27.3      0.038
3 III           35     56.2     28.2      0.029
4 IV            25     57.5     24.2      0.16
```

```r
# 5. Save
write.csv(summary_by_stage,
          file = "output/summary_by_stage.csv",
          row.names = FALSE)
```

2.6 Bonus: how the clinical data were created

If you want to understand how we generated the dataset for this lesson, download the R script Notice the use of `set.seed` to make results reproducible.

Chapter 3

Data visualization

Data visualization is a crucial step of data analysis, as it can help us in:

- Understanding the distribution of our data
- Evaluate outliers
- Visualize immediately the differences between groups
- Communicate our results to peers

I suggest you to visualize data as soon as you have finished to prepare them, prior to any statistical test (even prior to test for parametric assumptions), that's why today we are doing it.

3.1 Data visualization in R with ggplot2

`ggplot2` is a package built for data visualization in R. It is simple to use and it works mainly on dataframes.

To create a plot, we need to use different functions, that act as single building blocks to change single aspects of the plot. Here an example of mock code to create a plot with all the functions:

```
ggplot() +
  geom_<type>() +
  labs() +
  scale_<type>() +
  facet_<type>() +
  coord_<type>() +
  theme_<type>
```

But, do we need all of them? No, the basic plots can be created with just `ggplot() + geom_<type>()`.

3.2 Geom

The `geom` function specifies the "geometry" we want to give to the data: basically, which type of graph we want (barplot, scatter, histogram, etc). It is the necessary and sufficient function to create a plot.

Here the list of all possible geom to use with the basic package.

First of all, let's load the data from previous session, and keep only the rows with all the information:

```
suppressPackageStartupMessages(library(tidyverse))
df <- read.csv("output/clinical_data_enriched.csv", header = T,
 ↪  stringsAsFactors = T)
df <- na.omit(df)
str(df)
```

```
'data.frame':   160 obs. of  26 variables:
 $ patient_id       : Factor w/ 240 levels "PT-001","PT-002",..:
 ↪  2 3 4 5 7 8 9 10 11 12 ...
 $ site             : Factor w/ 3 levels "Milan","Rome",..: 2 2 3
 ↪  3 2 2 1 2 1 2 ...
 $ visit_date       : Factor w/ 218 levels
 ↪  "2018-01-14","2018-01-24",..: 75 150 52 104 80 214 57 146 125
 ↪  170 ...
 $ visit_year       : int  2020 2023 2020 2023 2020 2025 2020
 ↪  2023 2023 2025 ...
 $ visit_month      : int  8 10 3 3 9 10 4 9 6 3 ...
 $ visit_day        : int  15 15 6 30 9 7 19 27 29 20 ...
 $ age              : int  40 59 46 58 51 34 43 60 65 52 ...
 $ sex              : Factor w/ 2 levels "Female","Male": 2 1 2 2
 ↪  2 1 2 2 1 2 ...
 $ smoking          : Factor w/ 3 levels "current","former",..: 1
 ↪  2 1 2 2 3 3 2 1 3 ...
 $ treatment        : Factor w/ 3 levels "Control","Drug_A",..: 3
 ↪  2 2 3 2 1 3 2 1 2 ...
 $ stage            : Factor w/ 4 levels "I","II","III",..: 1 1 1
 ↪  3 1 1 3 2 2 3 ...
 $ weight_kg        : num  100.9 65 95.9 75.7 69.7 ...
 $ height_cm        : num  185 166 182 168 170 ...
 $ systolic_bp_mmhg : int  143 152 142 160 137 146 151 159 125
 ↪  147 ...
```

```
 $ diastolic_bp_mmhg: int  102 98 88 96 96 106 94 104 76 96 ...
 $ glucose_mgdl     : int  105 110 120 131 108 93 106 110 94 86
 ↪  ...
 $ cholesterol_mgdl : int  239 146 232 192 163 199 200 185 209
 ↪  222 ...
 $ crp_ngml         : num  6.13 7.81 3.35 21.72 5.41 ...
 $ height_m         : num  1.85 1.66 1.81 1.68 1.7 ...
 $ BMI              : num  29.4 23.7 29.1 26.8 24.2 29.8 24.8
 ↪  26.9 30.5 29.6 ...
 $ BMI_class        : Factor w/ 4 levels "Normal","Obese",..: 3 4
 ↪  3 3 4 3 4 3 1 3 ...
 $ hemoglobin_gdl   : num  15 11.8 16 15.4 14 16.2 10.5 15.6 12.4
 ↪  14.5 ...
 $ wbc_10e9l        : num  7.49 7.46 7.09 6.63 6.99 9.26 4.79
 ↪  5.65 8.39 7.87 ...
 $ platelets_10e9l  : int  270 252 233 192 195 188 163 269 165
 ↪  263 ...
 $ alt_ul           : int  56 27 20 16 61 51 50 18 47 51 ...
 $ creatinine_mgdl  : num  1.58 1.09 1.17 1.06 1.15 1.06 1.21
 ↪  1.08 1.12 1.03 ...
 - attr(*, "na.action")= 'omit' Named int [1:80] 1 6 14 21 25 33
 ↪  46 49 52 54 ...
  ..- attr(*, "names")= chr [1:80] "1" "6" "14" "21" ...
```

```r
summary(df)
```

```
   patient_id     site         visit_date    visit_year
 ↪  visit_month
 PT-002 :  1   Milan:65   2020-07-21:  3   Min.   :2018   Min.
 ↪  : 1.000
 PT-003 :  1   Rome :50   2018-03-03:  2   1st Qu.:2020   1st
 ↪  Qu.: 3.750
 PT-004 :  1   Turin:45   2018-07-18:  2   Median :2023   Median
 ↪  : 6.000
 PT-005 :  1              2020-05-23:  2   Mean   :2022   Mean
 ↪  : 5.969
 PT-007 :  1              2020-10-16:  2   3rd Qu.:2025   3rd
 ↪  Qu.: 8.000
 PT-008 :  1              2023-03-06:  2   Max.   :2025   Max.
 ↪  :11.000
 (Other):154              (Other)   :147
 ↪
   visit_day          age            sex        smoking
 ↪  treatment  stage
```

```
Min.   : 1.00   Min.   :33.00   Female:83   current:44
↪  Control:51   I  :47
1st Qu.:10.00   1st Qu.:48.00   Male  :77   former :55   Drug_A
↪  :62   II :53
Median :18.00   Median :56.50               never  :61   Drug_B
↪  :47   III:35
Mean   :17.16   Mean   :56.41
↪  IV :25
3rd Qu.:24.00   3rd Qu.:64.00
↪
Max.   :31.00   Max.   :85.00
↪

  weight_kg        height_cm     systolic_bp_mmhg
↪  diastolic_bp_mmhg
Min.   : 39.40   Min.   :148.9   Min.   :120.0    Min.   : 73.00
↪
1st Qu.: 66.17   1st Qu.:162.4   1st Qu.:138.0    1st Qu.: 89.00
↪
Median : 75.75   Median :168.2   Median :146.0    Median : 96.00
↪
Mean   : 77.59   Mean   :169.2   Mean   :145.9    Mean   : 95.46
↪
3rd Qu.: 88.33   3rd Qu.:174.6   3rd Qu.:153.0    3rd Qu.:101.00
↪
Max.   :132.90   Max.   :191.3   Max.   :178.0    Max.   :120.00
↪

 glucose_mgdl cholesterol_mgdl    crp_ngml         height_m
Min.   : 75   Min.   :113.0    Min.   : 1.930   Min.   :1.489
1st Qu.: 98   1st Qu.:187.8    1st Qu.: 5.300   1st Qu.:1.624
Median :109   Median :205.5    Median : 8.465   Median :1.682
Mean   :110   Mean   :206.1    Mean   :10.525   Mean   :1.692
3rd Qu.:119   3rd Qu.:223.2    3rd Qu.:13.165   3rd Qu.:1.746
Max.   :147   Max.   :307.0    Max.   :42.090   Max.   :1.913

     BMI              BMI_class  hemoglobin_gdl    wbc_10e9l
↪
Min.   :16.00   Normal     :46   Min.   : 9.20   Min.   : 2.310
↪
1st Qu.:24.18   Obese      : 9   1st Qu.:12.40   1st Qu.: 5.518
↪
Median :26.90   Overweight :65   Median :13.25   Median : 7.000
↪
Mean   :26.98   Underweight:40   Mean   :13.50   Mean   : 6.758
↪
```

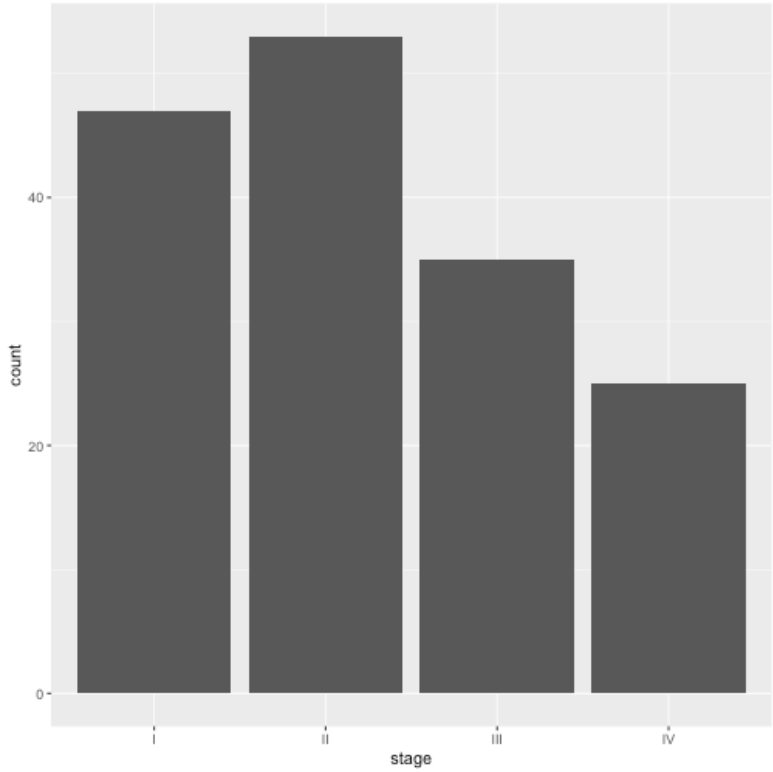
```
 3rd Qu.:30.50                     3rd Qu.:14.70   3rd Qu.: 7.902
↪
 Max.   :40.70                     Max.   :17.00   Max.   :11.380
↪

 platelets_10e9l     alt_ul       creatinine_mgdl
 Min.   : 60.0   Min.   : 10.00   Min.   :0.630
 1st Qu.:191.8   1st Qu.: 32.75   1st Qu.:1.018
 Median :232.0   Median : 46.50   Median :1.140
 Mean   :231.7   Mean   : 52.20   Mean   :1.138
 3rd Qu.:271.0   3rd Qu.: 61.00   3rd Qu.:1.252
 Max.   :391.0   Max.   :161.00   Max.   :1.580
```

3.2.1 Barplot

We create a basic barplot to show the number of patient for each stage using the `geom_bar` function:

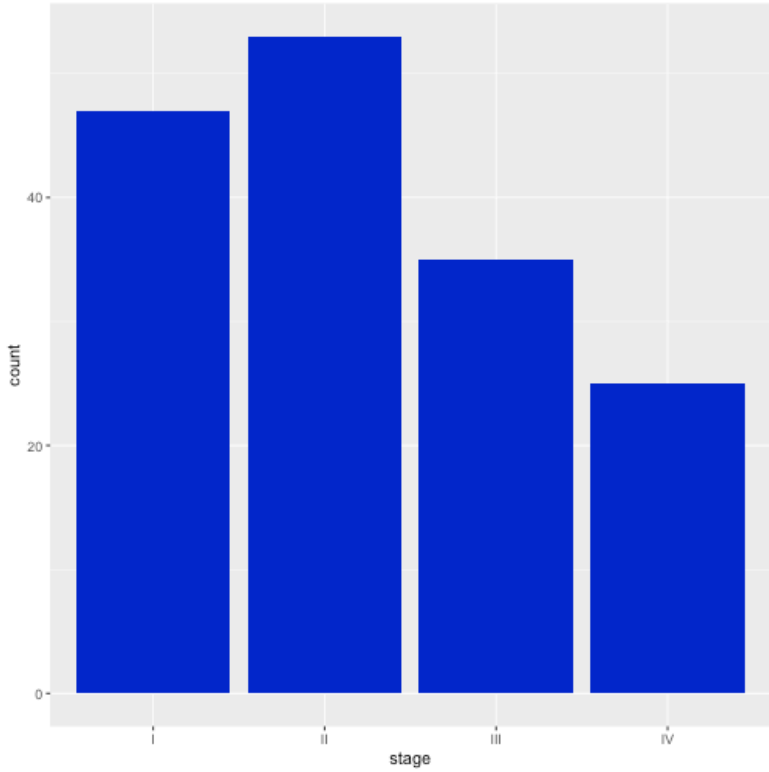
```
suppressPackageStartupMessages(library(ggplot2))
```

```
ggplot(df) +
  geom_bar(mapping = aes(x = stage))
```



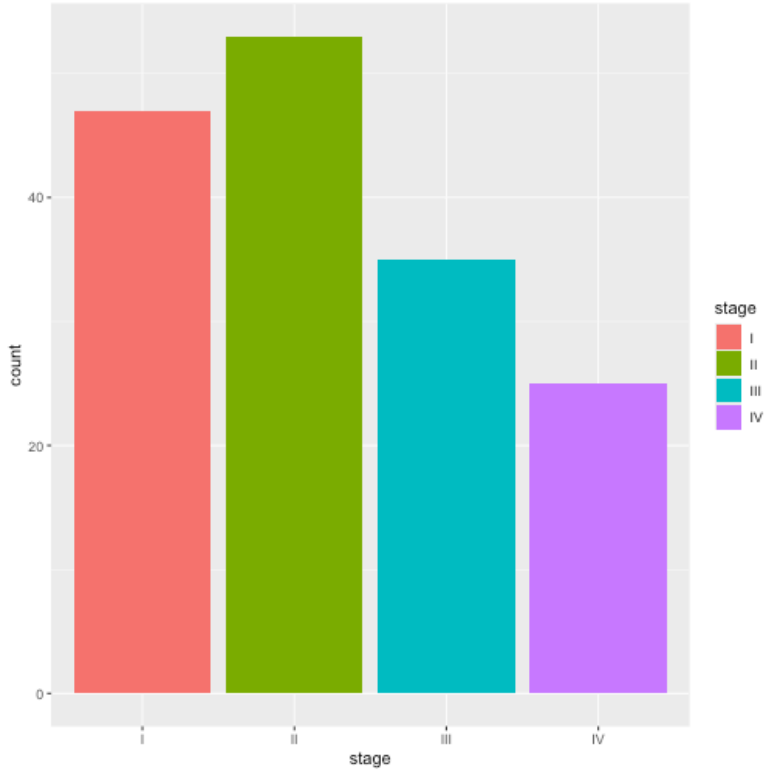
Let's add some colors. We can set colors for all bars:

```
ggplot(df) +
  geom_bar(mapping = aes(x = stage), fill = "blue3")
```



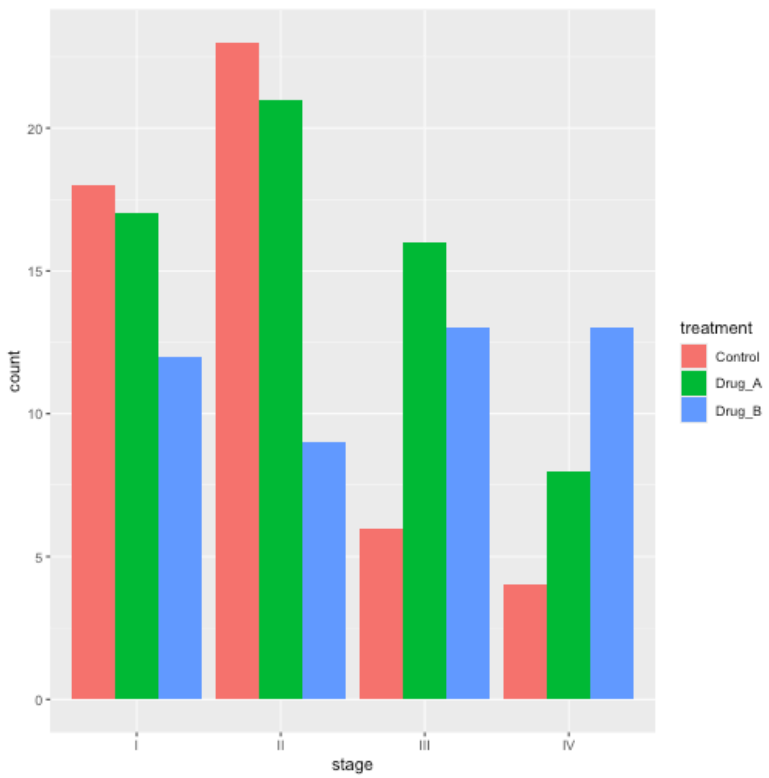
Or based on stage:

```
ggplot(df) +
  geom_bar(mapping = aes(x = stage, fill = stage))
```



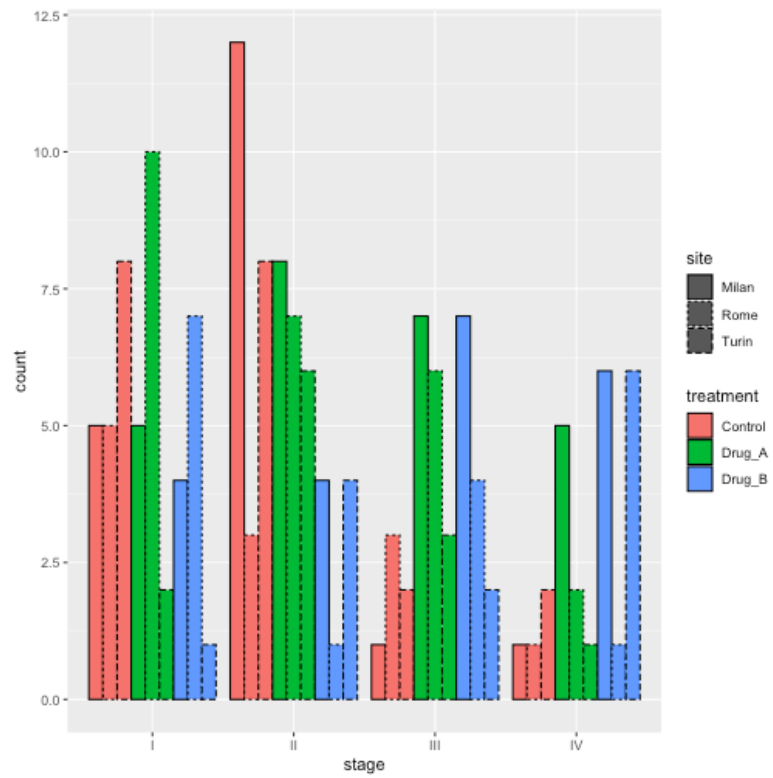
One of the main features of ggplot is that we can easily split data on x axis based on another column. Let's say we want a bar for each treatment within each stage; we can achieve it setting the column in another aes option:

```
ggplot(df) +
  geom_bar(mapping = aes(x = stage, fill = treatment), position =
   ↪  "dodge")
```



We can set different options in geom bar (both in aes or not), like color (the color of the outline of the bar), linewidth (the size of the outline), linetype (type of the outline of the bars, like dotted or continuous). Here you can find all the options. Here an example:

```
all_n_barplot <- ggplot(df) +
  geom_bar(mapping = aes(x = stage, fill = treatment, linetype =
   ↪  site), position = "dodge", color = "black", linewidth =
   ↪  0.5)
all_n_barplot
```



3.2.1.1 Stat identity

In this case, geom_bar counts the occurrence of each label of stage and plot them. But what if we have a dataframe with the actual data we want to plot? Let's say that we want to plot the mean values of "diastolic_bp_mmhg" of Stage I females for each treatment; first, we have to create the dataframe for it by filtering the data and calculating the mean:

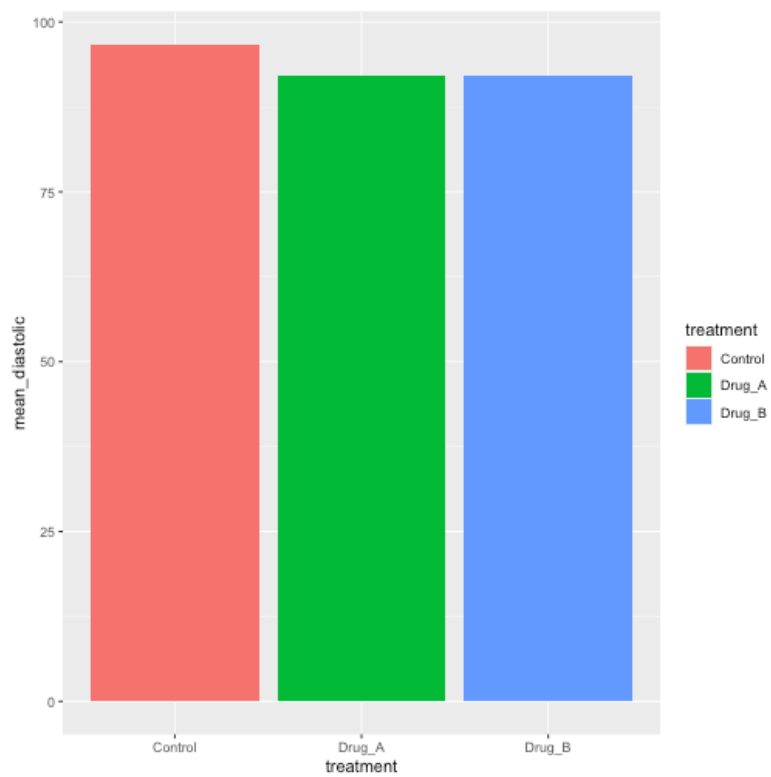
```
stageI_femalse_diastolic_means <- df %>%
  filter(stage == "I" & sex == "Female") %>%
  group_by(treatment) %>%
  summarise("mean_diastolic" = mean(diastolic_bp_mmhg))
stageI_femalse_diastolic_means
```

```
# A tibble: 3 x 2
  treatment mean_diastolic
  <fct>              <dbl>
1 Control             96.7
2 Drug_A              92
```

```
3 Drug_B              92
```

Then we can use the function specifying both x and y as aes:

```
stageI_female_diastolic_means_barplot <-
 ↪  ggplot(stageI_femalse_diastolic_means) +
  geom_bar(mapping = aes(x = treatment, y = mean_diastolic, fill
   ↪  = treatment), stat = "identity")
stageI_female_diastolic_means_barplot
```

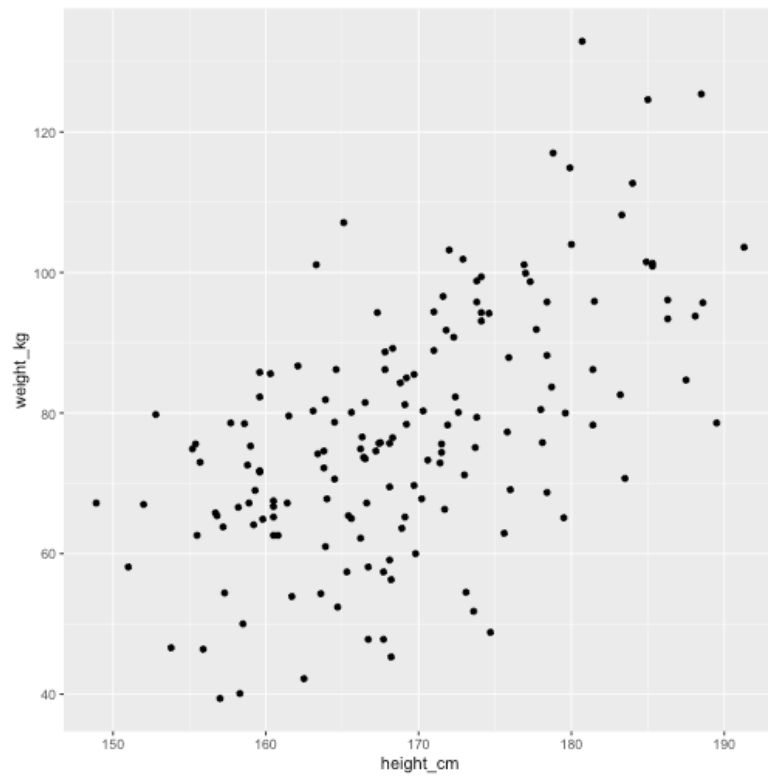


3.2.2 Scatter plot

If we want to create a scatter plot, we can use `geom_point` function.

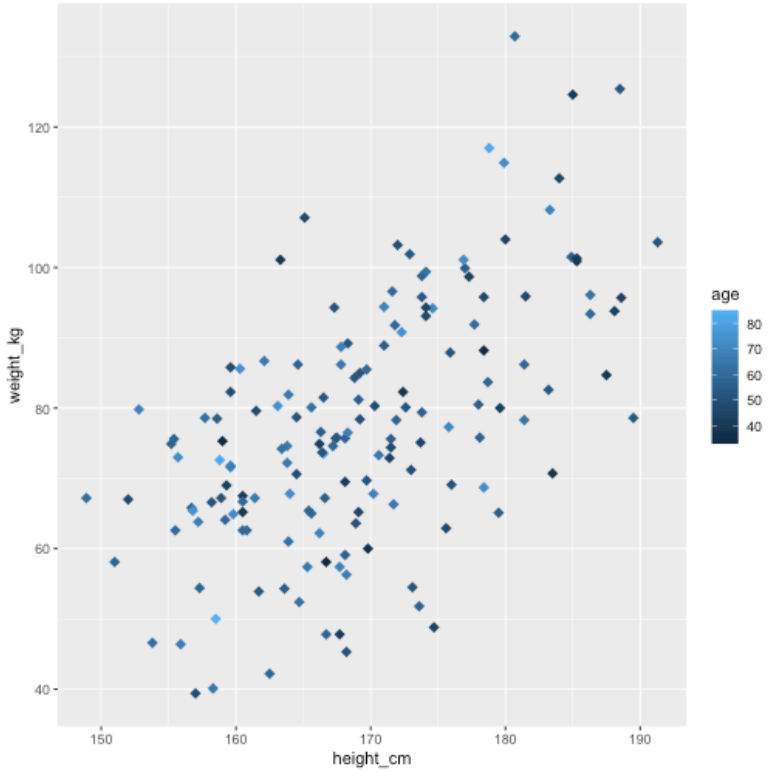
We can create the plot between weight and height:

```
ggplot(df) +
  geom_point(aes(x = height_cm, y = weight_kg))
```



Just from this plot we can see that there is a positive correlation. We can also change color, shape size of the dots. As before, all that is inserted in aes will apply to each dot based on the value of the corresponding column, while outside aes is applied to all dots:

```
corr_plot <- ggplot(df) +
  geom_point(aes(x = height_cm, y = weight_kg, color = age), size
   ↪  = 3, shape = 18)
corr_plot
```

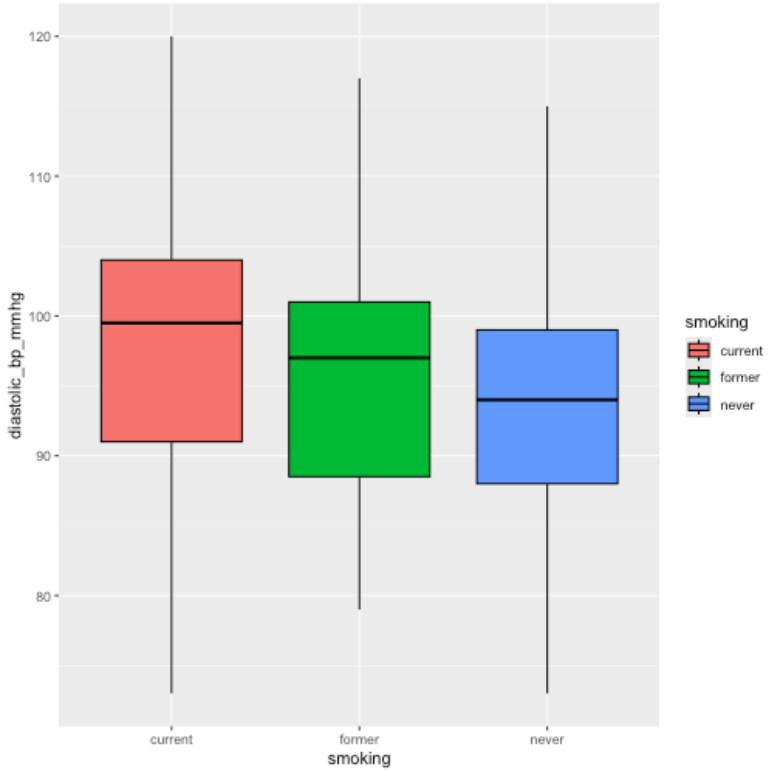


Here the complete list of options for geom_point.

3.2.3 Boxplot

Next, we will see how to create a boxplot, that is one of the most used plot when showing data distribution. Here we look at the diastolic bp in different smoking levels.

```
diastolic_bp_boxplot <- ggplot(df) +
  geom_boxplot(aes(x = smoking, y = diastolic_bp_mmhg, fill =
   ↪  smoking), color = "black", outlier.color = "black")
diastolic_bp_boxplot
```

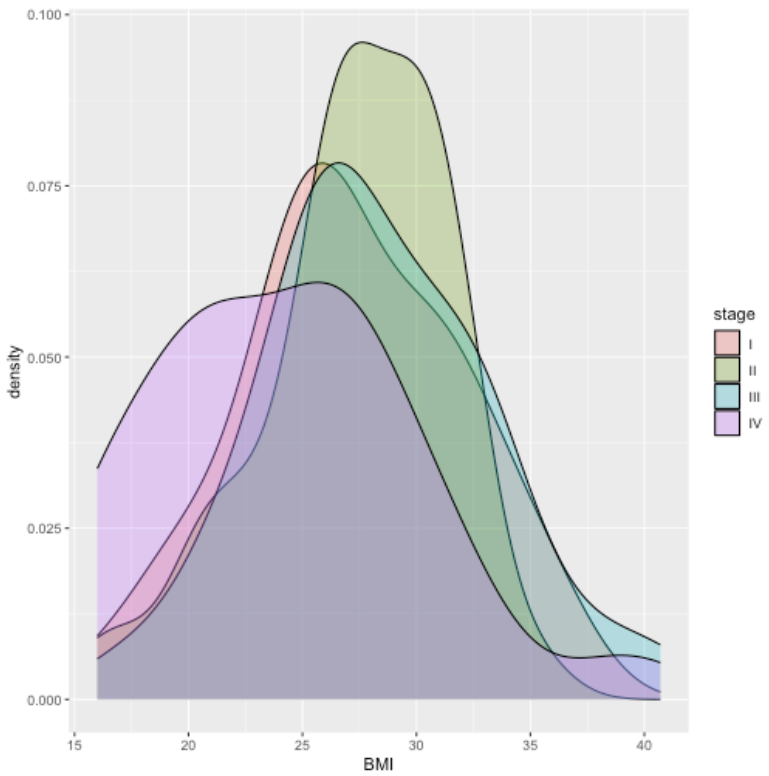


Here the complete list of options for geom_boxplot.

3.2.4 Density plot

Another way to visualize the distribution of the data is the density plot. Let's see the distribution of BMI among stage groups.

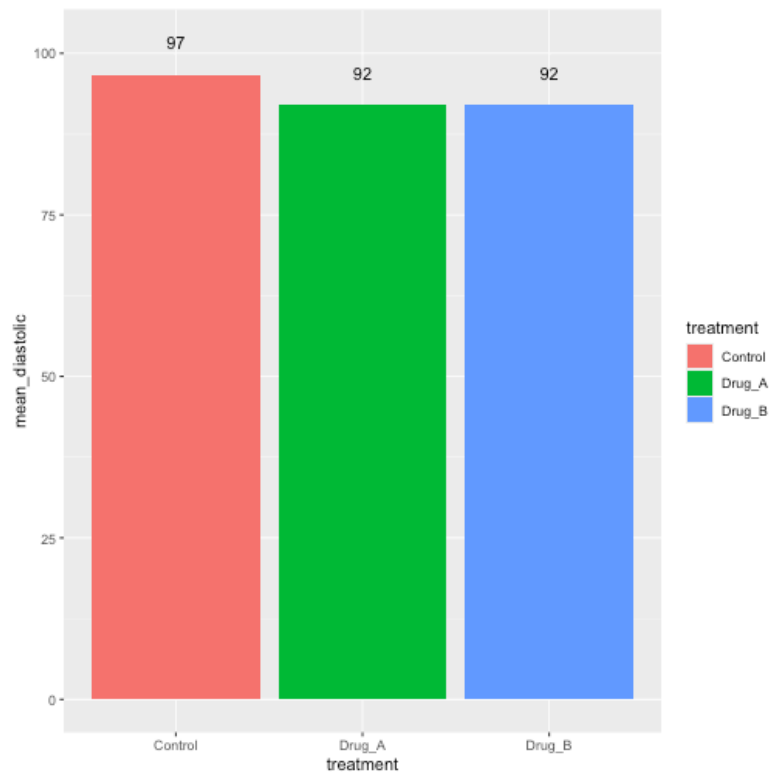
```
BMI_stages_density <- ggplot(df) +
  geom_density(aes(x = BMI, fill = stage), alpha = 0.3)
BMI_stages_density
```



3.2.5 Combine geom

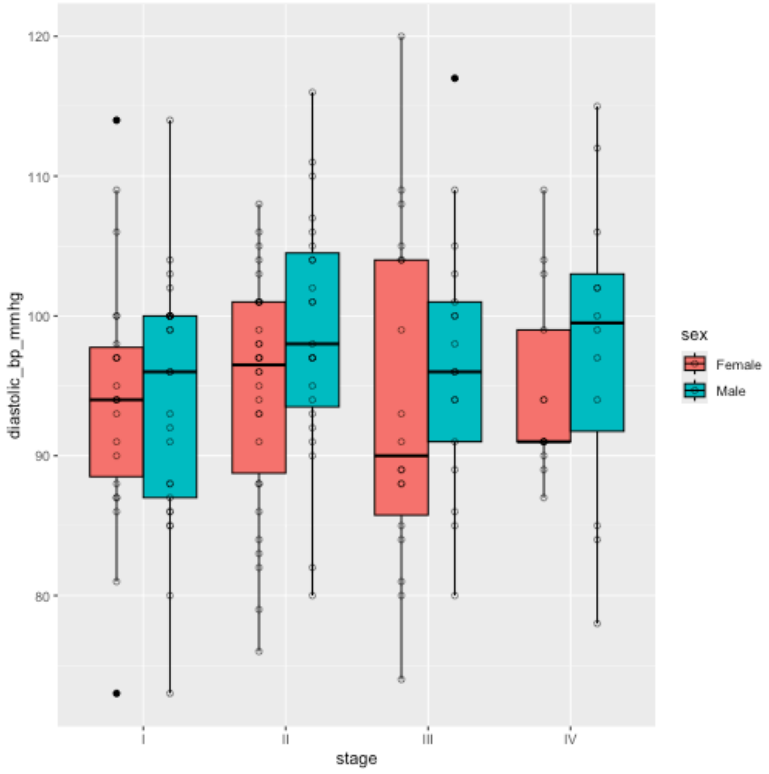
We can also combine geometries. For example, let's add to the *stageI_female_diastolic_means_barplot* the actual values with `geom_text`:

```
stageI_female_diastolic_means_barplot <-
 ↪  stageI_female_diastolic_means_barplot +
  geom_text(aes(x = treatment, y = mean_diastolic, label =
   ↪  round(mean_diastolic)), nudge_y = 5)
stageI_female_diastolic_means_barplot
```



We can combine as many as we want, another example is combine geom_boxplot with geom_point to add individual data points on the plot.

```
ggplot(df) +
  geom_boxplot(aes(x = stage, y = diastolic_bp_mmhg, fill = sex),
  ↪
               color = "black", outlier.color = "black", position
                ↪  = position_dodge(width = 0.75)) +
  geom_jitter(aes(x = stage, y = diastolic_bp_mmhg, fill = sex),
             position = position_dodge(width = 0.75), color =
              ↪  "black", alpha = 0.6, shape = 1)
```



3.3 Labs

All these plots are a good starting point, but we can now start to personalize them. The first thing we can change are labs.

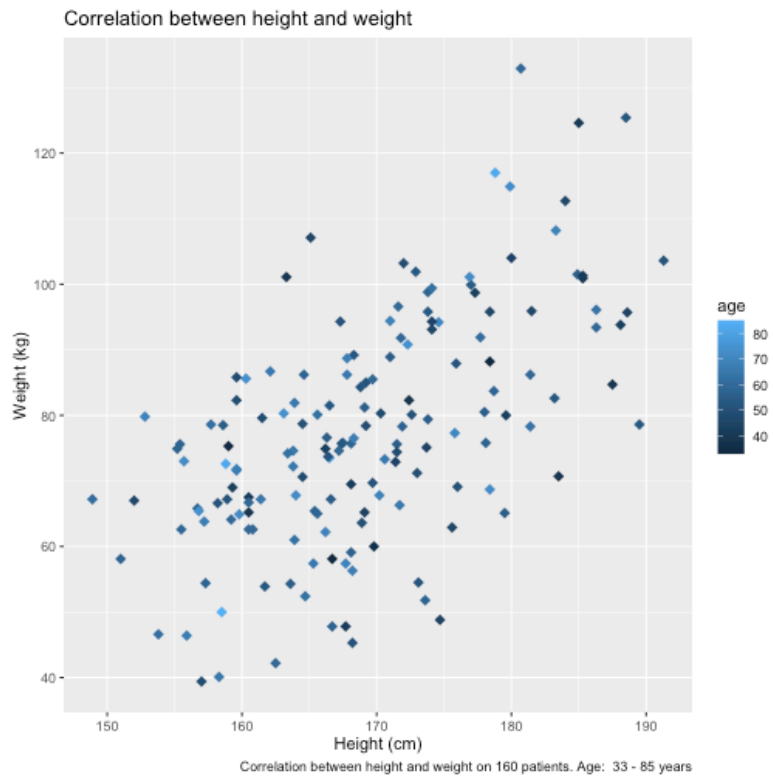
Labs comprehends axis title, plot title and subtitle, legend titles, caption and tag (here the reference page with examples).

```
ggplot() +
  geom_<type>() +
  labs()
```

Let's change the labels of the correlation plot:

```
corr_plot <- corr_plot +
  labs(title = "Correlation between height and weight",
       x = "Height (cm)",
       y = "Weight (kg)",
```

```
        caption = paste("Correlation between height and weight
        ↪  on", dim(df)[1], "patients. Age: ",
                        min(df$age), "-", max(df$age), "years"))
corr_plot
```



3.4 Scales

The definition of scales from the official documentation is "Scales control the details of how data values are translated to visual properties". Practically speaking, with scales we can control the data on the axis, the colors and the other parameters of the aesthetics (linetype, size, fill) by calling the correspondent function (that's what "" meant in the mock code).

```
ggplot() +
  geom_<type>() +
  labs() +
  scale_<type>()
```

We will see the main application of it.

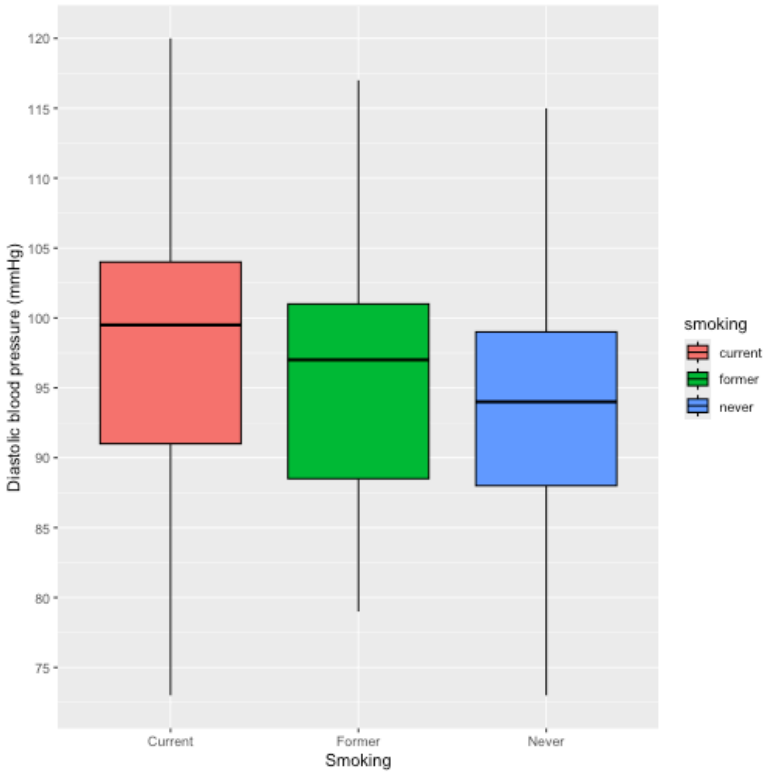
3.4.1 Axis

With `scales_x/y_<type>` (if continuous, binned, manual or discrete) we can control the details of the axis, such as breaks, labels, limits, mathematical transformation and so on.

For example, we can change the breaks on both x and y axis of diastolic_bp_boxplot:

- on y axis, the distance between breaks is now 10, we change it to 5
- on x axis, we change the labels of the axis with the first letter uppercase

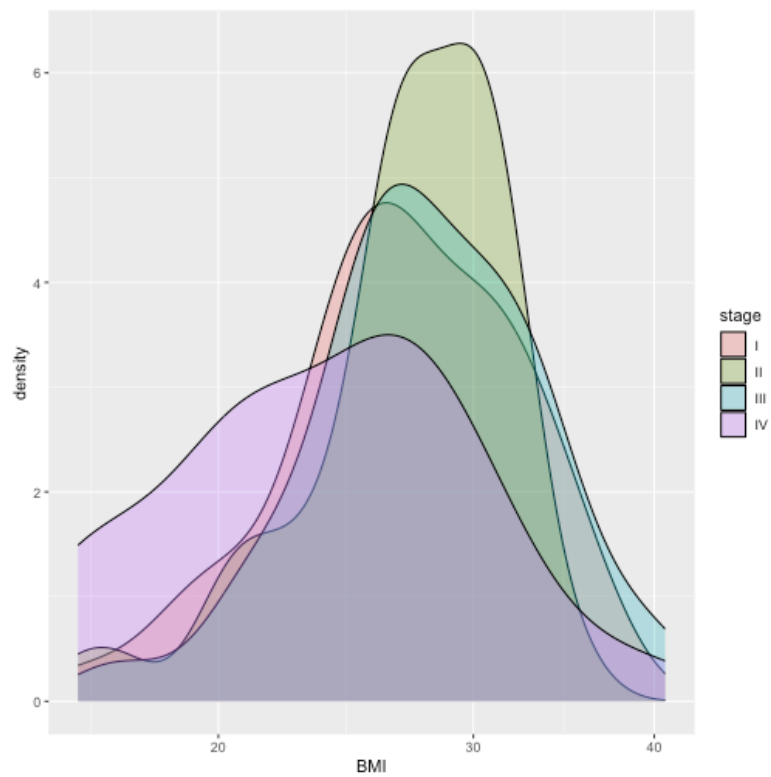
```
diastolic_bp_boxplot <- diastolic_bp_boxplot +
  labs(x = "Smoking", y = "Diastolic blood pressure (mmHg)") +
  scale_y_continuous(breaks = seq(0, max(df$diastolic_bp_mmhg),
   ↪  5)) +
  scale_x_discrete(labels = c("Current", "Former", "Never"))
diastolic_bp_boxplot
```



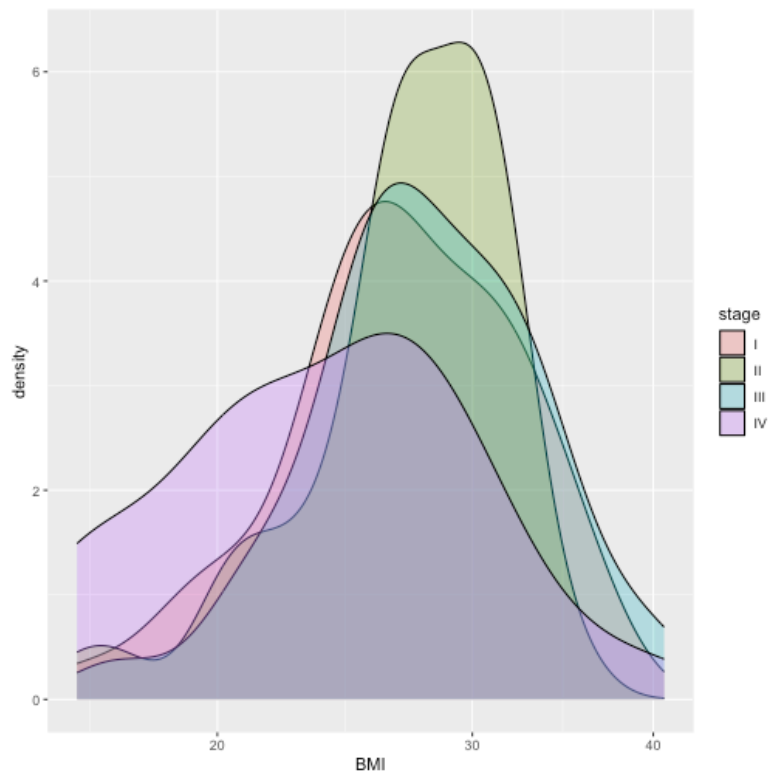
QUESTO SOLO NEL SITO

We can also apply mathematical transformation to continuous axis (like log scales):

```
BMI_stages_density +
  scale_x_continuous(trans = "log10")
```




```
BMI_stages_density +
  scale_x_log10()
```

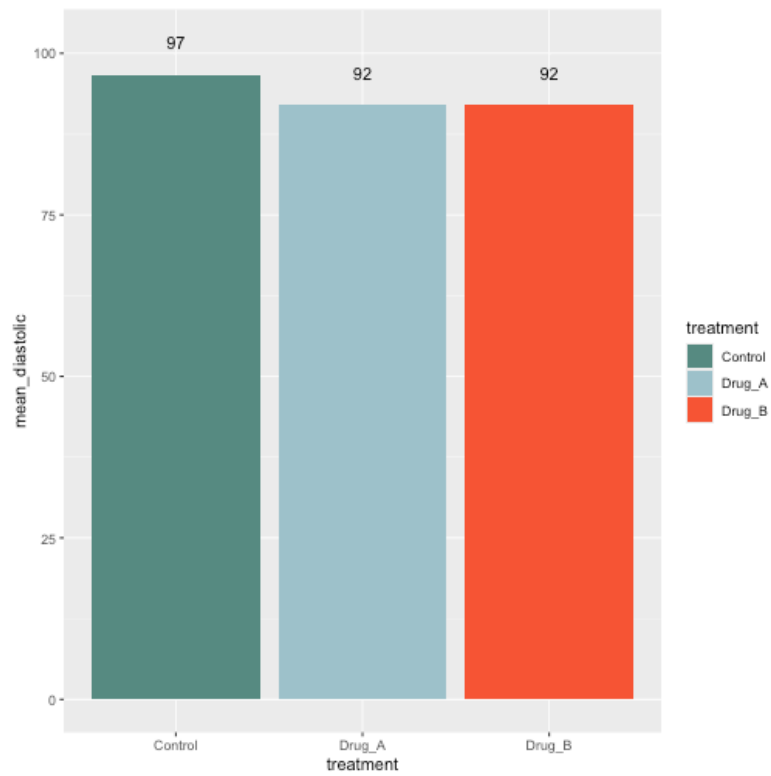


3.4.2 Colors

When we set `fill` or `color` aesthetics, we actually do not choose the colors that ggplot apply, we just say to it to make them different. With scales, we can set the colors we want. This site is very useful to choose colors.

For example, in stageI_female_diastolic_means_barplot, we want the Control to be gray, Drug_A to be blueish and Drug_B to be red. We can set it up like this:

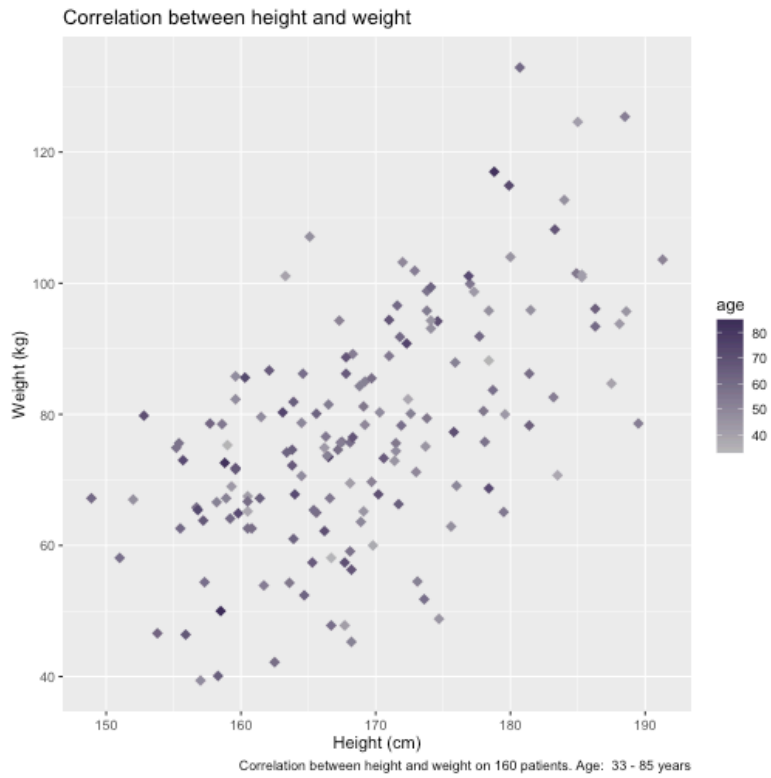
```
stageI_female_diastolic_means_barplot <-
 ↪  stageI_female_diastolic_means_barplot +
  scale_fill_manual(values = c("Control" = "#568A81",
                               "Drug_A" = "#9FC2CC",
                               "Drug_B" = "#F95738"))
stageI_female_diastolic_means_barplot
```



See this insite.) on hex color codes.

We can also change the color bar of corr_plot:

```
corr_plot <- corr_plot +
  scale_color_gradient(low = "#B8B7BA", high = "#3E2F5B")
corr_plot
```



See this stackoverflow post for some useful tips and explanation on color scales in ggplot.

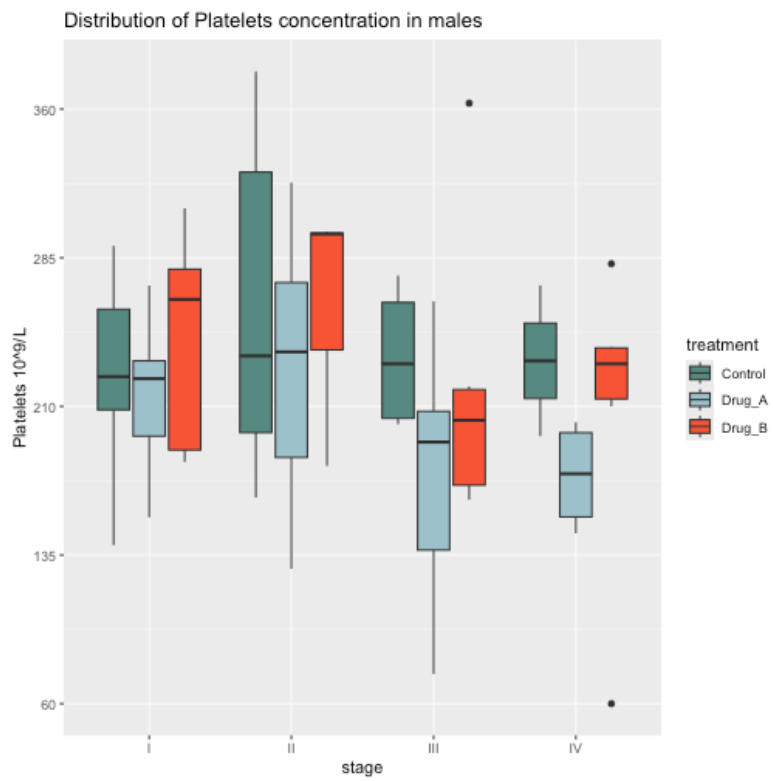
Exercise 3.1. Create a boxplot of platelets_10e9l in males, divided by stage (x axis) and treatment (fill). Then:

- y axis label should be "Platelets 10^9/L", with breaks every 75 starting from the minum value of platelets_10e9l up to the maximum value of platelets_10e9l
- Treatment colors should be the same as the ones in stageI_female_diastolic_means_barplot
- The title of the plot shoul be meaningful

Solution

```
males_df <- df |>
    filter(sex == "Male")

ggplot(males_df) +
    geom_boxplot(aes(x = stage, y = platelets_10e9l, fill =
     ↪  treatment)) +
```

```
    labs(y = "Platelets 10^9/L",
         title = "Distribution of Platelets concentration in
          ↪  males") +
    scale_fill_manual(values = c("Control" = "#568A81",
                                 "Drug_A" = "#9FC2CC",
                                 "Drug_B" = "#F95738")) +
    scale_y_continuous(breaks =
     ↪  seq(min(males_df$platelets_10e9l),
     ↪  max(males_df$platelets_10e9l), 75))
```



3.5 Facet

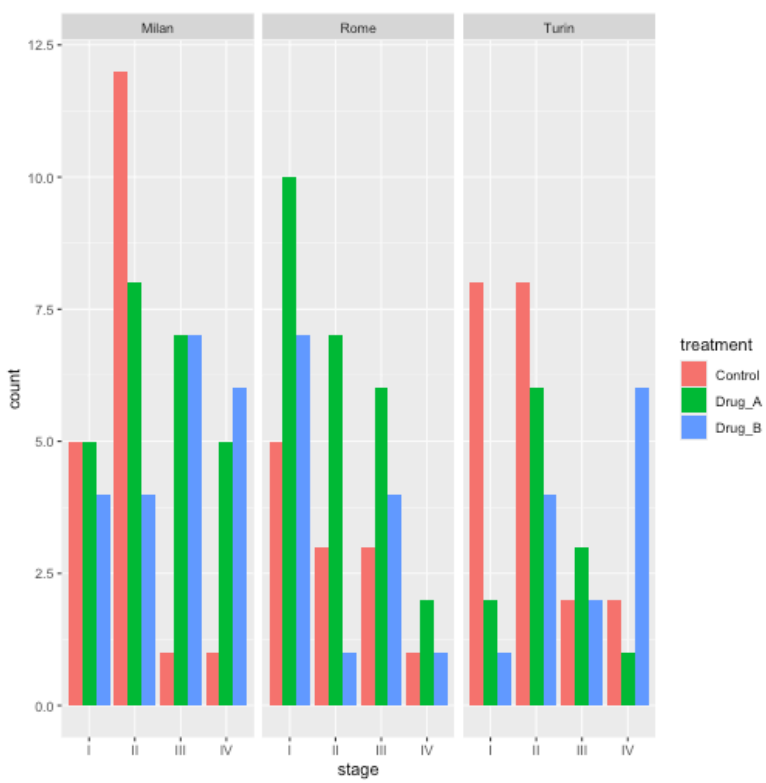
Faceting allow us to create a grid of panels, splitting data based on categorical variables.

```
ggplot() +
  geom_<type>() +
```

```
  labs() +
  scale_<type>() +
  facet_<type>() +
```

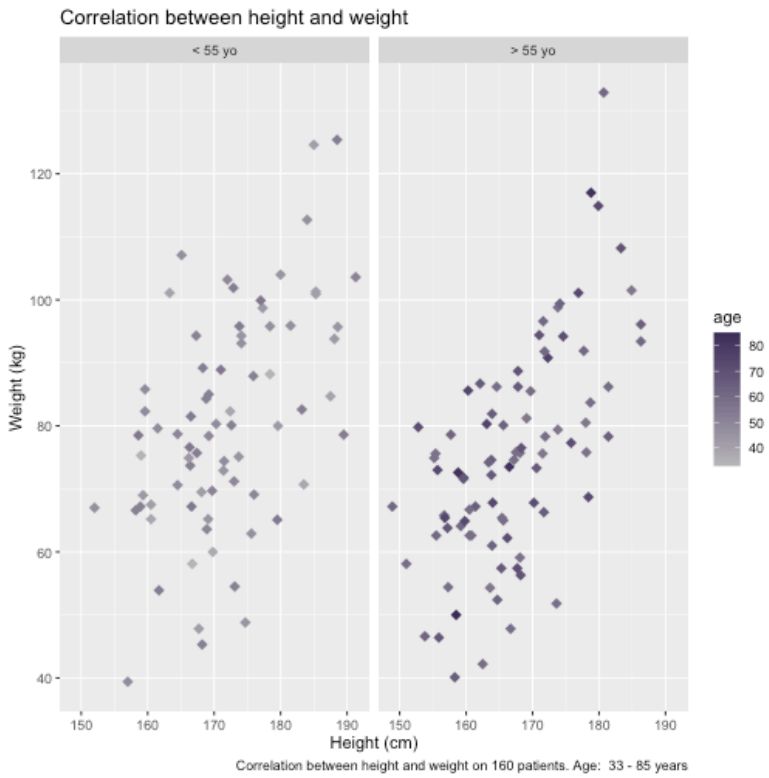
Let's see how we can rearrange the same data as in all_n_barplot but using multiple panels:

```
ggplot(df) +
  geom_bar(mapping = aes(x = stage, fill = treatment), position =
   ↪  "dodge") +
    facet_wrap(~ site)
```



We can also customize the labels, similar to using scales:

```
corr_plot +
  facet_wrap(~ age > 55, labeller = labeller(`age > 55` =
   ↪  c("TRUE" = "> 55 yo", "FALSE" = "< 55 yo")))
```



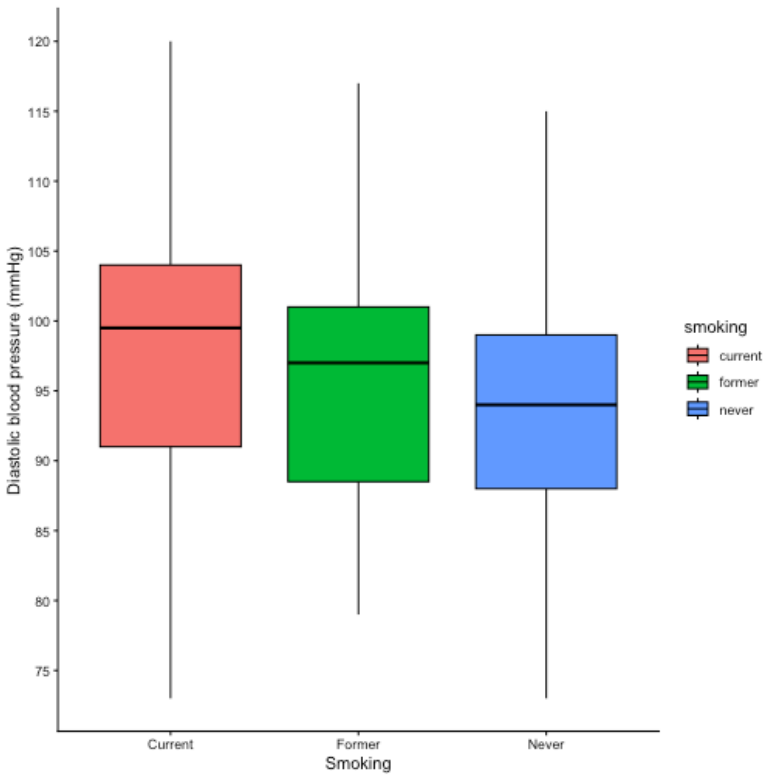
3.6 Theme

Ok, let's now see the last part (and maybe the most useful) of code. Theme allow us to control practically every single elements of the plot (except for the data): lines, font, labels, breaks, etc.

```
ggplot() +
  geom_<type>() +
  labs() +
  scale_<type>() +
  facet_<type>() +
  theme_<type>
```

There are few pre-installed theme, and we can further customize things with `theme`. Let's firt apply classic theme to diastolic_bp_boxplot:

```
diastolic_bp_boxplot <- diastolic_bp_boxplot +
  theme_classic()
diastolic_bp_boxplot
```

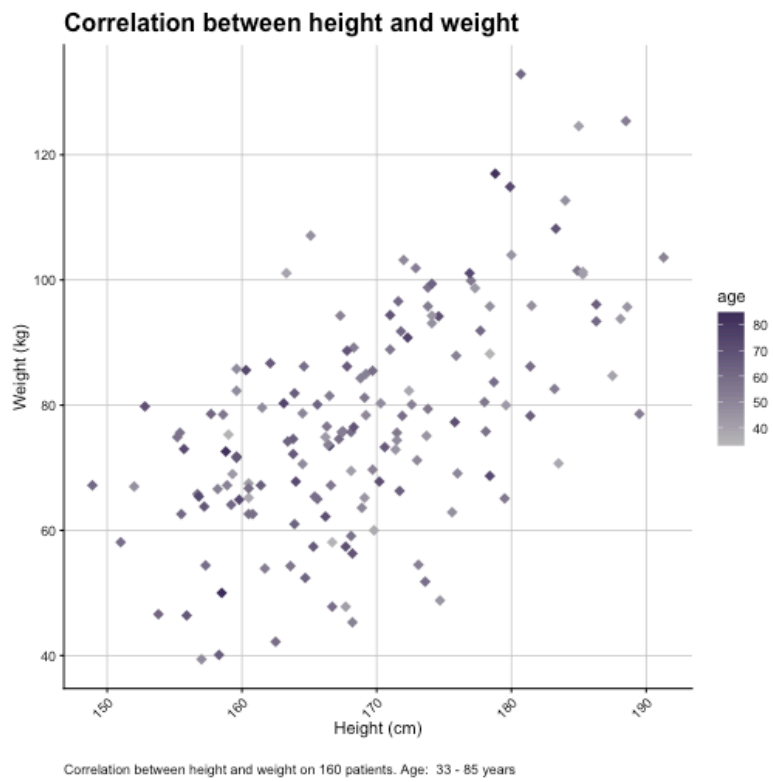


And now let's customize corr_plot:

- Title in bold and 16 pt size
- Add major grid lines in gray and 0.3 pt size
- Rotate x axis labels to 45°
- Change the adjustment of plot caption to left
- Add bottom margin to x title

```
corr_plot <- corr_plot +
  theme_classic() +
  theme(plot.title = element_text(face = "bold", size = 16),
        panel.grid.major = element_line(colour = "gray",
         ↪  linewidth = 0.3),
        axis.text.x = element_text(angle = 45, hjust = 1),
```

```
        plot.caption = element_text(hjust = 0),
        axis.title.x = element_text(margin = margin(b = 12)))
corr_plot
```



3.7 Save the plots

To save a plot to a file, we can use the function `ggsave` like this:

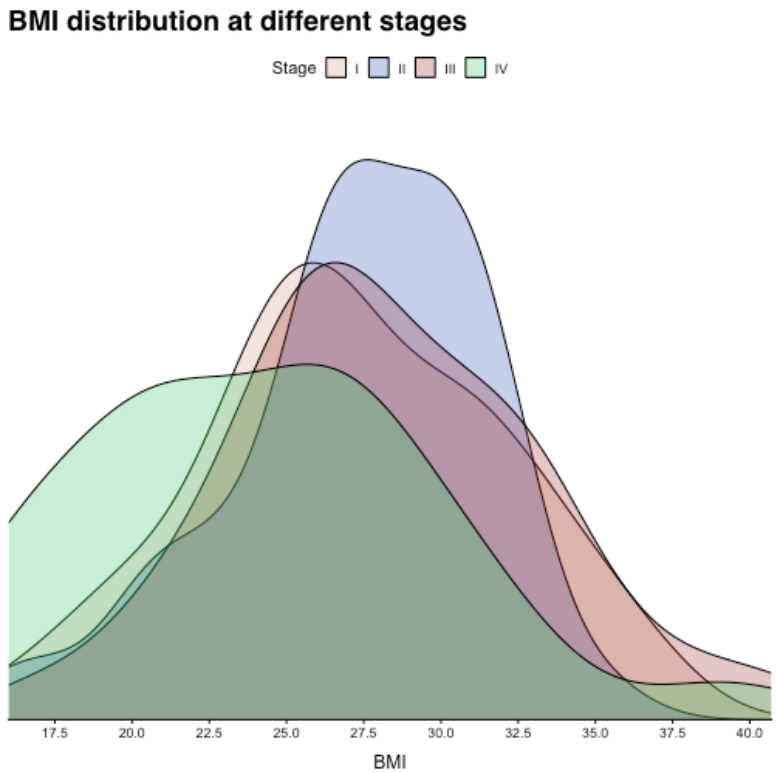
```
ggsave(plot = corr_plot, filename =
 ↪  "output/height_weight_corr-plot.pdf", units = "mm", width =
 ↪  200, height = 140)
```

We can set units, width and height. See this article for further details about dimensions, dpi, dots etc.

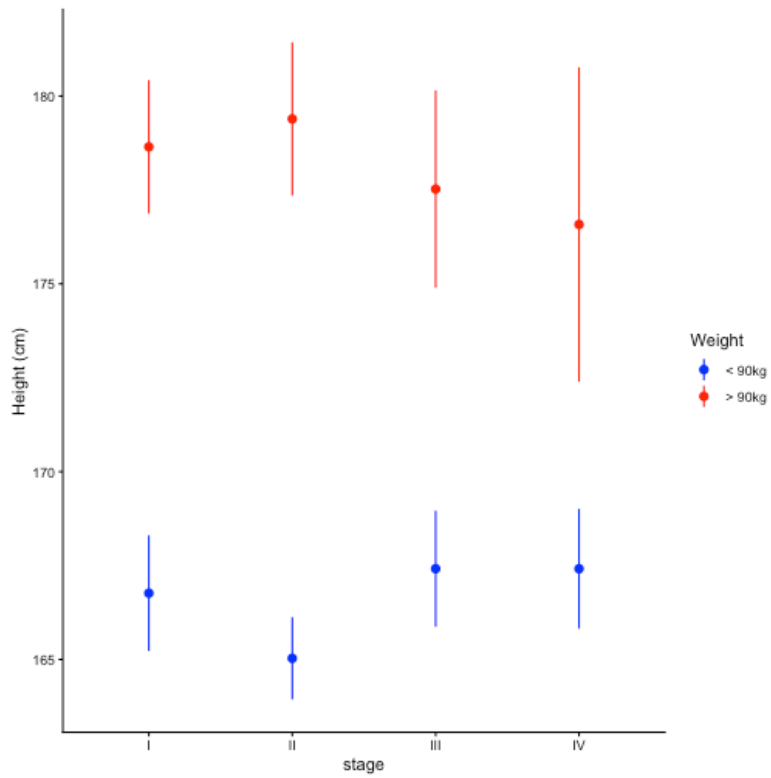
3.8 Advanced examples

Here are some more advanced examples:

```
BMI_stages_density +
  labs(title = "BMI distribution at different stages", fill =
   ↪  "Stage") + # change labels
  scale_y_continuous(expand = expansion(mult = c(0, 0.1))) + #
   ↪  remove space at the bottom of y axis
  scale_x_continuous(expand = expansion(mult = c(0, 0)), breaks =
   ↪  seq(0, max(df$BMI), 2.5)) + # reduce outer space of x axis
  scale_fill_manual(values = c("I" = "#E0A890", "II" = "#2B59C3",
   ↪  "III" = "#9E2B25", "IV" = "#23CE6B")) + # change fill
   ↪  colors
  theme_classic() +
  theme(axis.line.y = element_blank(), # remove y-axes line
        axis.text.y = element_blank(), # remove y-axes labels
        axis.ticks.y = element_blank(), # remove y-axes ticks
        axis.title.y = element_blank(), # remove y-axes title
        legend.position = "top", # change legend position
        plot.title = element_text(size = 18, family = "sans",
         ↪  face = "bold", margin = margin(b = 8)), # change
         ↪  title style
        axis.title.x = element_text(size = 12, margin = margin(t
         ↪  = 10)),
        legend.key.size = unit(x = 5, units = "mm") # change
         ↪  legens key size
        )
```




```
ggplot(df) +
  stat_summary(aes(x = stage, y = height_cm, color = weight_kg >
   ↪  90), fun.data = "mean_se", geom = "pointrange", size = .4)
   ↪  +
  labs(y = "Height (cm)", color = "Weight") +
  scale_color_manual(labels = c("TRUE" = "> 90kg", "FALSE" = "<
   ↪  90kg"), values = c("TRUE" = "red", "FALSE" = "blue")) +
  theme_classic()
```



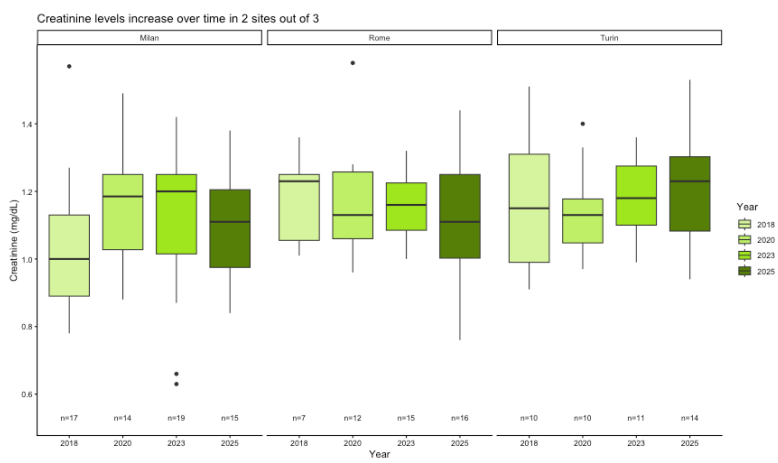
Exercise 3.2. Create a plot that shows if the levels of creatinine_mgdl changes depending on the year in each site. Set a proper scale for y axis, split the plot in panels, change the colors of the years, add a title. **Tip**: visit_year now is numeric, while we want it to be……..

bonus add the information about the numerosity in each group (year/site) at the bottom of x axis.

Solution

```
df$visit_year <- factor(df$visit_year)

df_n <- df |>
    group_by(site, visit_year) |>
    summarize(n = n(), .groups = "drop")

ggplot(df) +
    geom_boxplot(aes(x = visit_year, y = creatinine_mgdl, fill =
     ↪  visit_year),
                 position = position_dodge(width = 0.75)) +
    geom_text(data = df_n,
```

```
            aes(x = visit_year, y = min(df$creatinine_mgdl) -
             ↪  0.1, label = paste0("n=", n)),
            size = 3) +
  labs(y = "Creatinine (mg/dL)",
       title = "Creatinine levels increase over time in 2 sites
        ↪  out of 3",
       x = "Year",
       fill = "Year") +
  scale_fill_manual(values = c("2018" = "#D6F49D",
                               "2020" = "#C1EF6C",
                               "2023" = "#A2E723",
                               "2025" = "#59810E")) +
  scale_y_continuous(breaks = seq(0, max(df$creatinine_mgdl),
   ↪  0.2)) +
  facet_wrap(~ site) +
  theme_classic()
```

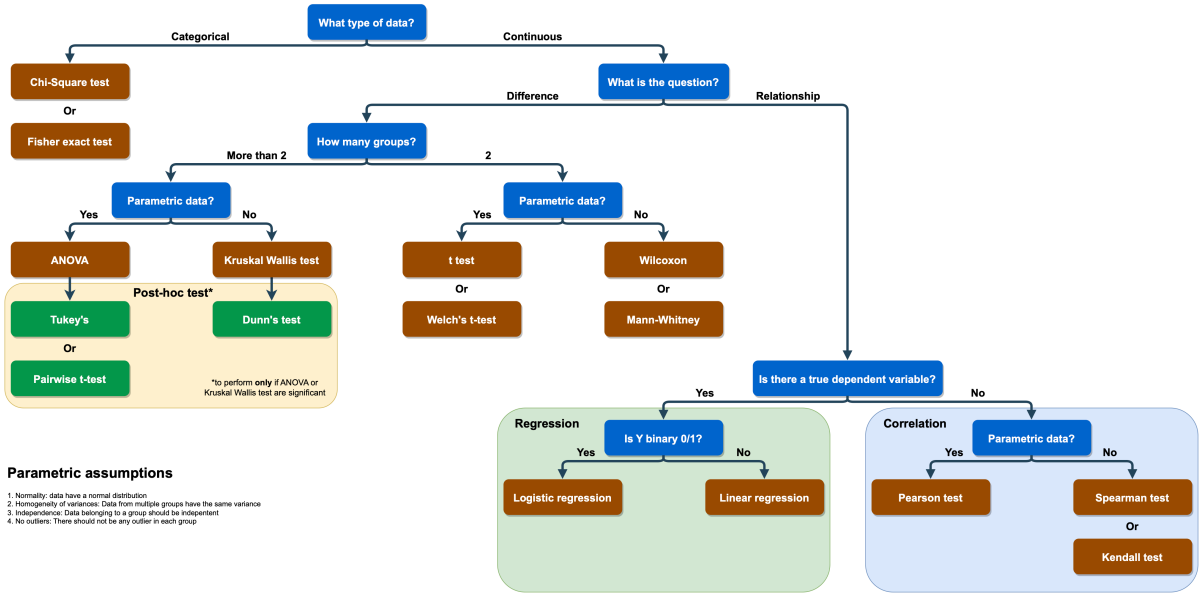


Chapter 4

Statistics

```
library(tidyverse)
```

First of all the choice of the right test is fundamental: here, you can see a scheme describing how to choose the right test depending on you design:



4.1 Categorical data

Measure the association between the different categories of categorical variables, so if the two variables are independent or not (e.g. smoking and lung cancer).

The two test we will see are the **Chi-squared test** and the **Fisher exact test** (only 2x2 contingency table and with low numerosity).

4.2 Contingency table

Contingency table is used to count the occurrence of the combination of the different levels of categorical variables. In R, to get a contingency table we use the function `table`.

We will load our patient data and see the correlation between stage and site, to see whether there is any difference in distribution.

```
df <- read.csv(file = "output/clinical_data_enriched.csv", header
 ↪  = T, stringsAsFactors = T)
summary(df)
```

```
   patient_id     site         visit_date    visit_year
 ↪  visit_month
 PT-001 :  1   Milan:96   2020-07-21:  3   Min.   :2018   Min.
 ↪  : 1.000
 PT-002 :  1   Rome :77   2025-03-20:  3   1st Qu.:2020   1st
 ↪  Qu.: 3.000
 PT-003 :  1   Turin:67   2018-03-03:  2   Median :2023   Median
 ↪  : 6.000
 PT-004 :  1              2018-05-18:  2   Mean   :2022   Mean
 ↪  : 5.754
 PT-005 :  1              2018-07-18:  2   3rd Qu.:2025   3rd
 ↪  Qu.: 8.000
 PT-006 :  1              2020-05-21:  2   Max.   :2025   Max.
 ↪  :11.000
 (Other):234              (Other)   :226
 ↪
   visit_day          age           sex         smoking
 ↪  treatment  stage
 Min.   : 1.00   Min.   :26.0   Female:118   current:63
 ↪  Control:74   I  :76
 1st Qu.: 9.00   1st Qu.:49.0   Male  :122   former :90   Drug_A
 ↪  :88   II :73
 Median :16.00   Median :57.5                never  :87   Drug_B
 ↪  :78   III:53
```

```
Mean   :16.23   Mean   :57.3
↪  IV :38
3rd Qu.:23.00   3rd Qu.:65.0
↪
Max.   :31.00   Max.   :85.0
↪

  weight_kg        height_cm     systolic_bp_mmhg
↪  diastolic_bp_mmhg
Min.   : 39.40   Min.   :148.9   Min.   :115.0    Min.   : 73.00
↪
1st Qu.: 65.35   1st Qu.:162.2   1st Qu.:139.0    1st Qu.: 90.75
↪
Median : 75.65   Median :168.5   Median :146.5    Median : 97.00
↪
Mean   : 76.87   Mean   :169.2   Mean   :146.7    Mean   : 96.11
↪
3rd Qu.: 87.83   3rd Qu.:174.8   3rd Qu.:153.2    3rd Qu.:102.00
↪
Max.   :132.90   Max.   :200.1   Max.   :192.0    Max.   :120.00
↪

 glucose_mgdl   cholesterol_mgdl    crp_ngml        height_m
Min.   : 60.0   Min.   :113.0    Min.   : 1.92   Min.   :1.489
1st Qu.: 98.0   1st Qu.:189.8    1st Qu.: 5.30   1st Qu.:1.623
Median :109.0   Median :207.0    Median : 8.37   Median :1.685
Mean   :108.8   Mean   :207.9    Mean   :10.35   Mean   :1.692
3rd Qu.:118.0   3rd Qu.:226.0    3rd Qu.:13.02   3rd Qu.:1.748
Max.   :149.0   Max.   :307.0    Max.   :42.09   Max.   :2.001

     BMI              BMI_class   hemoglobin_gdl    wbc_10e9l
↪
Min.   :16.00   Normal     : 58   Min.   : 9.20   Min.   : 2.310
↪
1st Qu.:23.77   Obese      : 13   1st Qu.:12.40   1st Qu.: 5.518
↪
Median :26.85   Overweight :102   Median :13.25   Median : 7.000
↪
Mean   :26.71   Underweight: 67   Mean   :13.50   Mean   : 6.758
↪
3rd Qu.:29.80                     3rd Qu.:14.70   3rd Qu.: 7.902
↪
Max.   :40.70                     Max.   :17.00   Max.   :11.380
↪
                                  NA's   :80      NA's   :80
↪
```

```
 platelets_10e9l     alt_ul       creatinine_mgdl
 Min.   : 60.0   Min.   : 10.00   Min.   :0.630
 1st Qu.:191.8   1st Qu.: 32.75   1st Qu.:1.018
 Median :232.0   Median : 46.50   Median :1.140
 Mean   :231.7   Mean   : 52.20   Mean   :1.138
 3rd Qu.:271.0   3rd Qu.: 61.00   3rd Qu.:1.252
 Max.   :391.0   Max.   :161.00   Max.   :1.580
 NA's   :80      NA's   :80       NA's   :80
```

```r
str(df)
```

```
'data.frame':   240 obs. of  26 variables:
 $ patient_id       : Factor w/ 240 levels "PT-001","PT-002",..:
 ↪  1 2 3 4 5 6 7 8 9 10 ...
 $ site             : Factor w/ 3 levels "Milan","Rome",..: 1 2 2
 ↪  3 3 1 2 2 1 2 ...
 $ visit_date       : Factor w/ 218 levels
 ↪  "2018-01-14","2018-01-24",..: 129 75 150 52 104 108 80 214 57
 ↪  146 ...
 $ visit_year       : int  2023 2020 2023 2020 2023 2023 2020
 ↪  2025 2020 2023 ...
 $ visit_month      : int  7 8 10 3 3 4 9 10 4 9 ...
 $ visit_day        : int  20 15 15 6 30 15 9 7 19 27 ...
 $ age              : int  40 40 59 46 58 53 51 34 43 60 ...
 $ sex              : Factor w/ 2 levels "Female","Male": 2 2 1 2
 ↪  2 1 2 1 2 2 ...
 $ smoking          : Factor w/ 3 levels "current","former",..: 3
 ↪  1 2 1 2 1 2 3 3 2 ...
 $ treatment        : Factor w/ 3 levels "Control","Drug_A",..: 2
 ↪  3 2 2 3 1 2 1 3 2 ...
 $ stage            : Factor w/ 4 levels "I","II","III",..: 2 1 1
 ↪  1 3 4 1 1 3 2 ...
 $ weight_kg        : num  85.9 100.9 65 95.9 75.7 ...
 $ height_cm        : num  173 185 166 182 168 ...
 $ systolic_bp_mmhg : int  164 143 152 142 160 150 137 146 151
 ↪  159 ...
 $ diastolic_bp_mmhg: int  101 102 98 88 96 107 96 106 94 104 ...
 $ glucose_mgdl     : int  113 105 110 120 131 78 108 93 106 110
 ↪  ...
 $ cholesterol_mgdl : int  172 239 146 232 192 232 163 199 200
 ↪  185 ...
 $ crp_ngml         : num  5.89 6.13 7.81 3.35 21.72 ...
 $ height_m         : num  1.73 1.85 1.66 1.81 1.68 ...
 $ BMI              : num  28.8 29.4 23.7 29.1 26.8 22.9 24.2
 ↪  29.8 24.8 26.9 ...
```

```
 $ BMI_class        : Factor w/ 4 levels "Normal","Obese",..: 3 3
 ↪  4 3 3 4 4 3 4 3 ...
 $ hemoglobin_gdl   : num  NA 15 11.8 16 15.4 NA 14 16.2 10.5
 ↪  15.6 ...
 $ wbc_10e9l        : num  NA 7.49 7.46 7.09 6.63 NA 6.99 9.26
 ↪  4.79 5.65 ...
 $ platelets_10e9l  : int  NA 270 252 233 192 NA 195 188 163 269
 ↪  ...
 $ alt_ul           : int  NA 56 27 20 16 NA 61 51 50 18 ...
 $ creatinine_mgdl  : num  NA 1.58 1.09 1.17 1.06 NA 1.15 1.06
 ↪  1.21 1.08 ...
```

And now we can use `table` to get the contingency table:

```
cont_table <- table(df$site, df$stage)
cont_table
```

```
         I II III IV
  Milan 23 35  19 19
  Rome  33 16  21  7
  Turin 20 22  13 12
```

To make it clearer, we can add header to it:

```
cont_table <- table("site" = df$site, "stage" = df$stage)
cont_table
```

```
       stage
site     I II III IV
  Milan 23 35  19 19
  Rome  33 16  21  7
  Turin 20 22  13 12
```

4.3 Chi-squared test

To perform the chi-squared test, we use the `chisq.test` on the contingency table.

It is very useful to store the results of any statistical test into a variable:

```
chisq_res <- chisq.test(cont_table)
chisq_res
```

```

    Pearson's Chi-squared test

data:  cont_table
X-squared = 13.277, df = 6, p-value = 0.03884
```

Here, we can see that the association between the two variables is significant (this does not mean it is a causative association).

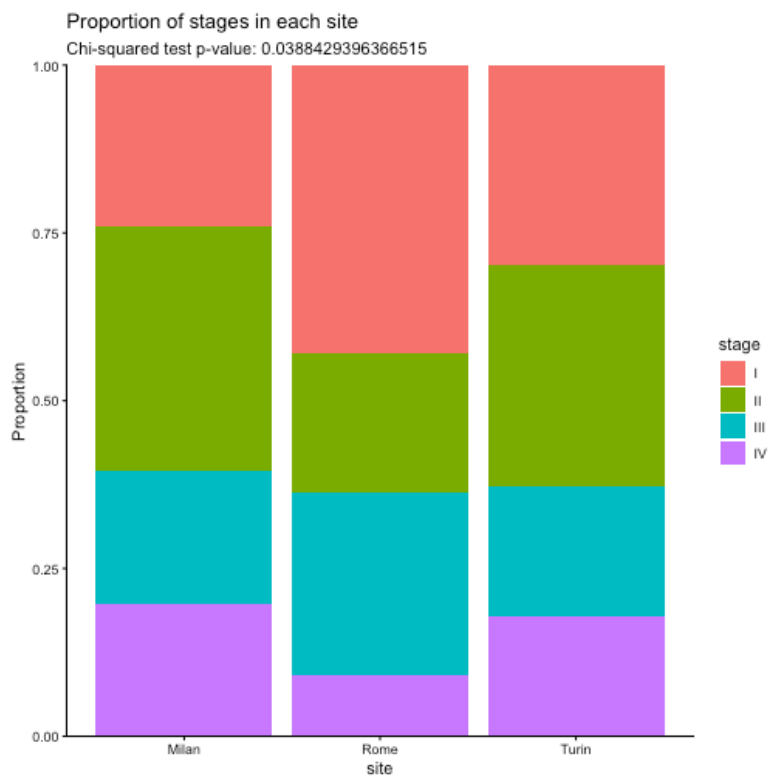
We can inspect the result object, if we want to then use the results (for example, print the p-value alone or insert it in a graph):

```
print(paste("P-value of the Chi-squared test is:",
 ↪  chisq_res$p.value))
```

```
[1] "P-value of the Chi-squared test is: 0.0388429396366515"
```

Let's create a figure for this result.

```
ggplot(df) +
    geom_bar(aes(x = site, fill = stage), position = "fill") +
    scale_y_continuous(expand = expansion(0, 0)) +
    labs(title = "Proportion of stages in each site",
         subtitle = paste0("Chi-squared test p-value: ",
          ↪  chisq_res$p.value),
         y = "Proportion") +
    theme_classic()
```



4.3.1 Fisher's Exact test

We can apply, just for example, the Fisher's exact test on sex and 2 sites: Turin and Rome (remember, Fisher's exact test wants a 2x2 table). The function to use is `fisher.test`:

```
filtered_df <- df |>
    filter_out(site == "Milan")

cont_table_fisher <- table(filtered_df$sex, filtered_df$site)

fisher_res <- fisher.test(cont_table_fisher)
fisher_res
```

```
    Fisher's Exact Test for Count Data

data:  cont_table_fisher
p-value = 0.6199
```

```
alternative hypothesis: two.sided
```

We can see that there is no significant correlation between those 2 sites and sex of the patients.

4.4 Comparing continuous variables

First thing to do is to check for parametric assumptions, which are:

- **Normality**: data in each group must be normally distributed
- **Homoschedasticity**: there must be equal variances between groups you are going to compare
- **Independency**: data belonging to a group should be independent
- **No outlier**s: there should not be any outlier in any group.

For sake of time, today we will see only the shapiro-test for normality. You can see a full reference of the tests to perform here.

4.4.1 Shapiro-Wilk test

To test for normality of the data, we use the Shapiro-Wilk test on a single vector (so we need to repeat it for each group).

The function to use is `shapiro.test`, and here we apply just on diastolic_bp_mmhg in smokers.

```
smoker_patients_diastolic_bp_mmhg <- df %>%
  filter(smoking == "current") %>%
  pull(diastolic_bp_mmhg)

head(smoker_patients_diastolic_bp_mmhg)
```

```
[1] 102  88 107  76  97  84
```

```
# Shapiro test
shap_res <- shapiro.test(smoker_patients_diastolic_bp_mmhg)
shap_res
```

```
    Shapiro-Wilk normality test

data:  smoker_patients_diastolic_bp_mmhg
W = 0.98022, p-value = 0.4045
```

Ok, this group of data has a normal distribution (p-value > 0.05). But how can we apply to all of type of smokers?

```
shap_df <- df %>%
  group_by(smoking) %>%
  summarise("shapiro_p-value" =
   ↪  shapiro.test(diastolic_bp_mmhg)$p.value,
            "shapiro-W" =
             ↪  shapiro.test(diastolic_bp_mmhg)$statistic)

shap_df
```

```
# A tibble: 3 x 3
  smoking `shapiro_p-value` `shapiro-W`
  <fct>               <dbl>       <dbl>
1 current             0.404       0.980
2 former              0.434       0.986
3 never               0.780       0.990
```

And we can check which of them are not normal by filtering this dataframe:

```
shap_df %>%
  filter(`shapiro_p-value` < 0.05)
```

```
# A tibble: 0 x 3
# i 3 variables: smoking <fct>, shapiro_p-value <dbl>, shapiro-W
 ↪  <dbl>
```

They are all normal, so we can use parametric tests to compare them.

4.4.2 Outlier detection

Outlier detection is a crucial point in the analyses, it helps to understand whether we have some wrong input data (very out of scale) and whether we can apply parametric or non parametric tests.

Exercise 4.1. Let's say that an outlier is a value which is < 1st quartile (25th quantile) - 1.5 IQR or > 3rd quartile (75th quantile) + 1.5 IQR.

To help:

- `quantile(x, 0.25)` finds the 25th quantile (1st quartile) of a vectors of values x
- `IQR(x)` finds the interquartile range of a vectors of values x
- there are NAs…

Find, if there are, outliers in creatinine levels in both stage I and stage IV patients.

Solution

```
# Extract data
stageI_creatinine <- df %>%
  filter(stage == "I") %>%
  na.omit() %>%
  pull(creatinine_mgdl)

stageIV_creatinine <- df %>%
  filter(stage == "IV") %>%
    na.omit() %>%
  pull(creatinine_mgdl)

# Calculate outliers
stageI_outliers <- (stageI_creatinine <
 ↪  quantile(stageI_creatinine, 0.25) - 1.5 *
 ↪  IQR(stageI_creatinine)) | (stageI_creatinine >
 ↪  quantile(stageI_creatinine, 0.75) + 1.5 *
 ↪  IQR(stageI_creatinine))

stageIV_outliers <- (stageIV_creatinine <
 ↪  quantile(stageIV_creatinine, 0.25) - 1.5 *
 ↪  IQR(stageIV_creatinine)) | (stageIV_creatinine >
 ↪  quantile(stageIV_creatinine, 0.75) + 1.5 *
 ↪  IQR(stageIV_creatinine))

# Get outliers
print(mean(stageI_creatinine))
```

```
[1] 1.138936
```

```
print(stageI_creatinine[stageI_outliers])
```

```
[1] 1.58 0.66
```

```
print(mean(stageIV_creatinine))
```

```
[1] 1.144
```

```
print(stageIV_creatinine[stageIV_outliers])
```

```
numeric(0)
```

4.4.3 T test

If we now want to compare diastolic_bp_mmhg between current smokers and patients who never smoke, we can apply standard t-test as both of them have normal distributions.

We can isolate the data from the dataframe:

```
t_test_df <- df %>%
  filter_out(smoking == "former")
str(t_test_df)
```

```
'data.frame':   150 obs. of  26 variables:
 $ patient_id       : Factor w/ 240 levels "PT-001","PT-002",..:
 ↪  1 2 4 6 8 9 11 12 13 15 ...
 $ site             : Factor w/ 3 levels "Milan","Rome",..: 1 2 3
 ↪  1 2 1 1 2 1 3 ...
 $ visit_date       : Factor w/ 218 levels
 ↪  "2018-01-14","2018-01-24",..: 129 75 52 108 214 57 125 170 8
 ↪  27 ...
 $ visit_year       : int  2023 2020 2020 2023 2025 2020 2023
 ↪  2025 2018 2018 ...
 $ visit_month      : int  7 8 3 4 10 4 6 3 2 7 ...
 $ visit_day        : int  20 15 6 15 7 19 29 20 23 10 ...
 $ age              : int  40 40 46 53 34 43 65 52 58 75 ...
 $ sex              : Factor w/ 2 levels "Female","Male": 2 2 2 1
 ↪  1 2 1 2 2 1 ...
 $ smoking          : Factor w/ 3 levels "current","former",..: 3
 ↪  1 1 1 3 3 1 3 3 1 ...
```

```
 $ treatment        : Factor w/ 3 levels "Control","Drug_A",..: 2
 ↪  3 2 1 1 3 1 2 2 1 ...
 $ stage            : Factor w/ 4 levels "I","II","III",..: 2 1 1
 ↪  4 1 3 2 3 2 1 ...
 $ weight_kg        : num  85.9 100.9 95.9 69.6 75.3 ...
 $ height_cm        : num  173 185 182 174 159 ...
 $ systolic_bp_mmhg : int  164 143 142 150 146 151 125 147 162
 ↪  149 ...
 $ diastolic_bp_mmhg: int  101 102 88 107 106 94 76 96 98 97 ...
 $ glucose_mgdl     : int  113 105 120 78 93 106 94 86 141 117
 ↪  ...
 $ cholesterol_mgdl : int  172 239 232 232 199 200 209 222 223
 ↪  204 ...
 $ crp_ngml         : num  5.89 6.13 3.35 32.42 5.08 ...
 $ height_m         : num  1.73 1.85 1.81 1.74 1.59 ...
 $ BMI              : num  28.8 29.4 29.1 22.9 29.8 24.8 30.5
 ↪  29.6 27 30.2 ...
 $ BMI_class        : Factor w/ 4 levels "Normal","Obese",..: 3 3
 ↪  3 4 3 4 1 3 3 1 ...
 $ hemoglobin_gdl   : num  NA 15 16 NA 16.2 10.5 12.4 14.5 13.5
 ↪  12.7 ...
 $ wbc_10e9l        : num  NA 7.49 7.09 NA 9.26 4.79 8.39 7.87
 ↪  3.95 6.37 ...
 $ platelets_10e9l  : int  NA 270 233 NA 188 163 165 263 193 288
 ↪  ...
 $ alt_ul           : int  NA 56 20 NA 51 50 47 51 95 36 ...
 $ creatinine_mgdl  : num  NA 1.58 1.17 NA 1.06 1.21 1.12 1.03
 ↪  1.27 1.18 ...
```

We can now apply t-test, using `t.test` function:

```
t_test_res <- t.test(t_test_df$diastolic_bp_mmhg ~
 ↪  t_test_df$smoking)
t_test_res
```

```

    Welch Two Sample t-test

data:  t_test_df$diastolic_bp_mmhg by t_test_df$smoking
t = 2.0406, df = 114.58, p-value = 0.04359
alternative hypothesis: true difference in means between group
 ↪  current and group never is not equal to 0
95 percent confidence interval:
 0.09540767 6.42238105
```

```
sample estimates:
mean in group current   mean in group never
             98.06349              94.80460
```

Also for this object, we can have access to each individual result:

```
print(t_test_res$p.value)
```

```
[1] 0.04358633
```

```
print(t_test_res$estimate)
```

```
mean in group current   mean in group never
             98.06349              94.80460
```

4.4.4 Mann-Whitney

Let's pretend that the distribution of diastolic_bp_mmhg in former smokers is not normal; if we want to perform a comparison between former and current, we cannot use standard t.test, but the Mann-Whitney test. The function to use is `wilcox.test`.

```
mann_df <- df %>%
  filter_out(smoking == "never")

wilcox_res <- wilcox.test(mann_df$diastolic_bp_mmhg ~
 ↪  mann_df$smoking)
wilcox_res
```

```

    Wilcoxon rank sum test with continuity correction

data:  mann_df$diastolic_bp_mmhg by mann_df$smoking
W = 3267.5, p-value = 0.1089
alternative hypothesis: true location shift is not equal to 0
```

We can see that there is no significant difference.

Exercise 4.2. Is there a significant difference in weight between current smokers and former smokers? Choose the right test and plot the results.

Solution

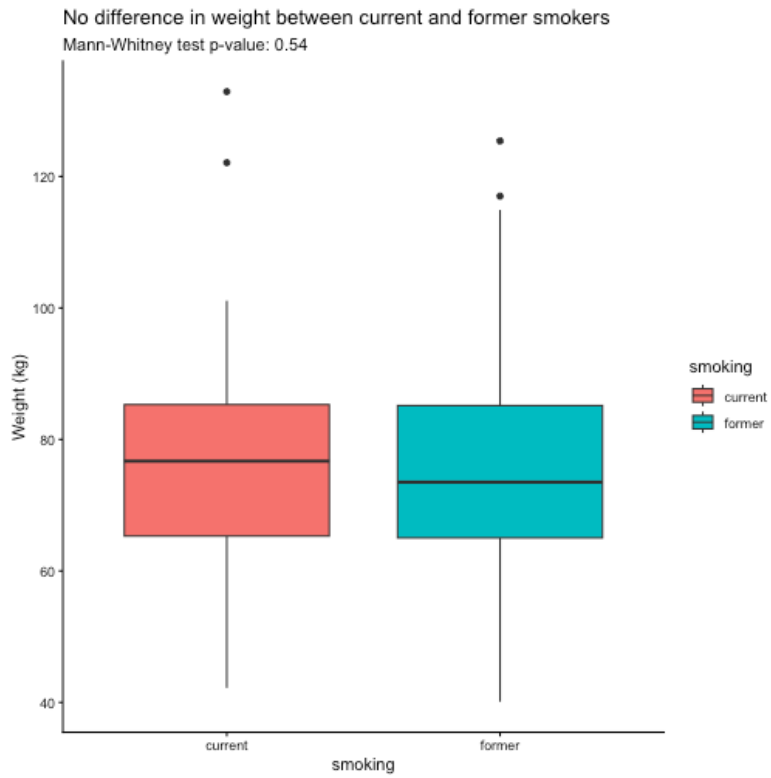
```
smoking_df <- df |>
    filter_out(smoking == "never")

smoking_df |>
    group_by(smoking) |>
    summarise("shapiro_p-value" =
     ↪  shapiro.test(weight_kg)$p.value,
              "shapiro-W" = shapiro.test(weight_kg)$statistic)
```

```
# A tibble: 2 x 3
  smoking `shapiro_p-value` `shapiro-W`
  <fct>               <dbl>       <dbl>
1 current            0.0773       0.966
2 former             0.0430       0.971
```

```
wilcox_res <- wilcox.test(smoking_df$weight_kg ~
 ↪  smoking_df$smoking)
wilcox_res
```

```

    Wilcoxon rank sum test with continuity correction

data:  smoking_df$weight_kg by smoking_df$smoking
W = 3001.5, p-value = 0.5383
alternative hypothesis: true location shift is not equal to 0
```

```
ggplot(smoking_df) +
    geom_boxplot(aes(x = smoking, y = weight_kg, fill = smoking))
     ↪  +
    labs(y = "Weight (kg)",
         title = "No difference in weight between current and
          ↪  former smokers",
         subtitle = paste0("Mann-Whitney test p-value: ",
          ↪  round(wilcox_res$p.value, 2))) +
    theme_classic()
```



4.4.5 ANOVA

When the groups are more than 2, we should first do the ANOVA and then, **only if it is significant** perform pairwise comparisons.

Let's check whether systolic_bp_mmhg is different between males and females and based on BMI class.

```
aov_res <- aov(df$systolic_bp_mmhg ~ df$BMI_class * df$sex)
summary(aov_res)
```

```
                     Df Sum Sq Mean Sq F value  Pr(>F)
df$BMI_class          3   1412   470.7   3.624 0.01378 *
df$sex                1   1151  1150.8   8.862 0.00322 **
df$BMI_class:df$sex   3    496   165.4   1.274 0.28396
Residuals           232  30126   129.9
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

So, both sex and BMI class influence systolic_bp_mmhg; but not the interaction. To get which BMI_class are different we perform the pairwise t test:

```
pairwise.t.test(x = df$systolic_bp_mmhg, g = df$BMI_class,
 ↪  p.adjust.method = "bonferroni")
```

```
    Pairwise comparisons using t tests with pooled SD

data:  df$systolic_bp_mmhg and df$BMI_class

            Normal Obese Overweight
Obese       0.022  -     -
Overweight  0.589  0.204 -
Underweight 0.165  0.590 1.000

P value adjustment method: bonferroni
```

The only BMI class comparison in which the systolic_bp_mmhg is different is Obese vs Normal.

4.4.6 Pearson correlation test

To perform a correlation test between continuous variables, we use `cor.test` with method "pearson" (if variables are normally distributed, otherwise "spearman" or "kendall").

REMEMBER: correlation is not causation.

Let's pretend all variable are normally distributed and see the correlation between *wbc_10e9l* and *crp_ngml*:

```
corr_res <- cor.test(df$wbc_10e9l, df$crp_ngml, method =
 ↪  "pearson")
corr_res
```

```
    Pearson's product-moment correlation

data:  df$wbc_10e9l and df$crp_ngml
t = 1.383, df = 158, p-value = 0.1686
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.04658334  0.26011079
```

```
sample estimates:
      cor
0.1093659
```

There is not a significant correlation.

Exercise 4.3. Is there a significant correlation between *glucose_mgdl* and *cholesterol_mgdl* in patients?

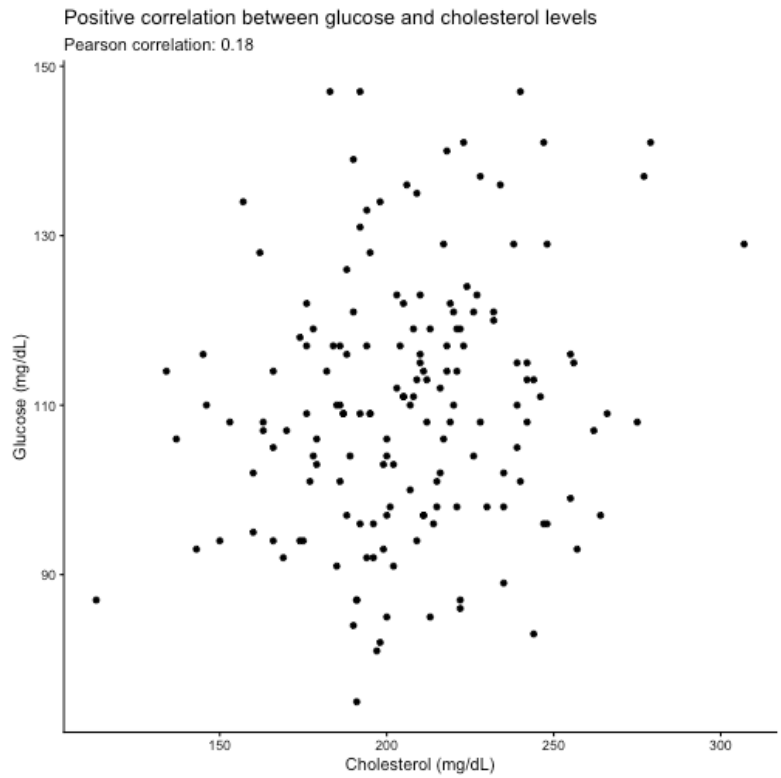
Plot the result.

Solution

```
corr_df <- na.omit(df)
corr_res <- cor.test(corr_df$glucose_mgdl,
 ↪  corr_df$cholesterol_mgdl, method = "pearson")
corr_res
```

```

    Pearson's product-moment correlation

data:  corr_df$glucose_mgdl and corr_df$cholesterol_mgdl
t = 2.281, df = 158, p-value = 0.02388
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.02406027 0.32471529
sample estimates:
      cor
0.1785525
```

```
corr_plot <- ggplot(corr_df) +
    geom_point(aes(y = glucose_mgdl, x = cholesterol_mgdl)) +
    labs(y = "Glucose (mg/dL)",
         x = "Cholesterol (mg/dL)",
         title = "Positive correlation between glucose and
          ↪  cholesterol levels",
         subtitle = paste0("Pearson correlation: ",
          ↪  round(corr_res$estimate, 2))) +
    theme_classic()

corr_plot
```



4.5 PCA

PCA (Principal Component Analysis) is useful to evaluate whether data can be clustered in groups based on any factor that we have (age, sex, stage, …).

To perform PCA, we should ouse only numeric scaled data. Se we first have to select only numeric variables and scale them.

```
data_n <- df |>
    na.omit() |>
    select(where(is.numeric))

str(data_n)
```

```
'data.frame':   160 obs. of  18 variables:
 $ visit_year       : int  2020 2023 2020 2023 2020 2025 2020
 ↪  2023 2023 2025 ...
 $ visit_month      : int  8 10 3 3 9 10 4 9 6 3 ...
```

```
 $ visit_day        : int  15 15 6 30 9 7 19 27 29 20 ...
 $ age              : int  40 59 46 58 51 34 43 60 65 52 ...
 $ weight_kg        : num  100.9 65 95.9 75.7 69.7 ...
 $ height_cm        : num  185 166 182 168 170 ...
 $ systolic_bp_mmhg : int  143 152 142 160 137 146 151 159 125
 ↪  147 ...
 $ diastolic_bp_mmhg: int  102 98 88 96 96 106 94 104 76 96 ...
 $ glucose_mgdl     : int  105 110 120 131 108 93 106 110 94 86
 ↪  ...
 $ cholesterol_mgdl : int  239 146 232 192 163 199 200 185 209
 ↪  222 ...
 $ crp_ngml         : num  6.13 7.81 3.35 21.72 5.41 ...
 $ height_m         : num  1.85 1.66 1.81 1.68 1.7 ...
 $ BMI              : num  29.4 23.7 29.1 26.8 24.2 29.8 24.8
 ↪  26.9 30.5 29.6 ...
 $ hemoglobin_gdl   : num  15 11.8 16 15.4 14 16.2 10.5 15.6 12.4
 ↪  14.5 ...
 $ wbc_10e9l        : num  7.49 7.46 7.09 6.63 6.99 9.26 4.79
 ↪  5.65 8.39 7.87 ...
 $ platelets_10e9l  : int  270 252 233 192 195 188 163 269 165
 ↪  263 ...
 $ alt_ul           : int  56 27 20 16 61 51 50 18 47 51 ...
 $ creatinine_mgdl  : num  1.58 1.09 1.17 1.06 1.15 1.06 1.21
 ↪  1.08 1.12 1.03 ...
 - attr(*, "na.action")= 'omit' Named int [1:80] 1 6 14 21 25 33
 ↪  46 49 52 54 ...
  ..- attr(*, "names")= chr [1:80] "1" "6" "14" "21" ...
```

We have to get rid of visit year, month and day, as they should not be considered numeric in this case, rather factors.

```
data_n <- data_n |>
    select(-starts_with("visit_"))

str(data_n)
```

```
'data.frame':   160 obs. of  15 variables:
 $ age              : int  40 59 46 58 51 34 43 60 65 52 ...
 $ weight_kg        : num  100.9 65 95.9 75.7 69.7 ...
 $ height_cm        : num  185 166 182 168 170 ...
 $ systolic_bp_mmhg : int  143 152 142 160 137 146 151 159 125
 ↪  147 ...
 $ diastolic_bp_mmhg: int  102 98 88 96 96 106 94 104 76 96 ...
 $ glucose_mgdl     : int  105 110 120 131 108 93 106 110 94 86
 ↪  ...
```

```
 $ cholesterol_mgdl : int  239 146 232 192 163 199 200 185 209
 ↪  222 ...
 $ crp_ngml         : num  6.13 7.81 3.35 21.72 5.41 ...
 $ height_m         : num  1.85 1.66 1.81 1.68 1.7 ...
 $ BMI              : num  29.4 23.7 29.1 26.8 24.2 29.8 24.8
 ↪  26.9 30.5 29.6 ...
 $ hemoglobin_gdl   : num  15 11.8 16 15.4 14 16.2 10.5 15.6 12.4
 ↪  14.5 ...
 $ wbc_10e9l        : num  7.49 7.46 7.09 6.63 6.99 9.26 4.79
 ↪  5.65 8.39 7.87 ...
 $ platelets_10e9l  : int  270 252 233 192 195 188 163 269 165
 ↪  263 ...
 $ alt_ul           : int  56 27 20 16 61 51 50 18 47 51 ...
 $ creatinine_mgdl  : num  1.58 1.09 1.17 1.06 1.15 1.06 1.21
 ↪  1.08 1.12 1.03 ...
 - attr(*, "na.action")= 'omit' Named int [1:80] 1 6 14 21 25 33
 ↪  46 49 52 54 ...
  ..- attr(*, "names")= chr [1:80] "1" "6" "14" "21" ...
```

Now we have to scale data:

```r
data_n_scaled <- scale(data_n)
```

To confirm that they have been scaled:

```r
# little exercise: confirm that for each variable mean ~ 0 and sd
 ↪  = 1

apply(data_n_scaled, 2, mean)
```

```
              age         weight_kg         height_cm
 ↪  systolic_bp_mmhg
    -1.307256e-16     -3.620592e-16      1.146123e-15
 ↪  -4.966372e-16
diastolic_bp_mmhg      glucose_mgdl  cholesterol_mgdl
 ↪  crp_ngml
     2.995596e-16      1.127231e-18     -3.578572e-16
 ↪  -3.238851e-17
         height_m               BMI    hemoglobin_gdl
 ↪  wbc_10e9l
     4.793675e-16     -1.910446e-16     -2.797160e-16
 ↪  1.546709e-16
  platelets_10e9l            alt_ul   creatinine_mgdl
    -1.187201e-18     -1.036389e-16      6.266217e-16
```

```r
apply(data_n_scaled, 2, sd)
```

```
              age         weight_kg         height_cm
 ↪  systolic_bp_mmhg
                1                 1                 1
 ↪  1
diastolic_bp_mmhg      glucose_mgdl  cholesterol_mgdl
 ↪  crp_ngml
                1                 1                 1
 ↪  1
         height_m               BMI    hemoglobin_gdl
 ↪  wbc_10e9l
                1                 1                 1
 ↪  1
  platelets_10e9l            alt_ul   creatinine_mgdl
                1                 1                 1
```

To perform PCA, we use `prcomp` function:

```r
pca_res <- prcomp(data_n_scaled, center = F, scale. = F)
pca_res
```

```
Standard deviations (1, .., p=15):
 [1] 1.912258e+00 1.513155e+00 1.198225e+00 1.148116e+00
 ↪  1.015913e+00
 [6] 9.597070e-01 9.321631e-01 9.017047e-01 8.814928e-01
 ↪  8.032314e-01
[11] 7.133833e-01 6.701814e-01 5.296499e-01 6.166601e-02
 ↪  9.127822e-16

Rotation (n x k) = (15 x 15):
                           PC1          PC2         PC3
 ↪  PC4
age                0.008330314 -0.434005567 -0.05421122
 ↪  0.235479055
weight_kg         -0.477734288  0.011178321 -0.09567052
 ↪  -0.210115072
height_cm         -0.365304360  0.390722311  0.02010322
 ↪  0.220378230
systolic_bp_mmhg  -0.223192721 -0.348380634  0.31736070
 ↪  0.349042281
diastolic_bp_mmhg -0.164838721 -0.333543008  0.40895939
 ↪  0.339825013
glucose_mgdl      -0.241516653 -0.216418273 -0.24080333
 ↪  0.007671028
```

```
cholesterol_mgdl  -0.104945115 -0.274894361 -0.37944086
 ↪  0.081850470
crp_ngml          -0.129743208  0.021505087  0.44081292
 ↪  -0.429896917
height_m          -0.365304360  0.390722311  0.02010322
 ↪  0.220378230
BMI               -0.364370368 -0.216549105 -0.13560650
 ↪  -0.397096477
hemoglobin_gdl    -0.297504455  0.137908795 -0.13646690
 ↪  0.196583824
wbc_10e9l          0.008459241  0.128814723  0.29511743
 ↪  -0.126508471
platelets_10e9l    0.082793288 -0.008592387 -0.44038790
 ↪  0.037973578
alt_ul            -0.117603922 -0.256685107  0.04630145
 ↪  -0.398935550
creatinine_mgdl   -0.315611319 -0.047259459 -0.05162191
 ↪  -0.038211446
                           PC5         PC6          PC7
 ↪  PC8         PC9
age                0.236338866 -0.41999664  0.101996500
 ↪  -0.09669130  0.29086350
weight_kg          0.024747252  0.05574043  0.032095684
 ↪  0.13246964 -0.14563527
height_cm          0.175233679  0.03255715 -0.033106657
 ↪  0.11449815  0.08663982
systolic_bp_mmhg  -0.162631293  0.17911526  0.076744860
 ↪  0.05053639 -0.11228380
diastolic_bp_mmhg -0.115842387  0.24927132  0.006192917
 ↪  0.17171252  0.06245884
glucose_mgdl       0.054530234 -0.21732486  0.485915233
 ↪  -0.30724573 -0.30610735
cholesterol_mgdl   0.130655961 -0.18578891 -0.405818323
 ↪  0.58257658 -0.09940667
crp_ngml           0.235883262 -0.05194062  0.395750869
 ↪  0.22832212  0.23695996
height_m           0.175233679  0.03255715 -0.033106657
 ↪  0.11449815  0.08663982
BMI               -0.092001580  0.04070725  0.063909718
 ↪  0.10919816 -0.22676613
hemoglobin_gdl    -0.437602610  0.01001426 -0.056650636
 ↪  -0.35956963 -0.06861943
wbc_10e9l         -0.619699305 -0.58149117 -0.074295609
 ↪  0.24023399 -0.06313973
platelets_10e9l   -0.411932917  0.29664230  0.441613574
 ↪  0.32569607  0.47072619
```

```
alt_ul            -0.113292712  0.34739457 -0.430660345
 ↪  -0.24955969  0.21554525
creatinine_mgdl   -0.007243073 -0.30189898 -0.172397833
 ↪  -0.24825899  0.61218679
                        PC10        PC11          PC12
 ↪  PC13
age                0.16163362 -0.24928742  0.5340819816
 ↪  -0.207365391
weight_kg         -0.13771123 -0.25816540  0.1098718353
 ↪  -0.132978903
height_cm          0.25249113 -0.10825117  0.0101489045
 ↪  -0.041847706
systolic_bp_mmhg   0.01017092 -0.23922452  0.0182918726
 ↪  0.684845129
diastolic_bp_mmhg -0.09863487  0.21772001 -0.2343978658
 ↪  -0.598159203
glucose_mgdl       0.39160327  0.18757599 -0.4163160098
 ↪  -0.022466399
cholesterol_mgdl   0.08709701  0.40086145 -0.0380195920
 ↪  0.165651557
crp_ngml           0.04359699  0.44822308  0.2115814342
 ↪  0.181013559
height_m           0.25249113 -0.10825117  0.0101489045
 ↪  -0.041847706
BMI               -0.31900769 -0.24816854  0.1140144723
 ↪  -0.135712304
hemoglobin_gdl    -0.12366052  0.50814131  0.4822604936
 ↪  0.017783375
wbc_10e9l          0.24612723 -0.14285067 -0.0690764039
 ↪  -0.070497229
platelets_10e9l    0.10829681 -0.07365816  0.0003121005
 ↪  0.008273408
alt_ul             0.57431722 -0.01800349  0.0274637770
 ↪  -0.018342739
creatinine_mgdl   -0.36814365  0.02023977 -0.4202024057
 ↪  0.155520201
                         PC14          PC15
age               -0.0077207637  3.712802e-16
weight_kg          0.7479755644  2.524768e-16
height_cm         -0.1847014652 -7.071068e-01
systolic_bp_mmhg  -0.0024058682  1.126616e-18
diastolic_bp_mmhg -0.0007140495 -6.330386e-17
glucose_mgdl       0.0026863190 -9.027537e-17
cholesterol_mgdl   0.0053727625 -2.792104e-16
crp_ngml           0.0010498380 -4.024681e-18
```

```
height_m          -0.1847014652  7.071068e-01
BMI               -0.6098707954  2.154088e-16
hemoglobin_gdl    -0.0028866137  3.022242e-17
wbc_10e9l          0.0114953445 -3.120194e-17
platelets_10e9l    0.0098323540  1.924663e-16
alt_ul            -0.0026862862 -4.819710e-16
creatinine_mgdl    0.0036667895 -5.456474e-18
```

We can now create another df in which we have all the info about our patients + the coordinate of the PCs, so that we can plot the data.

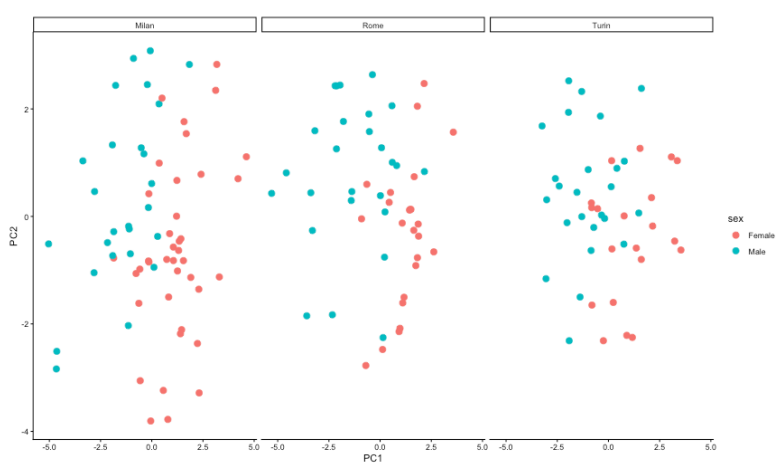
```
pc_data_df <- df |>
    na.omit() |>
    cbind( data.frame(pca_res$x))

head(pc_data_df)
```

```
  patient_id  site visit_date visit_year visit_month visit_day
 ↪  age    sex
2     PT-002  Rome 2020-08-15       2020           8        15
 ↪  40   Male
3     PT-003  Rome 2023-10-15       2023          10        15
 ↪  59 Female
4     PT-004 Turin 2020-03-06       2020           3         6
 ↪  46   Male
5     PT-005 Turin 2023-03-30       2023           3        30
 ↪  58   Male
7     PT-007  Rome 2020-09-09       2020           9         9
 ↪  51   Male
8     PT-008  Rome 2025-10-07       2025          10         7
 ↪  34 Female
  smoking treatment stage weight_kg height_cm systolic_bp_mmhg
2 current    Drug_B     I     100.9     185.3              143
3  former    Drug_A     I      65.0     165.6              152
4 current    Drug_A     I      95.9     181.5              142
5  former    Drug_B   III      75.7     168.1              160
7  former    Drug_A     I      69.7     169.7              137
8   never   Control     I      75.3     159.0              146
  diastolic_bp_mmhg glucose_mgdl cholesterol_mgdl crp_ngml
 ↪  height_m  BMI
2               102          105              239     6.13
 ↪  1.853 29.4
3                98          110              146     7.81
 ↪  1.656 23.7
```

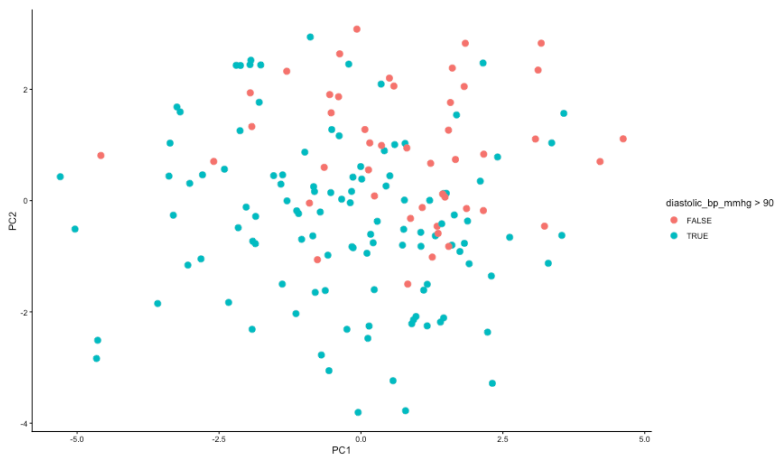
```
4                88          120              232     3.35
 ↪  1.815 29.1
5                96          131              192    21.72
 ↪  1.681 26.8
7                96          108              163     5.41
 ↪  1.697 24.2
8               106           93              199     5.08
 ↪  1.590 29.8
    BMI_class hemoglobin_gdl wbc_10e9l platelets_10e9l alt_ul
 ↪  creatinine_mgdl
2  Overweight           15.0      7.49             270     56
 ↪  1.58
3 Underweight           11.8      7.46             252     27
 ↪  1.09
4  Overweight           16.0      7.09             233     20
 ↪  1.17
5  Overweight           15.4      6.63             192     16
 ↪  1.06
7 Underweight           14.0      6.99             195     61
 ↪  1.15
8  Overweight           16.2      9.26             188     51
 ↪  1.06
         PC1        PC2        PC3         PC4        PC5
 ↪  PC6
2 -3.1848752  1.5940551 -0.8330956  0.42893639 -0.7556020
 ↪  0.15162987
3  1.5043769  0.1310580  1.0588467  0.72295656 -0.2805843
 ↪  -0.02007899
4 -1.9499679  1.9368075 -1.6129773  0.82280702 -0.3562495
 ↪  -0.49865886
5 -0.7172053 -0.2052340  0.9639881  0.60533057  0.1025643
 ↪  -0.62852241
7  0.5943474  1.0061787  0.4715918  0.02971275 -0.2755777
 ↪  0.20823014
8  0.4423734  0.2620532  1.0162111 -0.34260605 -2.5520409
 ↪  0.60211481
         PC7        PC8         PC9        PC10       PC11
 ↪  PC12
2 -1.3256103  0.7741280  1.30447738 -0.64066623  0.2798982
 ↪  -1.41411138
3  1.3199058 -0.5254682  0.16217081 -0.24345315 -1.2285633
 ↪  -0.53511971
4 -0.1358980  0.2480322 -1.07321376 -0.30823140  0.3172393
 ↪  -0.01790244
5  1.8563071 -0.5588202 -1.09376445 -0.29296465  1.1500676
 ↪  0.54120182
```

```
7 -0.3861093 -1.3948701 -0.03683629  0.06404422 -0.1240594
 ↪  -0.31877450
8 -1.2551166 -0.1767724 -1.47110004 -1.50399528  1.0036042
 ↪  -0.11140713
         PC13         PC14          PC15
2 -0.20183004  0.077333696 -2.722210e-15
3 -0.08353174  0.008189665  2.880921e-16
4  0.15939163  0.036984617 -1.381056e-15
5  0.93463483 -0.026541502  1.010757e-15
7 -0.68253066 -0.025616541  5.606148e-16
8 -0.48527809 -0.024698988  1.271017e-16
```

```
ggplot(pc_data_df) +
    geom_point(aes(x = PC1, y = PC2, color = sex), size = 3) +
    facet_wrap(~ site) +
    theme_classic()
```




```
ggplot(pc_data_df) +
    geom_point(aes(x = PC1, y = PC2, color = diastolic_bp_mmhg >
     ↪  90), size = 3) +
    theme_classic()
```



4.6 Linear models

Lastly, let's see linear models, that determine which features are important in predicting or influencing an outcome feature (e.g. BMI/age and diabetes).

4.6.1 Linear model with continuous outcome

If the outcome is continuous, we can use `lm` function.

Let's see if we can create a model that predicts glucose levels starting from all the others variables:

```
lm_data <- df |>
    na.omit() |>
    select(-starts_with("visit_"), - "patient_id")

lm_res <- lm(glucose_mgdl ~ ., data = lm_data)
summary(lm_res)
```

```
Call:
lm(formula = glucose_mgdl ~ ., data = lm_data)

Residuals:
    Min      1Q  Median      3Q     Max 
-30.315  -6.941  -0.241   6.351  33.009 

Coefficients: (1 not defined because of singularities)
```

```
                       Estimate Std. Error t value Pr(>|t|)
(Intercept)          111.128909 130.213184   0.853 0.394951
siteRome              -4.343176   2.797351  -1.553 0.122894
siteTurin             -2.870328   2.862795  -1.003 0.317859
age                    0.466927   0.123516   3.780 0.000236 ***
sexMale                4.224451   3.702029   1.141 0.255871
smokingformer         -6.108882   3.248254  -1.881 0.062204 .
smokingnever           1.023849   3.222333   0.318 0.751184
treatmentDrug_A        2.273550   2.766178   0.822 0.412599
treatmentDrug_B        3.304591   2.991571   1.105 0.271313
stageII               -3.407747   2.874672  -1.185 0.237958
stageIII              -2.920170   3.417514  -0.854 0.394380
stageIV               -9.152404   4.229189  -2.164 0.032243 *
weight_kg              0.466022   0.759459   0.614 0.540512
height_cm             -0.448431   0.742911  -0.604 0.547129
systolic_bp_mmhg       0.077726   0.134848   0.576 0.565323
diastolic_bp_mmhg     -0.127067   0.158403  -0.802 0.423882
cholesterol_mgdl      -0.008258   0.040264  -0.205 0.837802
crp_ngml               0.061043   0.206082   0.296 0.767535
height_m                     NA         NA      NA       NA
BMI                    0.290750   2.305197   0.126 0.899821
BMI_classObese         6.946945  11.355943   0.612 0.541751
BMI_classOverweight    4.870855   4.463734   1.091 0.277155
BMI_classUnderweight  11.372951   7.377695   1.542 0.125563
hemoglobin_gdl         0.577618   0.953527   0.606 0.545700
wbc_10e9l             -0.808985   0.620294  -1.304 0.194419
platelets_10e9l        0.013806   0.020099   0.687 0.493357
alt_ul                -0.004121   0.044427  -0.093 0.926234
creatinine_mgdl       -0.973014   7.784103  -0.125 0.900712
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 13.24 on 133 degrees of freedom
Multiple R-squared:  0.3527,    Adjusted R-squared:  0.2261
F-statistic: 2.787 on 26 and 133 DF,  p-value: 7.005e-05
```

From this we can see that age is significantly associated with the outcome (every year results in a predicted increase of 0.466927), as well as being of stage IV (compared to stage I, -9.152404).

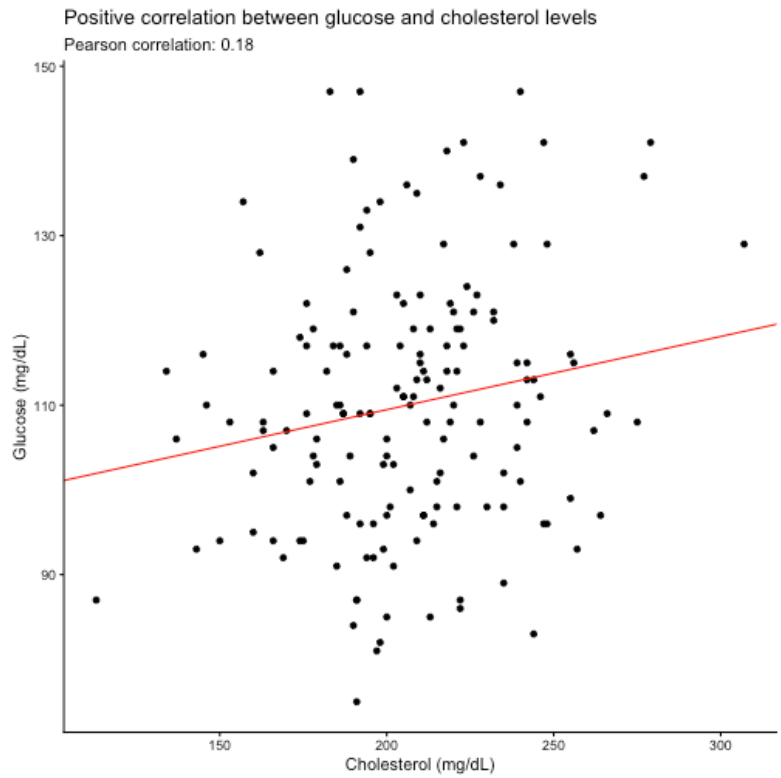
We can also use the same function to create a simple linear model that returns us intercept and slope of the line that correlates *glucose_mgdl* and *cholesterol_mgdl* in stage III patients (remember [#correlation]?):

```
gl_chol_lm <- lm(glucose_mgdl ~ cholesterol_mgdl, data = corr_df)
summary(gl_chol_lm)
```

```

Call:
lm(formula = glucose_mgdl ~ cholesterol_mgdl, data = corr_df)

Residuals:
    Min      1Q  Median      3Q     Max 
-33.691 -11.546   0.438   8.858  39.000 

Coefficients:
                 Estimate Std. Error t value Pr(>|t|)    
(Intercept)      92.18423    7.89821  11.672   <2e-16 ***
cholesterol_mgdl  0.08642    0.03789   2.281   0.0239 *  
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 14.85 on 158 degrees of freedom
Multiple R-squared:  0.03188,	Adjusted R-squared:  0.02575 
F-statistic: 5.203 on 1 and 158 DF,  p-value: 0.02388
```

We got the same results, but we also have *Intercept* and *Estimate*, that we can use to plot a line in corr_plot:

```
corr_plot +
      geom_abline(aes(slope =
       ↪  gl_chol_lm$coefficients["cholesterol_mgdl"],
                      intercept =
                       ↪  gl_chol_lm$coefficients["(Intercept)"]),
                       ↪
                  color = "red")
```



4.6.1.1 Predict

We can then use this model to make predictions. For example, which is the expected value of Glucose given a Cholesterol level of 170?

```
predict(gl_chol_lm, newdata = data.frame("cholesterol_mgdl" =
 ↪  160), type = "response")
```

```
       1
106.0121
```

4.6.2 Generalized linear model with dicotomic outcome

We can also do the other way round (which is more meaningful for us): check which predictors are associated with diabetes, using a generalized linear model:

```r
library(MASS)
```

```
Attaching package: 'MASS'

The following object is masked from 'package:dplyr':

    select
```

```r
df_poli <- lm_data |>
    dplyr::select(-height_m, -BMI_class, -weight_kg, -height_cm,
 ↪  -site, -treatment)


polr_res <- polr(stage ~ ., data = df_poli, method = "logistic")
summary(polr_res)
```

```
Re-fitting to get Hessian
```

```
Call:
polr(formula = stage ~ ., data = df_poli, method = "logistic")

Coefficients:
                       Value Std. Error  t value
age                0.0232525   0.017088  1.36076
sexMale            0.4197909   0.446624  0.93992
smokingformer      0.2306958   0.449945  0.51272
smokingnever       0.7644046   0.466764  1.63767
systolic_bp_mmhg  -0.0068315   0.019111 -0.35746
diastolic_bp_mmhg  0.0132073   0.022251  0.59355
glucose_mgdl      -0.0237946   0.011726 -2.02925
cholesterol_mgdl   0.0032712   0.005369  0.60928
crp_ngml           0.1707731   0.029469  5.79491
BMI               -0.1134834   0.040660 -2.79106
hemoglobin_gdl    -0.2752424   0.129483 -2.12571
wbc_10e9l          0.0151978   0.086915  0.17486
platelets_10e9l    0.0001578   0.002627  0.06005
alt_ul             0.0084260   0.006276  1.34248
creatinine_mgdl    0.9294965   1.049239  0.88588

Intercepts:
       Value   Std. Error t value
```

```
I|II   -4.4825  3.0120    -1.4882
II|III -2.6685  3.0035    -0.8885
III|IV -0.8728  2.9869    -0.2922

Residual Deviance: 361.4394
AIC: 397.4394
```

Important predictors are the ones with t value > 2 or < -2.

Higher levels of CRP are strongly associated with increased odds of being in a more advanced stage; conversely, higher BMI, hemoglobin, and glucose levels are associated with lower odds of advanced stage.

Chapter 5

For loops and If/Else statements

```r
library(tidyverse)
```

So far, every time we needed to repeat an operation on a different subset of data, we either copy-pasted the code or used tidyverse functions like `group_by`. Both approaches have limits: copy-pasting is error-prone and does not scale, while `group_by` only works inside a tidy pipeline.

For loops and **if/else statements** are the building blocks of any programming language and will let you automate repetitive tasks in a much more flexible way.

5.1 For loops

The idea behind a for loop is simple: *repeat a block of code for each element of a sequence.*

The syntax in R is:

```
for (variable in sequence) {
    # code to repeat
}
```

- `sequence` is any vector or list: `c(1, 2, 3)`, `1:10`, a vector of filenames, …

- `variable` is a name you choose; at each iteration it takes the next value of the sequence
- the code inside the `{}` is executed once per element

5.1.1 A first example

The simplest possible loop: print each number from 1 to 5.

```
for (i in 1:5) {
    print(i)
}
```

```
[1] 1
[1] 2
[1] 3
[1] 4
[1] 5
```

`i` here is just a conventional name (short for *index*), but you can call it whatever you want.

5.1.2 Looping over a character vector

You can loop over any vector, not just numbers. Here we loop over a vector of stop codons:

```
stop_codons <- c("UAG", "UAA", "UGA")

for (stop_c in stop_codons) {
    print(paste("Stop codon in this iteration is:", stop_c))
    Sys.sleep(2)
}
```

```
[1] "Stop codon in this iteration is: UAG"
[1] "Stop codon in this iteration is: UAA"
[1] "Stop codon in this iteration is: UGA"
```

Let's say we have a vector of area values in mm^2 and we want to convert each one to μm^2:

```
areas_mm2 <- c(0.0042, 0.0031, 0.0058, 0.0049, 0.0037)

for (area in areas_mm2) {
    area_um2 <- area * 1e6
    print(paste(area, "mm² =", area_um2, "µm²"))
}
```

```
[1] "0.0042 mm² = 4200 µm²"
[1] "0.0031 mm² = 3100 µm²"
[1] "0.0058 mm² = 5800 µm²"
[1] "0.0049 mm² = 4900 µm²"
[1] "0.0037 mm² = 3700 µm²"
```

5.1.3 Collecting results: the importance of initialising a container

In the example above, we just printed the results. Most of the time you want to **store** them. The right way to do this is to create an empty vector *before* the loop and fill it at each iteration.

```
areas_um2 <- c()   # empty container

for (i in 1:length(areas_mm2)) {
  areas_um2[i] <- areas_mm2[i] * 1e6
}

areas_um2
```

```
[1] 4200 3100 5800 4900 3700
```

Notice that we are now looping over the *index* `i` (from 1 to the length of the vector) rather than the values directly. This is very useful when you need to refer to the position of an element, for example to store the result in the same position of another vector.

5.1.4 Looping over a list of dataframes

In biology, a very common scenario is having multiple dataframes — one per sample, one per experiment, etc. — and wanting to apply the same operation to each of them. We can store them in a **list** and loop over the list.

Let's create a small example with 3 mini-dataframes:

```
df1 <- data.frame(group = c("wt", "KO"), intensity = c(450,
 ↪  1420), area = c(0.005, 0.0048))
df2 <- data.frame(group = c("wt", "KO"), intensity = c(510,
 ↪  1380), area = c(0.0052, 0.0051))
df3 <- data.frame(group = c("wt", "KO"), intensity = c(490,
 ↪  1600), area = c(0.0049, 0.0047))

sample_list <- list(df1, df2, df3)
```

Now we loop over the list and calculate the normalised intensity for each dataframe:

```
for (i in 1:length(sample_list)) {

    df <- sample_list[[i]]   # [[]] extracts one element from a
 ↪  list
    df$norm_intensity <- df$intensity / (df$area * 1e6)
    sample_list[[i]] <- df   # store the modified df back in the
 ↪  list

}

# Let's check the first element
sample_list[[1]]
```

```
  group intensity   area norm_intensity
1    wt       450 0.0050      0.0900000
2    KO      1420 0.0048      0.2958333
```

Note the `[[]]` double brackets: to extract a single element from a list you always use `[[]]`, not `[]`.

5.1.5 Naming the elements of a list

It is good practice to name the elements of a list, so that you can refer to them by name later:

```
names(sample_list) <- c("sample_1", "sample_2", "sample_3")

sample_list[["sample_1"]]
```

```
  group intensity   area norm_intensity
1    wt       450 0.0050      0.0900000
2    KO      1420 0.0048      0.2958333
```

5.2 If/Else statements

An `if/else` statement lets you execute different code depending on whether a condition is `TRUE` or `FALSE`.

The syntax is:

```
if (condition) {
    # code if condition is TRUE
} else {
    # code if condition is FALSE
}
```

5.2.1 A simple example

```r
x <- 15

if (x > 10) {
    print("x is greater than 10")
} else {
    print("x is 10 or less")
}
```

```
[1] "x is greater than 10"
```

Change `x` to `5` and re-run to see the other branch.

5.2.2 else if: more than two cases

When you have more than two possible outcomes, you can chain conditions with `else if`:

```
p_value <- 0.03

if (p_value < 0.001) {
    print("Highly significant (***)")
} else if (p_value < 0.01) {
    print("Very significant (**)")
} else if (p_value < 0.05) {
    print("Significant (*)")
} else {
    print("Not significant")
}
```

```
[1] "Significant (*)"
```

R evaluates the conditions in order and executes the **first** block whose condition is `TRUE`. Once a match is found, the rest are skipped.

5.3 Putting it all together: reading files in a loop

In the next session we will do this for real with our immunofluorescence data, but let's preview the full pattern here. The typical workflow is:

1. Get the list of files to process
2. Create an empty list to collect results
3. Loop over the files: read each one, process it, store it in the list
4. Combine all dataframes into one big dataframe with `bind_rows`

```
# 1. Get the list of files
files <- list.files(path = "data/day-5", pattern = "*.csv",
 ↪  full.names = TRUE)
files
```

```
[1] "data/day-5/slide_1.csv" "data/day-5/slide_2.csv"
 ↪  "data/day-5/slide_3.csv"
[4] "data/day-5/slide_4.csv" "data/day-5/slide_5.csv"
```

```
# 2. Empty list to collect results
results_list <- list()

# 3. Loop
for (i in 1:length(files)) {
```

```
  # Read the file
  df <- read.csv(files[i])

  # Add a column with the sample name (useful to know where each
   ↪  row comes from)
  df$sample <- basename(files[i])

  # Convert area from mm2 to um2
  df$area_um2 <- df$area * 1e6

  # Calculate normalised intensity
  df$norm_intensity <- df$intensity / df$area_um2

  # Store in the list
  results_list[[i]] <- df

}

# 4. Combine into one dataframe
results_df <- bind_rows(results_list)

head(results_df)
```

```
  group     area intensity      sample area_um2 norm_intensity
1    wt 0.006371    540.43 slide_1.csv     6371     0.08482656
2    wt 0.004435    489.39 slide_1.csv     4435     0.11034724
3    KO 0.005363    901.15 slide_1.csv     5363     0.16803095
4    KO 0.005633    740.53 slide_1.csv     5633     0.13146281
5    wt 0.007018    361.11 slide_2.csv     7018     0.05145483
6    wt 0.004937    472.12 slide_2.csv     4937     0.09562892
```

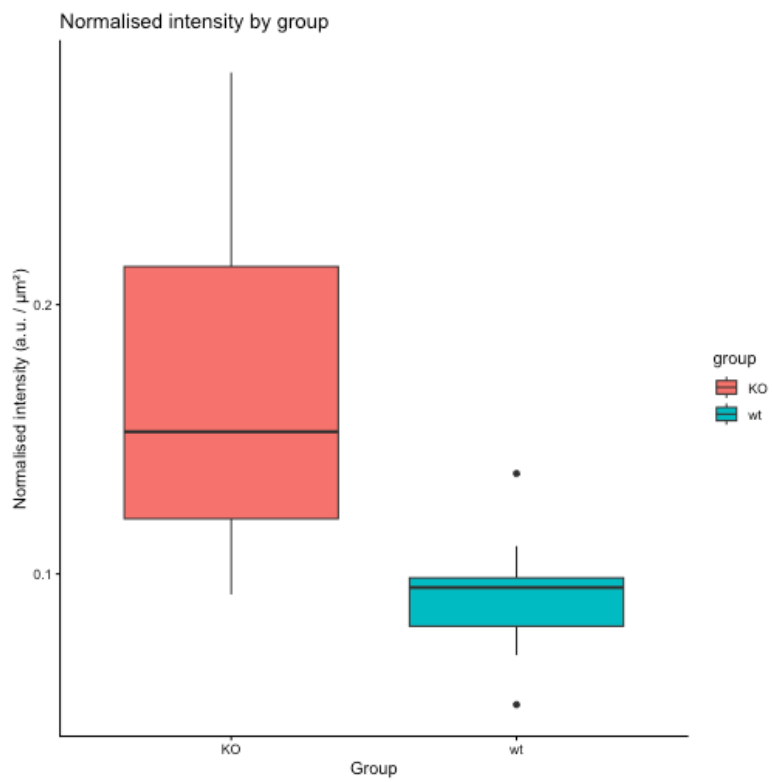
We now have a single tidy dataframe with all samples, ready for statistical analysis.

```
# Quick sanity check: how many rows per sample?
results_df %>%
  count(sample)
```

```
       sample n
1 slide_1.csv 4
2 slide_2.csv 4
3 slide_3.csv 4
4 slide_4.csv 4
```

```
5 slide_5.csv 4
```

```r
# And a first look at the normalised intensity by group
ggplot(results_df) +
  geom_boxplot(aes(x = group, y = norm_intensity, fill = group))
  ↪  +
  labs(title = "Normalised intensity by group",
       y = "Normalised intensity (a.u. / µm²)",
       x = "Group") +
  theme_classic()
```



Chapter 6

Extra: Table 1

Most of clinical paper have a descriptive Table 1 in which case groups are compared to check whether they are balanced for demographic and clinical variables.

Let's see how we can build our own Table 1 with R, starting from our previous table1_dfset.

6.1 Load necessary packages

For this, we need to load some packages:

```
library(tidyverse)
library(rstatix)
```

```
Attaching package: 'rstatix'

The following object is masked from 'package:MASS':

    select

The following object is masked from 'package:stats':

    filter
```

```
library(effsize)
library(knitr)
library(kableExtra)
```

```
Attaching package: 'kableExtra'

The following object is masked from 'package:dplyr':

    group_rows
```

6.2 Load table1_df

```r
df = read.csv("output/clinical_data_enriched.csv",
 ↪  stringsAsFactors = TRUE)
str(df)
```

```
'data.frame':   240 obs. of  26 variables:
 $ patient_id       : Factor w/ 240 levels "PT-001","PT-002",..:
 ↪  1 2 3 4 5 6 7 8 9 10 ...
 $ site             : Factor w/ 3 levels "Milan","Rome",..: 1 2 2
 ↪  3 3 1 2 2 1 2 ...
 $ visit_date       : Factor w/ 218 levels
 ↪  "2018-01-14","2018-01-24",..: 129 75 150 52 104 108 80 214 57
 ↪  146 ...
 $ visit_year       : int  2023 2020 2023 2020 2023 2023 2020
 ↪  2025 2020 2023 ...
 $ visit_month      : int  7 8 10 3 3 4 9 10 4 9 ...
 $ visit_day        : int  20 15 15 6 30 15 9 7 19 27 ...
 $ age              : int  40 40 59 46 58 53 51 34 43 60 ...
 $ sex              : Factor w/ 2 levels "Female","Male": 2 2 1 2
 ↪  2 1 2 1 2 2 ...
 $ smoking          : Factor w/ 3 levels "current","former",..: 3
 ↪  1 2 1 2 1 2 3 3 2 ...
 $ treatment        : Factor w/ 3 levels "Control","Drug_A",..: 2
 ↪  3 2 2 3 1 2 1 3 2 ...
 $ stage            : Factor w/ 4 levels "I","II","III",..: 2 1 1
 ↪  1 3 4 1 1 3 2 ...
 $ weight_kg        : num  85.9 100.9 65 95.9 75.7 ...
 $ height_cm        : num  173 185 166 182 168 ...
 $ systolic_bp_mmhg : int  164 143 152 142 160 150 137 146 151
 ↪  159 ...
 $ diastolic_bp_mmhg: int  101 102 98 88 96 107 96 106 94 104 ...
 $ glucose_mgdl     : int  113 105 110 120 131 78 108 93 106 110
 ↪  ...
 $ cholesterol_mgdl : int  172 239 146 232 192 232 163 199 200
 ↪  185 ...
```

```
 $ crp_ngml         : num  5.89 6.13 7.81 3.35 21.72 ...
 $ height_m         : num  1.73 1.85 1.66 1.81 1.68 ...
 $ BMI              : num  28.8 29.4 23.7 29.1 26.8 22.9 24.2
 ↪  29.8 24.8 26.9 ...
 $ BMI_class        : Factor w/ 4 levels "Normal","Obese",..: 3 3
 ↪  4 3 3 4 4 3 4 3 ...
 $ hemoglobin_gdl   : num  NA 15 11.8 16 15.4 NA 14 16.2 10.5
 ↪  15.6 ...
 $ wbc_10e9l        : num  NA 7.49 7.46 7.09 6.63 NA 6.99 9.26
 ↪  4.79 5.65 ...
 $ platelets_10e9l  : int  NA 270 252 233 192 NA 195 188 163 269
 ↪  ...
 $ alt_ul           : int  NA 56 27 20 16 NA 61 51 50 18 ...
 $ creatinine_mgdl  : num  NA 1.58 1.09 1.17 1.06 NA 1.15 1.06
 ↪  1.21 1.08 ...
```

For demonstration purposes, we will reduce the table1_dfset to only stage I and IV:

```
df = df %>%
    filter(stage %in% c("I", "IV")) %>%
    na.omit()
```

6.3 Select column of interest

As some of these columns are not of interest now (e.g. Patient ID, visit_*, height_m, weight_kg and BMI_class), we can remove them to have a clean table1_dfset:

```
table1_df = df %>%
    dplyr::select(-starts_with("visit_"), -patient_id, -height_m,
 ↪  weight_kg, -BMI_class)
str(table1_df)
```

```
'data.frame':   72 obs. of  19 variables:
 $ site             : Factor w/ 3 levels "Milan","Rome",..: 2 2 3
 ↪  2 2 3 2 1 1 2 ...
 $ age              : int  40 59 46 51 34 75 57 52 45 58 ...
 $ sex              : Factor w/ 2 levels "Female","Male": 2 1 2 2
 ↪  1 1 2 1 2 2 ...
 $ smoking          : Factor w/ 3 levels "current","former",..: 1
 ↪  2 1 2 3 1 2 1 1 1 ...
```

```
 $ treatment        : Factor w/ 3 levels "Control","Drug_A",..: 3
 ↪  2 2 2 1 1 2 2 3 1 ...
 $ stage            : Factor w/ 4 levels "I","II","III",..: 1 1 1
 ↪  1 1 1 1 1 1 4 ...
 $ weight_kg        : num  100.9 65 95.9 69.7 75.3 ...
 $ height_cm        : num  185 166 182 170 159 ...
 $ systolic_bp_mmhg : int  143 152 142 137 146 149 124 141 138
 ↪  130 ...
 $ diastolic_bp_mmhg: int  102 98 88 96 106 97 85 95 100 78 ...
 $ glucose_mgdl     : int  105 110 120 108 93 117 129 107 96 106
 ↪  ...
 $ cholesterol_mgdl : int  239 146 232 163 199 204 238 170 248
 ↪  137 ...
 $ crp_ngml         : num  6.13 7.81 3.35 5.41 5.08 ...
 $ BMI              : num  29.4 23.7 29.1 24.2 29.8 30.2 23.9
 ↪  18.2 24.9 26.2 ...
 $ hemoglobin_gdl   : num  15 11.8 16 14 16.2 12.7 15.4 12.9 14.7
 ↪  12.4 ...
 $ wbc_10e9l        : num  7.49 7.46 7.09 6.99 9.26 6.37 6.42
 ↪  6.07 9.23 7.77 ...
 $ platelets_10e9l  : int  270 252 233 195 188 288 241 260 275
 ↪  195 ...
 $ alt_ul           : int  56 27 20 61 51 36 41 32 29 42 ...
 $ creatinine_mgdl  : num  1.58 1.09 1.17 1.15 1.06 1.18 1.2 1.03
 ↪  1.38 1.34 ...
 - attr(*, "na.action")= 'omit' Named int [1:42] 4 7 10 18 21 23
 ↪  24 25 30 33 ...
  ..- attr(*, "names")= chr [1:42] "4" "7" "10" "18" ...
```

6.4 Define group levels

First thing first, let's define grouping variables and check that there are only 2 levels:

```
group.var = "stage"
table1_df[[group.var]] = droplevels(table1_df[[group.var]])
group_levels <- unique(na.omit(table1_df[[group.var]]))
if (length(group_levels) != 2)
    stop(sprintf(
        "`group.var` must have exactly 2 levels; '%s' has %d:
         ↪  %s",
        group.var, length(group_levels), paste(group_levels,
 ↪  collapse = ", ")
```

```
    ))

group_levels
```

```
[1] I  IV
Levels: I IV
```

6.4.1 Group sizes

Usually group sizes are at the top of the table, so we can start with them

```
sizes <- table1_df %>%
    dplyr::count(.data[[group.var]]) %>%
    dplyr::arrange(.data[[group.var]])


g1_name <- sprintf("%s (n = %d)", sizes[[1, 1]], sizes[[1, 2]])
g2_name <- sprintf("%s (n = %d)", sizes[[2, 1]], sizes[[2, 2]])

print(g1_name)
```

```
[1] "I (n = 47)"
```

```
print(g2_name)
```

```
[1] "IV (n = 25)"
```

6.5 Detect variable types

Now we can divide variables into numeric and categorical, so that we can treat them separately.

```
#   Auto-detect variable types
# Exclude the grouping variable from candidates
candidate_cols <- setdiff(colnames(table1_df), group.var)

# Everything else (factor, character, logical, or low-cardinality
 ↪  numeric) → categorical
cont.var <- candidate_cols[sapply(candidate_cols, function(col) {
```

```
    is.numeric(table1_df[[col]]) &&
     ↪  length(unique(na.omit(table1_df[[col]]))) > 3
})]


cat.var <- candidate_cols[sapply(candidate_cols, function(col) {
    is.factor(table1_df[[col]]) ||
        is.character(table1_df[[col]]) ||
        is.logical(table1_df[[col]]) ||
        (is.numeric(table1_df[[col]]) &&
 ↪  length(unique(na.omit(table1_df[[col]]))) <= 3)
})]

message(sprintf(
    "Auto-detected: %d continuous variable(s), %d categorical
     ↪  variable(s).\n  Continuous : %s\n  Categorical: %s",
    length(cont.var), length(cat.var),
    if (length(cont.var)) paste(cont.var, collapse = ", ") else
     ↪  "none",
    if (length(cat.var))  paste(cat.var,  collapse = ", ") else
     ↪  "none"
))
```

```
Auto-detected: 14 continuous variable(s), 4 categorical variable(s).
  Continuous : age, weight_kg, height_cm, systolic_bp_mmhg, diastolic_bp_mmhg, glucose_
  Categorical: site, sex, smoking, treatment
```

```
if (length(cont.var) == 0) cont.var <- NULL
if (length(cat.var)  == 0) cat.var  <- NULL
```

6.6 Continuous variables

Let's start with continuous variables. For each, we calculate p value of the differences in the two groups of the grouping variable, applying the right test based on parametric assumptions tests.

```
col_names <- c(
    "Variable", g1_name, g2_name, "p-value", "Method", "Effect
     ↪  Size"
)

# Helper: build an empty row with the right number of columns
```

```
empty_row <- function(var, g1 = "", g2 = "", p = NA_real_, method
 ↪  = "", es = "") {
    row <- list(var, g1, g2, p, method, list(es))
    setNames(as.data.frame(row, stringsAsFactors = FALSE),
     ↪  col_names)
}


result_rows <- list()

#   Continuous variables
if (!is.null(cont.var)) {
    for (vname in cont.var) {

        x    <- table1_df[[vname]]
        grp  <- table1_df[[group.var]]
        vals <- split(x, grp)

        # Summary: mean ± SD per group (ignoring NAs)
        fmt_mean_sd <- function(v) {
            v <- na.omit(v)
            sprintf("%.3f \u00b1 %.3f", mean(v), sd(v))   # ± as
             ↪  unicode
        }
        g1_val <- fmt_mean_sd(vals[[1]])
        g2_val <- fmt_mean_sd(vals[[2]])

        # Normality check (Shapiro-Wilk, only reliable for n
         ↪  3-5000)
        safe_shapiro <- function(v) {
            v <- na.omit(v)
            if (length(v) < 3 || length(v) > 5000)
             ↪  return(list(p.value = 0))  # force non-parametric
            shapiro.test(v)
        }
        sw1 <- safe_shapiro(vals[[1]])
        sw2 <- safe_shapiro(vals[[2]])
        both_normal <- sw1$p.value > 0.05 & sw2$p.value > 0.05

        if (both_normal) {
            # Bartlett test for equal variances
            bart_p <- bartlett.test(x ~ grp)$p.value
            equal_var <- bart_p > 0.05

            test_res <- t.test(x ~ grp, var.equal = equal_var)
            p_val    <- test_res$p.value
```

```
            method   <- trimws(test_res$method)

            # Cohen's d effect size
            es_val <- {
                d <- tryCatch(effsize::cohen.d(x ~ grp,
 ↪  )$estimate, error = function(e) NA)
                if (!is.na(d)) sprintf("d = %.3f", d) else "d =
                 ↪  NA"
            }

        } else {
            # Non-parametric: Wilcoxon / Mann-Whitney
            test_res <- suppressWarnings(wilcox.test(x ~ grp))
            p_val    <- test_res$p.value
            method   <- trimws(test_res$method)

            # Rank-biserial correlation as effect size
            es_val <- {
                rb <- tryCatch(
                    rstatix::wilcox_effsize(table1_df,
 ↪  as.formula(paste(vname, "~", group.var)))$effsize,
                    error = function(e) NA
                )
                if (!is.na(rb)) sprintf("r = %.3f", rb) else "r =
                 ↪  NA"
            }
        }

        result_rows <- c(result_rows, list(empty_row(vname,
 ↪  g1_val, g2_val, p_val, method, es_val)))
    }
}

result_rows[[1]]
```

```
  Variable      I (n = 47)     IV (n = 25)   p-value
 ↪  Method
1      age 55.234 ± 11.493 57.520 ± 10.170 0.4064722 Two Sample
 ↪  t-test
  Effect Size
1  d = -0.207
```

6.7 Categorical variables

We can do the same for categorical variables.

```
if (!is.null(cat.var)) {
    for (vname in cat.var) {

        x   <- table1_df[[vname]]
        grp <- table1_df[[group.var]]

        cont_tab <- table(x, grp)
        n_levels <- nrow(cont_tab)
        n_total  <- sum(cont_tab)

        # Percentages within each group
        pct <- prop.table(cont_tab, margin = 2) * 100
        g1_cells <- sprintf("%d (%.1f%%)", cont_tab[, 1], pct[,
↪   1])
        g2_cells <- sprintf("%d (%.1f%%)", cont_tab[, 2], pct[,
↪   2])

        # Test selection
        expected <- suppressWarnings(chisq.test(x, grp)$expected)
        use_fisher <- n_levels == 2 && n_total <= 100 &&
↪   any(expected < 5)

        if (use_fisher) {
            test_res   <- suppressWarnings(fisher.test(cont_tab))
            p_cat      <- test_res$p.value
            method_cat <- "Fisher's Exact Test"
        } else {
            use_yates  <- n_levels == 2 && n_total <= 100
            test_res   <- suppressWarnings(chisq.test(x, grp,
↪   correct = use_yates))
            p_cat      <- test_res$p.value
            method_cat <- trimws(test_res$method)
        }

        # Cramér's V effect size
        es_cat <- {
            v <- tryCatch(rstatix::cramer_v(cont_tab), error =
↪   function(e) NA)
            if (!is.na(v)) sprintf("V = %.3f", v) else "V = NA"
        }

        # Header row (variable name + p-value)
```

```
        result_rows <- c(result_rows, list(
            empty_row(vname, "", "", p_cat, method_cat, es_cat)
        ))

        # Level rows (indented with 2 spaces for readability)
        for (j in seq_len(n_levels)) {
            result_rows <- c(result_rows, list(
                empty_row(paste0("  ", rownames(cont_tab)[j]),
                          g1_cells[j], g2_cells[j])
            ))
        }
    }
}
```

6.8 Combine everything

We can combine everything now to have one table

```
out <- dplyr::bind_rows(result_rows)

# Format p-values: < 0.001 notation, round to 3 decimal places
 ↪  elsewhere
out[["p-value"]] <- sapply(out[["p-value"]], function(p) {
    if (is.na(p)) return("")
    if (p < 0.001) return("< 0.001")
    formatC(p, digits = 3, format = "f")
})

out
```

```
            Variable       I (n = 47)      IV (n = 25) p-value
1                age  55.234 ± 11.493  57.520 ± 10.170   0.406
2          weight_kg  78.836 ± 17.151  70.016 ± 19.384   0.051
3          height_cm  170.051 ± 9.863  169.248 ± 8.294   0.730
4   systolic_bp_mmhg 146.000 ± 10.807 144.080 ± 11.694   0.488
5  diastolic_bp_mmhg   94.000 ± 8.925   96.280 ± 9.062   0.308
6       glucose_mgdl 112.660 ± 14.918 102.480 ± 12.007   0.004
7   cholesterol_mgdl 202.234 ± 29.909 190.760 ± 33.686   0.143
8           crp_ngml    7.484 ± 5.133   16.605 ± 8.665 < 0.001
9                BMI   27.160 ± 4.846   24.228 ± 5.740   0.025
10    hemoglobin_gdl   13.796 ± 1.576   12.412 ± 0.909 < 0.001
11         wbc_10e9l    6.879 ± 1.871    6.736 ± 2.092   0.767
12   platelets_10e9l 229.085 ± 44.603 209.480 ± 56.791   0.111
```

```
13            alt_ul  49.043 ± 26.280  55.040 ± 33.768   0.500
14   creatinine_mgdl    1.139 ± 0.178    1.144 ± 0.172   0.908
15              site                                     0.035
16             Milan       14 (29.8%)       12 (48.0%)
17              Rome       22 (46.8%)        4 (16.0%)
18             Turin       11 (23.4%)        9 (36.0%)
19               sex                                     0.863
20            Female       22 (46.8%)       13 (52.0%)
21              Male       25 (53.2%)       12 (48.0%)
22           smoking                                     0.346
23           current       10 (21.3%)        9 (36.0%)
24            former       17 (36.2%)        6 (24.0%)
25             never       20 (42.6%)       10 (40.0%)
26         treatment                                     0.049
27           Control       18 (38.3%)        4 (16.0%)
28            Drug_A       17 (36.2%)        8 (32.0%)
29            Drug_B       12 (25.5%)       13 (52.0%)
                                                         Method
 ↪  Effect Size
1                                          Two Sample t-test
 ↪  d = -0.207
2                                          Two Sample t-test
 ↪  d = 0.491
3                                          Two Sample t-test
 ↪  d = 0.086
4                                          Two Sample t-test
 ↪  d = 0.173
5                                          Two Sample t-test
 ↪  d = -0.254
6                                          Two Sample t-test
 ↪  d = 0.728
7                                          Two Sample t-test
 ↪  d = 0.367
8            Wilcoxon rank sum test with continuity correction
 ↪  r = NA
9                                          Two Sample t-test
 ↪  d = 0.567
10                                    Welch Two Sample t-test
 ↪  d = 1.000
11                                         Two Sample t-test
 ↪  d = 0.074
12                                         Two Sample t-test
 ↪  d = 0.399
13           Wilcoxon rank sum test with continuity correction
 ↪  r = NA
```

```
14                                                  Two Sample t-test
 ↪  d = -0.029
15                                       Pearson's Chi-squared test
 ↪  V = 0.306
16
 ↪
17
 ↪
18
 ↪
19 Pearson's Chi-squared test with Yates' continuity correction
 ↪  V = 0.020
20
 ↪
21
 ↪
22                                       Pearson's Chi-squared test
 ↪  V = 0.172
23
 ↪
24
 ↪
25
 ↪
26                                       Pearson's Chi-squared test
 ↪  V = 0.289
27
 ↪
28
 ↪
29
 ↪
```

6.9 Save it

We can now save it as html, and later on open it and save it as pdf.

This is the helper function needed to create it.

```
format_desc_table <- function(tbl, output = "html", save_pdf =
 ↪  FALSE, pdf_path = "table.pdf") {

  #   Extract methods for footnotes
  is_header      <- !is.na(tbl[["p-value"]]) & tbl[["p-value"]]
 ↪  != ""
```

```r
methods_raw    <- tbl[["Method"]][is_header]
unique_methods <- unique(methods_raw[methods_raw != ""])

#   Inject superscript letters next to p-values
pval_col <- tbl[["p-value"]]
pval_col[is_header] <- mapply(function(p, m) {
  if (m == "" || is.na(m)) return(p)
  idx <- which(unique_methods == m)
  if (output == "html") {
    paste0(p, "<sup>", letters[idx], "</sup>")
  } else {
    paste0(p, "$^", letters[idx], "$")
  }
}, tbl[["p-value"]][is_header], methods_raw, SIMPLIFY = TRUE)

#   Drop Method column
out              <- tbl[, !colnames(tbl) %in% "Method"]
out[["p-value"]] <- pval_col

#   Significance colouring
p_numeric <- suppressWarnings(as.numeric(tbl[["p-value"]]))
p_sig     <- p_numeric < 0.05 | tbl[["p-value"]] == "< 0.001"
p_color   <- ifelse(p_sig & !is.na(p_sig), "red", "black")

#   Identify categorical header rows (p-value present, group
 ↪  cols empty)
# These are variable name rows for categorical vars - need bold
 ↪  separately
g1_col       <- colnames(out)[2]
is_cat_header <- is_header & (is.na(out[[g1_col]]) |
↪  out[[g1_col]] == "")

#   Alignment
align <- c("l", rep("c", ncol(out) - 1))

#   Build table
tab <- kbl(out,
           escape = FALSE,
           align  = align,
           format = output) |>
  # Bold all header rows (continuous vars + categorical var
   ↪  names)
  row_spec(seq_len(nrow(out)), extra_css = "height: 2em;") |>
  column_spec(1, width = "18em") |>
  column_spec(which(colnames(out) == "p-value"), color =
   ↪  p_color)
```

```
# Bold categorical variable name rows in column 1
# (row_spec bold applies to whole row; column_spec lets us
 ↪  target just col 1
#  for the level rows that should NOT be bold)
if (any(is_cat_header)) {
  tab <- tab |>
    column_spec(1,
                width = "18em",
                bold  = is_cat_header)   # TRUE only for
                 ↪  categorical headers
}

tab <- tab |>
  footnote(
    general           = "Continuous variables expressed as mean
     ↪  ± SD; categorical as n (%).",
    # One test per line via <br> in HTML, \\\\ newline in LaTeX
     ↪  # ← CHANGE 3
    alphabet          = if (output == "html") {
                          paste0(unique_methods, "<br>")
                        } else {
                          unique_methods                  #
↪  LaTeX handles line breaks natively
                        },
    general_title     = "",
    alphabet_title    = "Statistical tests: ",
    footnote_as_chunk = FALSE,     # FALSE so each alphabet
     ↪  entry stays separate
    escape            = FALSE
  )

if (output == "html") {
  tab <- tab |> kable_classic(full_width = FALSE)
} else {
  tab <- tab |> kable_styling(
    latex_options = c("striped", "hold_position",
     ↪  "scale_down"),
    font_size     = 9
  )
}

#   Optionally save as HTML
if (save_pdf) {
  save_kable(tab, file = gsub("\\.pdf$", ".html", pdf_path),
   ↪  self_contained = TRUE)
```

```
    message("Saved as HTML (open in browser → Print → Save as
     ↪  PDF): ",
            gsub("\\.pdf$", ".html", pdf_path))
  }

  tab
}
```

And here is how we can call it.

```
#format_desc_table(out, output = "html", save_pdf = TRUE,
 ↪  pdf_path = "my_table.pdf")
```